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Inventor(s)

HASHIMOTO; Yasunobu et al.

DISPLAY CONTROLLING METHOD, TIME CONTROLLING METHOD, TIME MANAGING METHOD, AND BROADCAST RECEIVING APPARATUS

Abstract

A display controlling method for a broadcast receiving apparatus allowed to receive IP simultaneous broadcasting of a 4K broadcasting service via a communication transmission line, in which simultaneous broadcasting of the 4K broadcasting service and a 2K broadcasting service is executed by a broadcasting wave, is configured to: receive and demodulate the 4K broadcasting service, and display a broadcasting program of the 4K broadcasting service; and select and execute one display switching process of first and second display switching processes in accordance with priority order in a case where a defect occurs in reception, demodulation, or display of the 4K broadcasting service, the 2K broadcasting service being received and demodulated to display a broadcasting program of the 2K broadcasting service in the first display switching process, the IP simultaneous broadcasting being received and demodulated to display a program of the IP simultaneous broadcasting in the second display switching process.

Inventors: HASHIMOTO; Yasunobu (Kyoto, JP), YOSHIZAWA; Kazuhiko (Kyoto, JP), SHIMIZU; Takuya (Kyoto, JP), AKIYAMA; Hitoshi (Kyoto, JP)

Applicant: MAXELL, LTD. (Kyoto, JP)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation under 35 U.S.C. § 111(a) of U.S. patent application Ser. No. 18/491,098, filed on Oct. 20, 2023, which is a continuation under 35 U.S.C. § 111(a) of U.S. patent application Ser. No. 17/779,461, filed on May 24, 2022, now U.S. Pat. No. 11,831,951 issued on Nov. 28, 2023, which in turn is the U.S. National Phase under 35 U.S.C. § 371 of International Application No. PCT/JP2020/042666, filed on Nov. 16, 2020, which claims the benefit of Japanese Application No. 2019-212000, filed on Nov. 25, 2019, the entire contents of each of which applications are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The present invention relates to a broadcast transmission technique or a broadcast reception technique.

BACKGROUND ART

[0003] A digital broadcasting service has been started in each country since the latter half of the 1990s in place of a conventional analog broadcasting service. The digital broadcasting service realized improvement of broadcasting quality using an error correction technique, multi-channel and HD (High Definition) using a compression coding technique, and multimedia of a service using a BML (Broadcast Markup Language) or HTML5 (Hyper Text Markup Language version 5).

[0004] In recent years, for the purpose of further improvement of a frequency usage efficiency, higher resolution, and higher functionality, consideration of an advanced digital broadcasting method is caused to proceed in each country.

RELATED ART DOCUMENTS

Patent Documents

[0005] Patent document 1: Japanese Patent Application Publication No. 2016-144020

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

[0006] More than ten years have already passed since the service of the existing digital broadcasting was started, and broadcast receiving apparatuses capable of receiving an existing digital broadcasting service are widespread sufficiently. For this reason, it is necessary to consider compatibility with the existing digital broadcasting service when an advanced digital broadcasting service, which is currently considered, is to be started. Namely, it is preferable that UHD (Ultra High Definition) of a video signal is realized while maintaining viewing environment of the existing digital broadcasting service.

[0007] There is a system described in Patent document 1 as a technique for realizing UHD

broadcasting in the digital broadcasting service. However, the system described in Patent document 1 is one by which the existing digital broadcasting is replaced, and does not take into consideration the maintenance of the viewing environment of the existing digital broadcasting service.

[0008] It is an object of the present invention to a technique for transmitting or receiving a higher functional advanced digital broadcasting service more suitably in consideration of compatibility with an existing digital broadcasting service.

Means for Solving the Problem

[0009] Techniques described in the following claims are used as means for solving the problem.

[0010] As one example, a display controlling method for a broadcast receiving apparatus allowed to receive a broadcasting wave via a broadcast transmission path and allowed to receive IP simultaneous broadcasting of the 4K broadcasting service via a communication transmission line, in which simultaneous broadcasting of a 4K broadcasting service and a 2K broadcasting service is executed by the broadcasting wave, may be configured to: receive the 4K broadcasting service included in the broadcasting wave, demodulate the 4K broadcasting service, and display a broadcasting program of the 4K broadcasting service; and select and execute one display switching process of a two display switching processes of a first display switching process and a second display switching process in accordance with a predetermined priority order in a case where a defect occurs in reception, demodulation, or display of the 4K broadcasting service, the 2K broadcasting service that is simultaneous broadcasting of the 4K broadcasting service via the broadcast transmission path being received and demodulated to display a broadcasting program of the 2K broadcasting service in the first display switching process, the IP simultaneous broadcasting via the communication transmission line that is simultaneous broadcasting of the 4K broadcasting service being received and demodulated to display a program of the IP simultaneous broadcasting in the second display switching process.

Effects of the Invention

[0011] According to the present invention, it is possible to provide a technique for transmitting or receiving an advanced digital broadcasting service more suitably.

Description

BRIEF DESCRIPTIONS OF THE DRAWINGS

[0012] FIG. 1 is a system configuration diagram of a broadcasting system according to one embodiment of the present invention;

[0013] FIG. 2A is a block diagram of a broadcast receiving apparatus according to one embodiment of the present invention;

[0014] FIG. 2B is a detailed block diagram of a first tuner/demodulator in the broadcast receiving apparatus according to one embodiment of the present invention;

[0015] FIG. 2C is a detailed block diagram of a second tuner/demodulator in the broadcast receiving apparatus according to one embodiment of the present invention;

[0016] FIG. 2D is a detailed block diagram of a third tuner/demodulator in the broadcast receiving apparatus according to one embodiment of the present invention;

[0017] FIG. 2E is a detailed block diagram of a fourth tuner/demodulator in the broadcast receiving apparatus according to one embodiment of the present invention;

[0018] FIG. 2F is a detailed block diagram of a first decoder in the broadcast receiving apparatus according to one embodiment of the present invention;

[0019] FIG. 2G is a detailed block diagram of a second decoder in the broadcast receiving apparatus according to one embodiment of the present invention;

[0020] FIG. 2H is a software configuration diagram of the broadcast receiving apparatus according to one embodiment of the present invention;

[0021] FIG. 3A is a configuration diagram of a broadcast station server according to one embodiment of the present invention;

[0022] FIG. 3B is a configuration diagram of a service provider server according to one embodiment of the present invention;

[0023] FIG. 4A is a diagram for explaining a segment configuration for digital broadcasting according to one embodiment of the present invention;

[0024] FIG. 4B is a diagram for explaining layer allocation in layer transmission for the digital broadcasting according to one embodiment of the present invention;

[0025] FIG. 4C is a diagram for explaining a generating process for an OFDM transmission wave for the digital broadcasting according to one embodiment of the present invention;

[0026] FIG. 4D is a diagram for explaining a basic configuration of a transmission path coder for the digital broadcasting according to one embodiment of the present invention;

[0027] FIG. 4E is a diagram for explaining an OFDM type segment parameter for the digital broadcasting according to one embodiment of the present invention;

[0028] FIG. 4F is a diagram for explaining a transmission signal parameter for the digital broadcasting according to one embodiment of the present invention;

[0029] FIG. 4G is a diagram for explaining arrangement of a pilot signal of a synchronous modulation segment for the digital broadcasting according to one embodiment of the present invention;

[0030] FIG. 4H is a diagram for explaining arrangement of a pilot signal of a differential modulation segment for the digital broadcasting according to one embodiment of the present invention;

[0031] FIG. 5A is a diagram for explaining bit allocation of a TMCC carrier for the digital broadcasting according to one embodiment of the present invention;

[0032] FIG. 5B is a diagram for explaining bit allocation of TMCC information for the digital broadcasting according to one embodiment of the present invention;

[0033] FIG. 5C is a diagram for explaining transmission parameter information of the TMCC information for the digital broadcasting according to one embodiment of the present invention;

[0034] FIG. 5D is a diagram for explaining system identification of the TMCC information for the digital broadcasting according to one embodiment of the present invention;

[0035] FIG. 5E is a diagram for explaining a carrier modulation mapping method of the TMCC information for the digital broadcasting according to one embodiment of the present invention;

[0036] FIG. 5F is a diagram for explaining frequency converting process identification of the TMCC information for the digital broadcasting according to one embodiment of the present invention;

[0037] FIG. 5G is a diagram for explaining physical channel number identification of the TMCC information for the digital broadcasting according to one embodiment of the present invention;

[0038] FIG. 5H is a diagram for explaining one example of main signal identification of the TMCC information for the digital broadcasting according to one embodiment of the present invention;

[0039] FIG. 5I is a diagram for explaining 4K signal transmission layer identification of the TMCC information for the digital broadcasting according to one embodiment of the present invention;

[0040] FIG. 5J is a diagram for explaining additional layer transmission identification of the TMCC information for the digital broadcasting according to one embodiment of the present invention;

[0041] FIG. 5K is a diagram for explaining identification of a code rate of an inner code of the TMCC information for the digital broadcasting according to one embodiment of the present invention;

[0042] FIG. 6A is a diagram for explaining bit allocation of an AC signal for the digital broadcasting according to one embodiment of the present invention;

[0043] FIG. 6B is a diagram for explaining configuration identification of the AC signal for the

digital broadcasting according to one embodiment of the present invention;

[0044] FIG. 6C is a diagram for explaining seismic motion warning information of the AC signal for the digital broadcasting according to one embodiment of the present invention;

[0045] FIG. 6D is a diagram for explaining signal identification of the seismic motion warning information of the AC signal for the digital broadcasting according to one embodiment of the present invention;

[0046] FIG. 6E is a diagram for explaining seismic motion warning detailed information of the seismic motion warning information of the AC signal for the digital broadcasting according to one embodiment of the present invention;

[0047] FIG. 6F is a diagram for explaining the seismic motion warning detailed information of the seismic motion warning information of the AC signal for the digital broadcasting according to one embodiment of the present invention;

[0048] FIG. 6G is a diagram for explaining additional information regarding a transmission control of a modulation wave of the AC signal for the digital broadcasting according to one embodiment of the present invention;

[0049] FIG. 6H is a diagram for explaining transmission parameter additional information of the AC signal for the digital broadcasting according to one embodiment of the present invention;

[0050] FIG. 6I is a diagram for explaining an error correcting method of the AC signal for the digital broadcasting according to one embodiment of the present invention;

[0051] FIG. 6J is a diagram for explaining an NUC form of the AC signal for the digital broadcasting according to one embodiment of the present invention;

[0052] FIG. 7A is a diagram for explaining a dual-polarized transmission method according to one embodiment of the present invention;

[0053] FIG. 7B is a system configuration diagram of the broadcasting system using the dual-polarized transmission method according to one embodiment of the present invention;

[0054] FIG. 7C is a system configuration diagram of the broadcasting system using the dual-polarized transmission method according to one embodiment of the present invention;

[0055] FIG. 7D is a diagram for explaining a frequency converting process according to one embodiment of the present invention;

[0056] FIG. 7E is a diagram for explaining a configuration of a pass-through transmission method according to one embodiment of the present invention;

[0057] FIG. 7F is a diagram for explaining a pass-through transmission band according to one embodiment of the present invention;

[0058] FIG. 7G is a diagram for explaining a configuration of the pass-through transmission method according to one embodiment of the present invention;

[0059] FIG. 7H is a diagram for explaining the pass-through transmission band according to one embodiment of the present invention;

[0060] FIG. 7I is a diagram for explaining the pass-through transmission band according to one embodiment of the present invention;

[0061] FIG. 8A is a diagram for explaining a layered division multiplexing transmission method according to one embodiment of the present invention;

[0062] FIG. 8B is a system configuration diagram of the broadcasting system using the layered division multiplexing transmission method according to one embodiment of the present invention;

[0063] FIG. 8C is a diagram for explaining a frequency conversion amplification process according to one embodiment of the present invention;

[0064] FIG. 9A is a diagram for explaining a protocol stack of an MPEG-2 TS;

[0065] FIG. 9B is a diagram for explaining names and functions of tables used in the MPEG-2 TS;

[0066] FIG. 9C is a diagram for explaining names and functions of the tables used in the MPEG-2 TS;

[0067] FIG. 9D is a diagram for explaining names and functions of descriptors used in the MPEG-2

TS;

[0068] FIG. **9E** is a diagram for explaining names and functions of descriptors used in the MPEG-2 TS;

[0069] FIG. **9F** is a diagram for explaining names and functions of descriptors used in the MPEG-2 TS;

[0070] FIG. **9G** is a diagram for explaining names and functions of descriptors used in the MPEG-2 TS;

[0071] FIG. **9H** is a diagram for explaining names and functions of descriptors used in the MPEG-2 TS;

[0072] FIG. **9I** is a diagram for explaining names and functions of descriptors used in the MPEG-2 TS;

[0073] FIG. **10A** is a diagram for explaining a protocol stack in a broadcast transmission path of MMT;

[0074] FIG. **10B** is a diagram for explaining the protocol stack in a communication line of the MMT;

[0075] FIG. **10C** is a diagram for explaining names and functions of tables used in TLV-SI of the MMT;

[0076] FIG. **10D** is a diagram for explaining names and functions of descriptors used in the TLV-SI of the MMT;

[0077] FIG. **10E** is a diagram for explaining names and functions of messages used in MMT-SI of the MMT;

[0078] FIG. **10F** is a diagram for explaining names and functions of tables used in the MMT-SI of the MMT;

[0079] FIG. **10G** is a diagram for explaining names and functions of descriptors used in the MMT-SI of the MMT;

[0080] FIG. **10H** is a diagram for explaining names and functions of descriptors used in the MMT-SI of the MMT;

[0081] FIG. **10I** is a diagram for explaining names and functions of descriptors used in the MMT-SI of the MMT;

[0082] FIG. **10J** is a diagram for explaining a relation between data transmission of an MMT method and respective tables;

[0083] FIG. **11A** is an operation sequence diagram of a channel setting process by a broadcast receiving apparatus **100** according to one embodiment of the present invention;

[0084] FIG. **11B** is a diagram for explaining a data structure of a network information table;

[0085] FIG. **11C** is a diagram for explaining a data structure of a terrestrial delivery system descriptor;

[0086] FIG. **11D** is a diagram for explaining a data structure of a service list descriptor;

[0087] FIG. **11E** is a diagram for explaining a data structure of a TS information descriptor;

[0088] FIG. **12A** is an appearance diagram of a remote controller according to one embodiment of the present invention;

[0089] FIG. **12B** is a diagram for explaining banner display at the time of channel selection according to one embodiment of the present invention;

[0090] FIG. **13** is a diagram for explaining a system clock configuration according to one embodiment of the present invention;

[0091] FIG. **14** is a diagram for explaining a data format of a TS packet in an MPEG2-TS method;

[0092] FIG. **15A** is a diagram for explaining a configuration of an NTP format;

[0093] FIG. **15B** is a diagram for explaining a data structure of an MPU timestamp descriptor;

[0094] FIG. **16** is a diagram for explaining guidance for switching into backup broadcasting when a reception state deteriorates;

[0095] FIG. **17** is a diagram for explaining a case where a portable information terminal executes

the guidance for switching into backup broadcasting when the reception state deteriorates;
[0096] FIG. **18A** is a diagram for explaining leap second adjustment for time insertion in an NTP system clock;
[0097] FIG. **18B** is a diagram for explaining leap second adjustment for time deletion in the NTP system clock;
[0098] FIG. **19A** is a diagram for explaining a data structure of TOT;
[0099] FIG. **19B** is a diagram for explaining a data structure of JST_time; and
[0100] FIG. **20** is a diagram for explaining a data structure of an EIT.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0101] Hereinafter, an example of an embodiment of the present invention will be described with reference to the drawings.

First Embodiment

[System Configuration]

[0102] FIG. **1** is a system configuration diagram illustrating one example of a configuration of a broadcasting system.

[0103] The broadcasting system is configured by, for example, a broadcast receiving apparatus **100** and an antenna **200**, a radio tower **300** and a broadcast station server **400** of a broadcast station, a service provider server **500**, a mobile phone communication server **600** and a base station **600B** for a mobile phone communication network, a portable information terminal **700**, and a broadband network **800** such as the Internet and a router device **800R**. Further, various kinds of server apparatuses and communication equipment may further be connected to the Internet **800**.

[0104] The broadcast receiving apparatus **100** is a television receiver provided with a receiving function for an advanced digital broadcasting service. The broadcast receiving apparatus **100** may further include a receiving function for an existing digital broadcasting service. Moreover, the broadcast receiving apparatus **100** is compatible with a broadcasting/communication cooperation system that combines acquisition of additional content via a broadband network, arithmetic processing by a server apparatus, a presenting process by cooperation with portable terminal equipment, and the like with the digital broadcasting service by linking functions using the broadband network to the digital broadcasting service (the existing digital broadcasting service or the advanced digital broadcasting service). The broadcast receiving apparatus **100** receives a digital broadcasting wave sent out from the radio tower **300** via the antenna **200**. The digital broadcasting wave may directly be transmitted from the radio tower **300** to the antenna **200**, or may be transmitted via a broadcast satellite or a communication satellite (whose illustration is omitted). A broadcasting signal retransmitted by a cable television station may be received via a cable line or the like. Further, the broadcast receiving apparatus **100** can be connected to the Internet **800** via the router device **800R**, and can transmit and receive data by communication with each server apparatus on the Internet **800**.

[0105] The router device **800R** is connected to the Internet **800** with wireless communication or wired communication; is connected to the broadcast receiving apparatus **100** with wired communication; and is connected to the portable information terminal **700** with wireless communication. This makes it possible to mutually execute transmission and reception of data among each server apparatus on the Internet **800**, the broadcast receiving apparatus **100**, and the portable information terminal **700** via the router device **800R**. The router device **800R**, the broadcast receiving apparatus **100**, and the portable information terminal **700** constitute a LAN (Local Area Network). Note that communication between the broadcast receiving apparatus **100** and the portable information terminal **700** may be executed directly by a method such as BlueTooth (registered trademark) or NFC (Near Field Communication) without the router device **800R**.

[0106] The radio tower **300** is a broadcasting facility of the broadcast station, and sends out the digital broadcasting wave including various kinds of control information and content data of a broadcasting program (such as moving image content or audio content) according to the digital

broadcasting service. Further, the broadcast station includes the broadcast station server **400**. The broadcast station server **400** stores metadata such as content data of broadcasting programs and program titles of the respective broadcasting programs, program IDs, program summaries, casts, broadcasting dates, or the like. The broadcast station server **400** provides the content data or the metadata with a service provider on the basis of a contract. Provision of the content data and the metadata to the service provider is executed through an API (Application Programming Interface) included in the broadcast station server **400**.

[0107] The service provider server **500** is a server apparatus that the service provider prepares for providing a service by the broadcasting/communication cooperation system. The service provider server **500** executes storage, management, distribution, and the like of the content data and the metadata provided from the broadcast station server **400** and content data and applications (such as at least one of operating program or various kinds of data) created for the broadcasting/communication cooperation system. Further, the service provider server **500** also has a function to execute search of providable applications or provision of a list in response to an inquiry from the television receiver. Note that different server apparatuses may respectively execute storage, management, and distribution of the content data and the metadata, and storage, management, and distribution of the applications. The broadcast station may be the same as the service provider, or may be a different provider. A plurality of service provider servers **500** may respectively be prepared for different services. Further, the broadcast station server **400** may be combined with the functions of the service provider server **500**.

[0108] The mobile phone communication server **600** is connected to the Internet **800**, and is also connected to the portable information terminal **700** via the base station **600B**. The mobile phone communication server **600** manages telephone communication (telephone call), and transmission and reception of data via the mobile phone communication network of the portable information terminal **700**, and allows transmission and reception of data by communication between the portable information terminal **700** and each server apparatus on the Internet **800**. Note that the communication between the portable information terminal **700** and the broadcast receiving apparatus **100** may be executed via the base station **600B**, the mobile phone communication server **600**, the Internet **800**, and the router device **800R**.

[Hardware Configuration of Broadcast Receiving Apparatus]

[0109] FIG. 2A is a block diagram illustrating one example of an internal configuration of the broadcast receiving apparatus **100**.

[0110] The broadcast receiving apparatus **100** is configured by a main controller **101**, a system bus **102**, a ROM **103**, a RAM **104**, a storage (accumulator) **110**, a LAN communication unit **121**, an extension interface **124**, a digital interface **125**, a first tuner/demodulator **130C**, a second tuner/demodulator **130T**, a third tuner/demodulator **130L**, a fourth tuner/demodulator **130B**, a first decoder **140S**, a second decoder **140U**, an operation input unit **180**, a video selector **191**, a monitor **192**, a video output unit **193**, an audio selector **194**, a speaker **195**, and an audio output unit **196**.

[0111] The main controller **101** is a microprocessor unit that controls the whole broadcast receiving apparatus **100** in accordance with a predetermined operating program. The system bus **102** is a communication path for executing transmission and reception of various kinds of data or commands between the main controller **101** and each of operating blocks in the broadcast receiving apparatus **100**.

[0112] The ROM (Read Only Memory) **103** is a non-volatile memory in which a basic operating program such as an operating system and other operating programs are stored, for example, and a rewritable ROM such as an EEPROM (Electrically Erasable Programmable ROM) or a flash ROM is used for the ROM **103**. Further, operating set values and the like necessary for operations of the broadcast receiving apparatus **100** are stored in the ROM **103**. The RAM (Random Access Memory) **104** becomes a work area when the basic operating program or the other operating program is executed. The ROM **103** and the RAM **104** may be configured integrally with the main

controller **101**. Further, the ROM **103** is not configured independently as illustrated in FIG. 2A, but a partial memory region in the storage (accumulator) **110** may be used instead.

[0113] An operating program and operating set values of the broadcast receiving apparatus **100**, and personal information of a user of the broadcast receiving apparatus **100** are stored in the storage (accumulator) **110**. Further, an operating program downloaded via the Internet **800** and various kinds of data created by the operating program can be stored in the storage (accumulator) **110**. Further, content such as moving images, still images, or audios, which are obtained from a broadcast wave or downloaded via the Internet **800** can be stored in the storage (accumulator) **110**. All or a part of functions of the ROM **103** may be substituted by a partial area of the storage (accumulator) **110**. Further, the storage (accumulator) **110** is required to hold stored information even in a state where power source is not supplied to the broadcast receiving apparatus **100** from the outside. Therefore, for example, a device including a semiconductor element memory such as a flash ROM or an SSD (Solid State Drive) or a magnetic disk drive such as an HDD (Hard Disc Drive) is used as the storage (accumulator) **110**.

[0114] Note that each of the operating programs stored in the ROM **103** or the storage (accumulator) **110** can be added, updated, and functionally extended by a downloading process from each server apparatus on the Internet **800** or the broadcast wave.

[0115] The LAN communication unit **121** is connected to the Internet **800** via the router device **800R**, and executes transmission and reception of data with each of the server apparatuses or the other communication equipment on the Internet **800**. Further, the LAN communication unit **121** executes acquisition of content data (or a part thereof) of a program, which are transmitted via a communication line. The LAN communication unit **121** may be connected to the router device **800R** through wired communication or wireless communication such as Wi-Fi (registered trademark). The LAN communication unit **121** includes a coding circuit and a decoding circuit. Further, the broadcast receiving apparatus **100** may further be provided with the other communication unit such as a Bluetooth (registered trademark) communication unit, an NFC communication unit, or an infrared communication unit.

[0116] Each of the first tuner/demodulator **130C**, the second tuner/demodulator **130T**, the third tuner/demodulator **130L**, and the fourth tuner/demodulator **130B** receives the broadcast wave of the digital broadcasting service, and executes a channel selecting process (channel selection) by tuning a channel for a predetermined service on the basis of a control of the main controller **101**.

Moreover, each of the first tuner/demodulator **130C**, the second tuner/demodulator **130T**, the third tuner/demodulator **130L**, and the fourth tuner/demodulator **130B** executes a demodulating process of a modulation wave from a received signal, a waveform shaping process, a reconfiguring process of a frame structure and a layer structure, energy dispersal removal, and error correction decoding process to reproduce a packet stream. Further, each of the first tuner/demodulator **130C**, the second tuner/demodulator **130T**, the third tuner/demodulator **130L**, and the fourth tuner/demodulator **130B** executes an extracting and decoding process for a transmission TMCC (Transmission Multiplexing Configuration Control) signal from the received signal.

[0117] Note that a digital broadcasting wave of an existing terrestrial digital broadcasting service received by an antenna **200C**, which is an antenna for receiving existing terrestrial digital broadcasting, can be inputted into the first tuner/demodulator **130C**. Further, one broadcasting signal of a polarized wave of a horizontal (H) polarization signal and a vertical (V) polarization signal of dual-polarized terrestrial digital broadcasting (will be described later) can also be inputted into the first tuner/demodulator **130C** to demodulate a segment of a layer adopting the same modulation method as that of the existing terrestrial digital broadcasting service. Further, a broadcasting signal of layered division multiplexing terrestrial digital broadcasting (will be described later) can also be inputted into the first tuner/demodulator **130C** to demodulate the layer adopting the same modulation method as that of the existing terrestrial digital broadcasting service. A digital broadcasting wave of an advanced terrestrial digital broadcasting service received by an

antenna **200T**, which is an antenna for receiving the dual-polarized terrestrial digital broadcasting, is inputted into the second tuner/demodulator **130T** via a converter **201T**. The digital broadcasting wave of the advanced terrestrial digital broadcasting service received by an antenna **200L**, which is an antenna for receiving the layered division multiplexing terrestrial digital broadcasting, is inputted into the third tuner/demodulator **130L** via a converter **201L**. A digital broadcasting wave of an advanced BS (Broadcasting Satellite) digital broadcasting service or an advanced CS (Communication Satellite) digital broadcasting service received by an antenna **200B**, which is an antenna for receiving both BS/CS, is inputted into the fourth tuner/demodulator **130B** via a converter **201B**.

[0118] Note that the expression “tuner/demodulator” means a component provided with a tuner function and a demodulator function.

[0119] Further, the antenna **200C**, the antenna **200T**, the antenna **200L**, the antenna **200B**, the converter **201T**, the converter **201L**, the converter **201B** do not constitute a part of the broadcast receiving apparatus **100**, and belong to a facility side such as a building in which the broadcast receiving apparatus **100** is installed.

[0120] Further, the existing terrestrial digital broadcasting described above is a broadcasting signal of a terrestrial digital broadcasting service for transmitting video whose maximum resolution is 1920 horizontal pixels×1080 vertical pixels.

[0121] Further, details of the dual-polarized terrestrial digital broadcasting (advanced terrestrial digital broadcasting adopting a dual-polarized transmission method) will be described later. However, it is a broadcasting signal of the terrestrial digital broadcasting service in which video whose maximum resolution is the number of pixels that exceeds 1920 horizontal pixels×1080 vertical pixels can be transmitted. The dual-polarized terrestrial digital broadcasting is terrestrial digital broadcasting using a plurality of polarized waves including a horizontal (H) polarized wave and a vertical (V) polarized wave. The terrestrial digital broadcasting service in which the video whose maximum resolution is the number of pixels that exceeds 1920 horizontal pixels×1080 vertical pixels can be transmitted is transmitted by a part of divided segments in both the polarized waves of the plurality of polarized waves.

[0122] Note that in the explanation of each embodiment of the present invention, in a case where the expression “a plurality of polarized waves” is used for the dual-polarized terrestrial digital broadcasting, it means the two polarized waves of the horizontal (H) polarized wave and the vertical (V) polarized wave unless otherwise stated. Further, even in case where the expression “polarized wave” is simply used, it means the “polarization signal”. Further, in one polarized wave or both the polarized waves of the plurality of polarized waves, the existing terrestrial digital broadcasting described above in which the video whose maximum resolution is 1920 horizontal pixels×1080 vertical pixels is transmitted by transmitted a part of divided segments can be transmitted by the same modulation method. Namely, in dual-polarized terrestrial digital broadcasting, by different segments of the plurality of polarized waves according to each embodiment of the present invention, the existing terrestrial digital broadcasting service in which the video whose maximum resolution is 1920 horizontal pixels×1080 vertical pixels is transmitted and the terrestrial digital broadcasting service in which the video whose maximum resolution is the number of pixels that exceeds 1920 horizontal pixels×1080 vertical pixels can be transmitted can be transmitted at the same time.

[0123] Further, although details of the layered division multiplexing terrestrial digital broadcasting (that is, the advanced terrestrial digital broadcasting adopting a layered division multiplexing transmission method) will be described later, it is a broadcasting signal of the terrestrial digital broadcasting service in which the video whose maximum resolution is the number of pixels that exceeds 1920 horizontal pixels×1080 vertical pixels can be transmitted. The layered division multiplexing terrestrial digital broadcasting multiplexes a plurality of digital broadcasting signals whose signal levels are different from each other. Note that the digital broadcasting signals whose

signal levels are different from each other means that electric powers with which the digital broadcasting signals are transmitted are different from each other. The layered division multiplexing terrestrial digital broadcasting according to each embodiment of the present invention can transmit, as the plurality of digital broadcasting signals whose signal levels are different from each other, a broadcasting signal of the existing terrestrial digital broadcasting service in which video whose maximum resolution is 1920 horizontal pixels×1080 vertical pixels is transmitted and a broadcasting signal of the terrestrial digital broadcasting service in which the video whose maximum resolution is the number of pixels that exceeds 1920 horizontal pixels×1080 vertical pixels can be transmitted by multiplexing these broadcasting signals as layers with a frequency band of the same physical channel. Namely, in the layered division multiplexing terrestrial digital broadcasting according to each embodiment of the present invention, by a plurality of layers whose signal levels are different from each other, it is possible to transmit the existing terrestrial digital broadcasting service in which the video whose maximum resolution is 1920 horizontal pixels×1080 vertical pixels is transmitted and the terrestrial digital broadcasting in which video whose maximum resolution is the number of pixels that exceeds 1920 horizontal pixels×1080 vertical pixels can be transmitted at the same time.

[0124] Note that the broadcast receiving apparatus according to each embodiment of the present invention may be configured so as to be capable of suitably receiving advanced digital broadcasting, and it is not essential to include all of the first tuner/demodulator **130C**, the second tuner/demodulator **130T**, the third tuner/demodulator **130L**, and the fourth tuner/demodulator **130B**. For example, the broadcast receiving apparatus may include at least one of the second tuner/demodulator **130T** or the third tuner/demodulator **130L**. Further, in order to realize a more advanced function, the broadcast receiving apparatus may be configured so as to one or plural of the four tuner/demodulators described above in addition to one of the second tuner/demodulator **130T** or the third tuner/demodulator **130L**.

[0125] Further, the antenna **200C**, the antenna **200T**, and the antenna **200L** may be used together appropriately. Further, among the first tuner/demodulator **130C**, the second tuner/demodulator **130T**, and the third tuner/demodulator **130L**, a plurality of tuners/demodulators may be used together (or integrated) appropriately.

[0126] A packet stream outputted from the first tuner/demodulator **130C**, the second tuner/demodulator **130T**, the third tuner/demodulator **130L**, and the fourth tuner/demodulator **130B** or a packet stream obtained from each server apparatus on the Internet **800** via the LAN communication unit **121** is inputted into each of the first decoder **140S** and the second decoder **140U**. The packet stream inputted by each of the first decoder **140S** and the second decoder **140U** may be a packet stream with any form such as an MPEG (Moving Picture Experts Group)-2 TS (Transport Stream), an MPEG-2 PS (Program Stream), a TLV (Type Length Value), or an MMT (MPEG Media Transport).

[0127] Each of the first decoder **140S** and the second decoder **140U** executes a conditional access (Conditional Access: CA) process, a demultiplexing process of demultiplexing and extracting various kinds of information data, such as video data or audio data, from the packet stream on the basis of various kinds of control information contained in the packet stream, a decoding process for the video data and the audio data, a process of obtaining program information and generating an EPG (Electronic Program Guide: electronic program guide), a reproducing process of a data broadcasting screen or multimedia data, and the like. Further, each of the first decoder **140S** and the second decoder **140U** also executes a process of superimposing the generated EPG or the reproduced multimedia data onto the decoded video data or the decoded audio data.

[0128] The video data outputted from the first decoder **140S** and the video data outputted from the second decoder **140U** are inputted into the video selector **191**, and the video selector **191** appropriately executes at least one of a selecting process or a superimposing process on the basis of a control of the main controller **101**. Further, the video selector **191** appropriately executes a

scaling process and a superimposing process of OSD (On Screen Display) data. The monitor **192** is a display device such as a liquid crystal panel for example. The monitor **192** displays video data that are subjected to the selecting/superimposing process in the video selector **191** to provide the user of the broadcast receiving apparatus **100** therewith. The video output unit **193** is a video output interface for outputting the video data that are subjected to the selecting/superimposing process in the video selector **191** to the outside.

[0129] The audio data outputted from the first decoder **140S** and the audio data outputted from the second decoder **140U** are inputted into the audio selector **194**, and the audio selector **194** appropriately executes at least one of a selecting process or a mixing process on the basis of a control of the main controller **101**. The speaker **195** outputs sound based on the audio data that are subjected to the selecting/mixing process in the audio selector **194** to provide the user of the broadcast receiving apparatus **100** therewith. The audio output unit **196** is an audio output interface for outputting the audio data that are subjected to the selecting/mixing process in the audio selector **194** to the outside.

[0130] The digital interface **125** is an interface for outputting or inputting the packet stream including at least one of coded digital video data or coded digital audio data. The digital interface **125** can output the packet stream that is inputted from any of the first tuner/demodulator **130C**, the second tuner/demodulator **130T**, the third tuner/demodulator **130L**, and the fourth tuner/demodulator **130B** by the first decoder **140S** or the second decoder **140U** as it is. Further, the digital interface **125** may input the packet stream inputted from the outside via the digital interface **125** into the first decoder **140S** or the second decoder **140U**, or may control the storage (accumulator) **110** to store the packet stream. Alternatively, the digital interface **125** may output the video data or the audio data, which are demultiplexed and extracted in the first decoder **140S** or the second decoder **140U**. Further, the digital interface **125** may input the video data or the audio data inputted from the outside via the digital interface **125** into the first decoder **140S** or the second decoder **140U**, or may control the storage (accumulator) **110** to store the video data or the audio data.

[0131] The extension interface **124** is a group of interfaces for extending the function of the broadcast receiving apparatus **100**, and is configured by an analog video/audio interface, a USB (Universal Serial Bus) interface, a memory interface, or the like. The analog video/audio interface executes an input of an analog video signal/audio signal from external video/audio output equipment, an output of an analog video signal/audio signal to the external video/audio input equipment. The USB interface executes transmission and reception of data to and from a PC by being connected to the PC. The USB interface may execute recording of broadcasting program or the other content data by being connected to an HDD. Further, the USB interface may execute connection to a keyboard or the other USB equipment. The memory interface executes transmission and reception of data by being connected to a memory card or the other memory medium.

[0132] The operation input unit **180** is an instruction input unit configured to execute an input of an operation instruction to the broadcast receiving apparatus **100**, and is configured by a remote controller receiver for receiving a command transmitted from a remote controller (remote controller) (whose illustration is omitted) and an operation key configured by arranging button switches. The operation input unit **180** may be configured by any one of the remote controller receiver and the operation key. Further, the operation input unit **180** can be substituted by a touch panel or the like disposed so as to be overlapped on the monitor **192**. The operation input unit **180** may be substituted by a keyboard or the like connected to the extension interface **124**. The remote controller can be substituted by the portable information terminal **700** provided with a remote controller command transmitting function.

[0133] Note that in a case where the broadcast receiving apparatus **100** is the television receiver or the like, the video output unit **193** and the audio output unit **196** are not essential configuration.

Further, the broadcast receiving apparatus **100** may be an optical disc drive recorder such as a DVD (Digital Versatile Disc) recorder, a magnetic disk drive recorder such as an HDD recorder, an STB (Set Top Box), or the like. The broadcast receiving apparatus **100** may be a PC (Personal Computer) provided with a receiving function of the digital broadcasting service, a tablet terminal, or the like. In a case where the broadcast receiving apparatus **100** is the DVD recorder, the HDD recorder, or the STB, the monitor **192** and the speaker **195** are not essential configuration. By connecting the video output unit **193** and the audio output unit **196** or the digital interface **125** to an external monitor and an external speaker, the broadcast receiving apparatus **100** can execute the similar operations to those of a television receiver or the like.

[0134] FIG. 2B is a block diagram illustrating one example of a detailed configuration of the first tuner/demodulator **130C**.

[0135] An existing digital broadcasting wave received by the antenna **200C** is inputted into the tuning/wave detector **131C**, and the tuning/wave detector **131C** executes channel selection on the basis of a channel selection control signal. A TMCC decoder **132C** extracts a TMCC signal from an output signal of the tuning/wave detector **131C**, and obtains various kinds of TMCC information. The obtained TMCC information is used for controlling respective latter processes. Details of the TMCC signal and the TMCC information will be described later.

[0136] A modulation wave modulated by using a method such as QPSK (Quadrature Phase Shift Keying), DQPSK (Differential QPSK), 16 QAM (Quadrature Amplitude Modulation), or 64 QAM is inputted into the demodulator **133C** on the basis of the TMCC information and the like, and the demodulator **133C** executes a demodulating process including frequency deinterleaving, time deinterleaving, and a carrier demapping process. The demodulator **133C** may further be compatible with a modulation method different from the respective modulation methods described above.

[0137] A stream reproducer **134C** executes a layer dividing process, an inner code error correcting process such as viterbi decoding, energy dispersal removal, a stream reproducing process, and an outer code error correcting process such as RS (Reed Solomon) decoding. Note that as the error correcting process, a method different from each method described above may be used. Further, a packet stream reproduced by and outputted from the stream reproducer **134C** may be an MPEG-2 TS or the like, for example. The packet stream may be the other form of packet stream.

[0138] FIG. 2C is a block diagram illustrating one example of a detailed configuration of the second tuner/demodulator **130T**.

[0139] The horizontal (H) polarization signal of the digital broadcasting wave received by the antenna **200T** is inputted into a tuning/wave detector **131H**, and the tuning/wave detector **131H** executes channel selection on the basis of the channel selection control signal. The vertical (V) polarization signal of the digital broadcasting wave received by the antenna **200T** is inputted into a tuning/wave detector **131V**, and the tuning/wave detector **131V** executes channel selection on the basis of the channel selection control signal. Note that an operation of the channel selecting process by the tuning/wave detector **131H** and an operation of the channel selecting process by the tuning/wave detector **131V** may be controlled in conjunction with each other, or may be controlled independently from each other. Namely, by assuming that the tuning/wave detector **131H** and the tuning/wave detector **131V** are one tuning/wave detector, it is possible to control the one tuning/wave detector to select one channel of the digital broadcasting service transmitted by using both horizontal/vertical polarized waves. By assuming that the tuning/wave detector **131H** and the tuning/wave detector **131V** are two independent tuning/wave detectors, it is possible to control the two independent tuning/wave detectors to respectively select two different channels of the digital broadcasting service transmitted by only the horizontal polarized wave (or only the vertical polarized wave).

[0140] Note that the horizontal (H) polarization signal and the vertical (V) polarization signal received by the second tuner/demodulator **130T** of the broadcast receiving apparatus according to each embodiment of the present invention may be a polarization signal by a broadcast wave whose

polarization direction is different by approximately 90 degrees, and a configuration regarding their reception between the horizontal (H) polarization signal and the vertical (V) polarization signal, which will be described below, may be reversed.

[0141] A TMCC decoder **132H** extracts the TMCC signal from an output signal of the tuning/wave detector **131H** to obtain various kinds of TMCC information. A TMCC decoder **132V** extracts the TMCC signal from an output signal of the tuning/wave detector **131V** to obtain various kinds of TMCC information. Any one of the TMCC decoder **132H** and the TMCC decoder **132V** may be provided. The obtained TMCC information is used for controlling respective latter processes.

[0142] A modulation wave modulated by using a method such as BPSK (Binary Phase Shift Keying), DBPSK (Differential BPSK), QPSK, DQPSK, 8PSK (Phase Shift Keying), 16APSK (Amplitude and Phase Shift Keying), 32 APSK, 16 QAM, 64 QAM, 256 QAM, or 1024 QAM is inputted into each of a demodulator **133H** and a demodulator **133V** on the basis of the TMCC information and the like, and each of the demodulator **133H** and the demodulator **133V** executes a demodulating process including frequency deinterleaving, time deinterleaving, and carrier demapping process. The demodulator **133H** and the demodulator **133V** may further be compatible with a modulation method different from the respective modulation methods described above.

[0143] Each of a stream reproducer **134H** and a stream reproducer **134V** executes a layer dividing process, an inner code error correcting process such as viterbi decoding or LDPC (Low Density Parity Check) decoding, energy dispersal removal, a stream reproducing process, and an outer code error correcting process such as RS decoding or BCH decoding. Note that as the error correcting process, one different from each of the methods described above may be used. Further, a packet stream reproduced by and outputted from the stream reproducer **134H** is an MPEG-2 TS or the like, for example. A packet stream reproduced by and outputted from the stream reproducer **134V** is a TLV including the MPEG-2 TS and an MMT packet stream, for example. Each of the packet streams may be the other form of packet stream.

[0144] FIG. 2D is a block diagram illustrating one example of detailed configuration of the third tuner/demodulator **130L**.

[0145] A digital broadcasting wave subjected to a layered division multiplexing (Layered Division Multiplexing: LDM) process is inputted into a tuning/wave detector **131L** from the antenna **200L**, and the tuning/wave detector **131L** executes channel selection on the basis of the channel selection control signal. The digital broadcasting wave subjected to the layered division multiplexing process may be used for transmission of a digital broadcasting service (or a different channel of the same broadcasting service) in which a modulation wave of an upper layer (Upper Layer: UL) is different from a modulation wave of a lower layer (Lower Layer: LL). Further, the modulation wave of the upper layer and the modulation wave of the lower layer are respectively outputted to a demodulator **133S** and a demodulator **133L**.

[0146] The modulation wave of the upper layer and the modulation wave of the lower layer outputted from the tuning/wave detector **131L** is inputted into a TMCC decoder **132L**, and the TMCC decoder **132L** extracts the TMCC signal to obtain various kinds of TMCC information. A signal inputted into the TMCC decoder **132L** may be any one of the modulation wave of the upper layer and the modulation wave of the lower layer.

[0147] The demodulator **133S** and the demodulator **133L** respectively execute the similar operations to those of the demodulator **133H** and the demodulator **133V**. For this reason, detailed explanation thereof is omitted. Further, a stream reproducer **134S** and a stream reproducer **134L** respectively execute the similar operations to those of the stream reproducer **134H** and the stream reproducer **134V**. For this reason, detailed explanation thereof is omitted.

[0148] FIG. 2E is a block diagram illustrating one example of a detailed configuration of the fourth tuner/demodulator **130B**.

[0149] A digital broadcasting wave of an advanced BS digital broadcasting service or an advanced CS digital broadcasting service received by the antenna **200B** is inputted into a tuning/wave

detector **131B**, and the tuning/wave detector **131B** executes channel selection on the basis of the channel selection control signal. The other operations are similar to those of the tuning/wave detector **131H** and the tuning/wave detector **131V**. For this reason, detailed explanation thereof is omitted. Further, a TMCC decoder **132B**, a demodulator **133B**, and a stream reproducer **134B** also execute the similar operations of the TMCC decoder **132H** or the TMCC decoder **132V**, the demodulator **133H** or the demodulator **133V**, and the stream reproducer **134V**, respectively. For this reason, detailed explanation thereof is omitted.

[0150] FIG. 2F is a block diagram illustrating one example of a detailed configuration of the first decoder **140S**.

[0151] A selector **141S** selects, on the basis of a control of the main controller **101**, one from the packet stream inputted from the first tuner/demodulator **130C**, the packet stream inputted from the second tuner/demodulator **130T**, and the packet stream inputted from the third tuner/demodulator **130L**, and outputs the selected one. The packet stream inputted from each of the first tuner/demodulator **130C**, the second tuner/demodulator **130T**, and the third tuner/demodulator **130L** is the MPEG-2 TS or the like, for example. A CA descrambler **142S** executes a releasing process of encoding algorithm with a predetermined scramble method on the basis of various kinds of control information regarding conditional access superimposed onto the packet stream.

[0152] A demultiplexer **143S** is a stream decoder, and demultiplexes and extracts video data, audio data, caption data, subtitle data, or program information data on the basis of various kinds of control information contained in the inputted packet stream. The demultiplexed and extracted video data are distributed to a video decoder **145S**; the demultiplexed and extracted audio data are distributed to an audio decoder **146S**; the demultiplexed and extracted caption data, subtitle data, and program information data are distributed to a data decoder **144S**. The packet stream obtained from the server apparatus on the Internet **800** via the LAN communication unit **121** (for example, the MPEG-2 PS or the like) may be inputted into the demultiplexer **143S**. Further, the demultiplexer **143S** can output the packet stream inputted from each of the first tuner/demodulator **130C**, the second tuner/demodulator **130T**, and the third tuner/demodulator **130L** via the digital interface **125** to the outside, and the packet stream obtained from the outside via the digital interface **125** can be inputted into the demultiplexer **143S**.

[0153] The video decoder **145S** executes a decoding process of video information subjected to compression coding, a colorimetry converting process to the decoded video information, and a dynamic range converting process with respect to the video data inputted from the demultiplexer **143S**. Further, the video decoder **145S** executes a process of resolution conversion (up/down conversion) based on a control of the main controller **101**, and appropriately outputs video data with resolution such as UHD (3840 horizontal pixels×2160 vertical pixels), HD (1920 horizontal pixels×1080 vertical pixels), or SD (720 horizontal pixels×480 vertical pixels). The output of the video data may be executed with the other resolution. The audio decoder **146S** executes a decoding process of audio information subjected to compression coding. Further, the audio decoder **146S** executes a downmixing process based on a control of the main controller **101**, and outputs the audio data by channel number such as 22.2 ch, 7.1 ch, 5.1 ch, or 2 ch. Note that the first decoder **140S** may include a plurality of video decoders **145S** and a plurality of audio decoders **146S** in order to execute a plurality of decoding processes for video data and audio data at the same time.

[0154] The data decoder **144S** executes a process of generating an EPG on the basis of the program information data, a data broadcasting screen generating process based on BML data, and a controlling process of a cooperation application based on a broadcasting/communication cooperating function. The data decoder **144S** includes a BML browser function of executing a BML document, and the data broadcasting screen generating process is executed by the BML browser function. Further, the data decoder **144S** executes a process of decoding the caption data and generating caption information, and a process of decoding the subtitle data and generating subtitle information.

[0155] Each of a superimposer **147S**, a superimposer **148S**, and a superimposer **149S** executes a superimposing process for the video data outputted from the video decoder **145S**, the EPG outputted from the data decoder **144S**, and the data broadcasting screen. A composer **151S** executes a process of synthesizing the audio data outputted from the audio decoder **146S** and the audio data reproduced by the data decoder **144S**. A selector **150S** executes resolution selection of the video data based on a control of the main controller **101**. Note that the functions of the superimposer **147S**, the superimposer **148S**, the superimposer **149S**, and the selector **150S** may be integrated with that of the video selector **191**. The function of the composer **151S** may be integrated with that of the audio selector **194**.

[0156] FIG. 2G is a block diagram illustrating one example of a detailed configuration of the second decoder **140U**.

[0157] A selector **141U** selects one from the packet stream inputted from the second tuner/demodulator **130T**, the packet stream inputted from the third tuner/demodulator **130L**, and the packet stream inputted from the fourth tuner/demodulator **130B** on the basis of a control of the main controller **101**, and outputs the selected one. The packet stream inputted from each of the second tuner/demodulator **130T**, the third tuner/demodulator **130L**, and the fourth tuner/demodulator **130B** is an MMT packet stream or TLV or the like including the MMT packet stream, for example. The packet stream may be a packet stream of an MPEG-2 TS form that adopts HEVC (High Efficiency Video Coding) to a video compressing method. A CA descrambler **142U** executes a releasing process of encoding algorithm with a predetermined scramble method on the basis of various kinds of control information regarding conditional access superimposed onto the packet stream.

[0158] A demultiplexer **143U** is a stream decoder, and demultiplexes and extracts video data, audio data, caption data, subtitle data, or program information data on the basis of various kinds of control information contained in the inputted packet stream. The demultiplexed and extracted video data are distributed to a video decoder **145U**; the demultiplexed and extracted audio data are distributed to an audio decoder **146U**; the demultiplexed and extracted caption data, subtitle data, and program information data are distributed to a multimedia decoder **144U**. The packet stream obtained from the server apparatus on the Internet **800** via the LAN communication unit **121** (for example, the MPEG-2 PS, the MMT packet stream, or the like) may be inputted into the demultiplexer **143U**. Further, the demultiplexer **143U** can output the packet stream inputted from each of the second tuner/demodulator **130T**, the third tuner/demodulator **130L**, and the fourth tuner/demodulator **130B** via the digital interface **125** to the outside, and the packet stream obtained from the outside via the digital interface **125** can be inputted into the demultiplexer **143U**.

[0159] The multimedia decoder **144U** executes a process of generating an EPG, a multimedia screen generating process based on multimedia data, and a controlling process of a cooperation application based on a broadcasting/communication cooperating function on the basis of the program information data. The multimedia decoder **144U** includes an HTML browser function of executing an HTML document, and the multimedia screen generating process is executed by the HTML browser function.

[0160] The video decoder **145U**, the audio decoder **146U**, a superimposer **147U**, a superimposer **148U**, a superimposer **149U**, a composer **151U**, and a selector **150U** are respectively components that have the similar functions to the video decoder **145S**, the audio decoder **146S**, the superimposer **147S**, the superimposer **148S**, the superimposer **149S**, the composer **151S**, and the selector **150S**. By changing the end of each of the reference signs “S” into “U” in the explanation about the video decoder **145S**, the audio decoder **146S**, the superimposer **147S**, the superimposer **148S**, the superimposer **149S**, the composer **151S**, and the selector **150S** illustrated in FIG. 2F, the video decoder **145U**, the audio decoder **146U**, the superimposer **147U**, the superimposer **148U**, the superimposer **149U**, the composer **151U**, and the selector **150U** illustrated in FIG. 2G may respectively be explained. For this reason, further detailed explanation thereof is omitted.

[Software Configuration of Broadcast Receiving Apparatus]

[0161] FIG. 2H is a software configuration diagram of the broadcast receiving apparatus **100**, and illustrates one example of a software configuration of the storage (accumulator) **110** (or the ROM **103**, the same applies hereinafter) and the RAM **104**. A basic operating program **1001**, a receiving function program **1002**, a browser program **1003**, a content managing program **1004**, and the other operating program **1009** are stored in the storage (accumulator) **110**. Further, the storage (accumulator) **110** includes: a content memory region **1011** configured to store content data such as moving images, still images, or audio; an authentication information memory region **1012** configured to store authentication information used when communicating or cooperating with external portable terminal equipment or the server apparatus; and a various kinds of information memory region **1019** configured to store various kinds of other information.

[0162] The basic operating program **1001** stored in the storage (accumulator) **110** is developed on the RAM **104**, and the main controller **101** further executes the developed basic operating program, thereby constituting a basic operating controller **1101**. Further, the receiving function program **1002**, the browser program **1003**, and the content managing program **1004**, which are stored in the storage (accumulator) **110**, are respectively developed on the RAM **104**, and the main controller **101** further executes the developed operating programs, thereby constituting a receiving function controller **1102**, a browser engine **1103**, and a content manager **1104**. Further, the RAM **104** includes a temporary memory region **1200** configured to temporarily hold data created at the time of execution of each of the operating programs as needed.

[0163] Note that hereinafter, in order to simplify explanation thereof, a case where the main controller **101** develops the basic operating program **1001** stored in the storage (accumulator) **110** on the RAM **104** and executes the developed basic operating program **1001**, whereby the basic operating controller **1101** executes a process of controlling each operating block will be described. The similar description is made with respect to the other operating programs.

[0164] The receiving function controller **1102** executes basic controls such as a broadcast receiving function or a broadcasting/communication cooperating function of the broadcast receiving apparatus **100**. In particular, a tuning/demodulator **1102a** mainly controls the channel selecting process, a TMCC information obtaining process, and the demodulating process in each of the first tuner/demodulator **130C**, the second tuner/demodulator **130T**, the third tuner/demodulator **130L**, and the fourth tuner/demodulator **130B**. A stream reproduction controller **1102b** mainly controls the layer dividing process, the error correction decoding process, the energy dispersal removal, and the stream reproducing process in each of the first tuner/demodulator **130C**, the second tuner/demodulator **130T**, the third tuner/demodulator **130L**, and the fourth tuner/demodulator **130B**. An AV decoder **1102c** mainly controls the demultiplexing process (or a stream decoding process), a video data decoding process, and an audio data decoding process in each of the first decoder **140S** and a second decoder **140H**. A multimedia (MM) data reproducer **1102d** mainly controls a BML data reproducing process, a caption data decoding process, a subtitle data decoding process, and a controlling process of a communication cooperation application in the first decoder **140S**, and an HTML data reproducing process, the multimedia screen generating process, and a controlling process of a communication cooperation application in the second decoder **140H**. An EPG generator **1102e** mainly controls an EPG generating process and a displaying process of the generated EPG in each of the first decoder **140S** and the second decoder **140H**. A presentation processor **1102f** executes controls of the colorimetry converting process, the dynamic range converting process, a resolution converting process, and the downmixing process for audio in each of the first decoder **140S** and the second decoder **140H**, and controls of the video selector **191** and the audio selector **194**.

[0165] At the time of the BML data reproducing process or the HTML data reproducing process described above, a BML browser **1103a** and an HTML browser **1103b** of the browser engine **1103** respectively execute interpretation of a BML document or an HTML document, and execute the

data broadcasting screen generating process or the multimedia screen generating process.

[0166] The content manager **1104** executes time schedule management and execution control at the time of execution of video recording reservation or timer recording of a broadcasting program, and expiration date management of the cooperation application obtained based on copyright management and the broadcasting/communication cooperating function when the broadcasting program or a recorded program is outputted from the digital interface **125** or the LAN communication unit **121**.

[0167] Each of the operating programs may be stored in advance in at least one of the storage (accumulator) **110** or the ROM **103** at the time of product shipment. Each of the operating programs may be obtained from the server apparatus on the Internet **800** via the LAN communication unit **121** or the like after the product shipment. Further, each of the operating programs stored in a memory card or an optical disc may be obtained via the extension interface **124**. Each of the operating programs may be newly obtained or updated via the broadcast wave.

[Configuration of Broadcast Station Server]

[0168] FIG. **3A** is one example of an internal configuration of the broadcast station server **400**. The broadcast station server **400** is configured by a main controller **401**, a system bus **402**, a RAM **404**, a storage **410**, a LAN communication unit **421**, and a digital broadcasting signal sender **460**.

[0169] The main controller **401** is a microprocessor unit configured to control the whole broadcast station server **400** in accordance with a predetermined operating program. The system bus **402** is a communication path for executing transmission and reception of various kinds of data or commands between the main controller **401** and each operating block in the broadcast station server **400**. The RAM **404** becomes a work area at the time of execution of each of operating programs.

[0170] A basic operating program **4001**, a content management/distribution program **4002**, and a content sending program **4003** are stored in the storage **410**, and the storage **410** further includes a content data memory region **4011** and a metadata memory region **4012**. Content data of each broadcasting program broadcast by the broadcast station are stored in the content data memory region **4011**. Metadata such as program titles of the respective broadcasting programs, program IDs, program summaries, casts, broadcasting dates, or the like are stored in the metadata memory region **4012**.

[0171] Further, each of the basic operating program **4001**, the content management/distribution program **4002**, and the content sending program **4003** stored in the storage **410** is developed on the RAM **404**, and the main controller **401** further executes the developed basic operating program, the developed content management/distribution program, the developed content sending program, thereby constituting a basic operating controller **4101**, a content management/distribution controller **4102**, and a content sending controller **4103**.

[0172] Note that hereinafter, in order to simplify explanation thereof, a case where the main controller **401** develops the basic operating program **4001** stored in the storage **410** on the RAM **404** and executed the developed basic operating program **4001**, whereby the basic operating controller **4101** executes a process of controlling each operating block will be described. The similar description is made with respect to the other operating programs.

[0173] The content management/distribution controller **4102** executes management of the content data and the metadata respectively stored in the content data memory region **4011** and the metadata memory region **4012**, and a control of the content data and the metadata at the time of provision of them to the service provider on the basis of contract. Moreover, the content management/distribution controller **4102** executes an authentication process of the service provider server **500** as needed at the time of provision of the content data or the metadata to the service provider.

[0174] The content sending controller **4103** executes time schedule management when the content data of the broadcasting program stored in the content data memory region **4011** and a stream

containing a program title or a program ID of the broadcasting program stored in the metadata memory region **4012** and copy control information of program content are sent out via the digital broadcasting signal sender **460**.

[0175] The LAN communication unit **421** is connected to the Internet **800**, and executes communication with the service provider server **500** or another communication equipment on the Internet **800**. The LAN communication unit **421** includes a coding circuit, a decoding circuit, and the like. The digital broadcasting signal sender **460** subjects a stream constituted by content data and program information data for each of the broadcasting programs stored in the content data memory region **4011** to a process such as modulation, and sends out the processed stream as a digital broadcasting wave via the radio tower **300**.

[Configuration of Service Provider Server]

[0176] FIG. **3B** is one example of an internal configuration of the service provider server **500**. The service provider server **500** is configured by a main controller **501**, a system bus **502**, a RAM **504**, a storage **510**, and a LAN communication unit **521**.

[0177] The main controller **501** is a microprocessor unit configured to control the whole service provider server **500** in accordance with a predetermined operating program. The system bus **502** is a communication path for executing transmission and reception of various kinds of data or commands between the main controller **501** and each operating block in the service provider server **500**. The RAM **504** becomes a work area at the time of execution of each of operating programs.

[0178] A basic operating program **5001**, a content management/distribution program **5002**, and an application management/distribution program **5003** are stored in the storage **510**, and the storage **510** further includes a content data memory region **5011**, a metadata memory region **5012**, and an application memory region **5013**. The content data and the metadata provided from the broadcast station server **400**, or content created by the service provider and metadata regarding the content are stored in the content data memory region **5011** and the metadata memory region **5012**.

Applications to be distributed in response to a request from each television receiver (at least one of operating programs or various kinds of data), which are required to realize each service of the broadcasting/communication cooperation system, are stored in the application memory region **5013**.

[0179] Further, the basic operating program **5001**, the content management/distribution program **5002**, and the application management/distribution program **5003**, which are stored in the storage **510**, are respectively developed on the RAM **504**, and the main controller **501** further executes the developed basic operating program, the developed content management/distribution program, and the developed application management/distribution program, thereby constituting a basic operating controller **5101**, a content management/distribution controller **5102**, and an application management/distribution controller **5103**.

[0180] Note that hereinafter, in order to simplify explanation thereof, a case where the main controller **501** develops the basic operating program **5001** stored in the storage **510** on the RAM **504** and executes the developed basic operating program **5001**, whereby the basic operating controller **5101** executes a process of controlling each operating block will be described. The similar description is made with respect to the other operating programs.

[0181] The content management/distribution controller **5102** executes acquisition of content data and metadata from the broadcast station server **400**, management of the content data and the metadata respectively stored in the content data memory region **5011** and the metadata memory region **5012**, and a control of distribution of the content data and the metadata to each television receiver. Further, the application management/distribution controller **5103** executes management of each application stored in the application memory region **5013**, and control when each application is distributed in response to a request from each television receiver. Moreover, the application management/distribution controller **5103** also executes an authentication process of the corresponding television receiver as needed when each application is distributed to each television

receiver.

[0182] The LAN communication unit **521** is connected to the Internet **800**, and executes communication with the broadcast station server **400** or the other communication equipment on the Internet **800**. Further, the LAN communication unit **521** executes communication with the broadcast receiving apparatus **100** or the portable information terminal **700** via the router device **800R**. The LAN communication unit **521** includes a coding circuit, a decoding circuit, and the like. [Broadcast Wave of Digital Broadcasting]

[0183] Here, one example of the broadcast wave of digital broadcasting received by the broadcast receiving apparatus according to the embodiment of the present invention will be described.

[0184] The broadcast receiving apparatus **100** can receive the terrestrial digital broadcasting service that shares at least a part of specifications with ISDB-T (Integrated Services Digital Broadcasting for Terrestrial Television Broadcasting) method. Specifically, the dual-polarized terrestrial digital broadcasting that the second tuner/demodulator **130T** can receive is advanced terrestrial digital broadcasting that shares the part of the specifications with the ISDB-T method. Further, the layered division multiplexing terrestrial digital broadcasting that the third tuner/demodulator **130L** can receive is the advanced terrestrial digital broadcasting shares the part of the specifications with the ISDB-T method. Note that the existing terrestrial digital broadcasting that the first tuner/demodulator **130C** can receive is terrestrial digital broadcasting with the ISDB-T method. Further, advanced BS digital broadcasting or advanced CS digital broadcasting that the fourth tuner/demodulator **130B** can receive is digital broadcasting with a method different from the ISDB-T method.

[0185] Here, the dual-polarized terrestrial digital broadcasting and the layered division multiplexing terrestrial digital broadcasting according to the present embodiment adopts OFDM (Orthogonal Frequency Division Multiplexing: orthogonal frequency division multiplexing) that is one of multi-carrier methods as a transmission method in the similar manner to the ISDB-T method. Since the OFDM is a multi-carrier method, a symbol length thereof is long, and it is effective to add a redundant portion in a time axis direction called a guard interval. This makes it possible to reduce an influence on multipath within a range of the guard interval. For this reason, it is possible to realize an SFN (Single Frequency Network: single frequency network), and this makes it possible to use the frequency effectively.

[0186] In the dual-polarized terrestrial digital broadcasting and the layered division multiplexing terrestrial digital broadcasting according to the present embodiment, a carrier of the OFDM is divided into groups called segments in the similar manner to the ISDB-T method. As illustrated in FIG. 4A, one channel bandwidth of the digital broadcasting service is configured by 13 segments. The central part of the band is set to a position of segment 0, and segment numbers (0 to 12) are assigned above and below this position in turn. Transmission path coding of the dual-polarized terrestrial digital broadcasting and the layered division multiplexing terrestrial digital broadcasting according to the present embodiment is executed in units of OFDM segments. For this reason, it is possible to define layer transmission. For example, in the bandwidth of one television channel, it is possible to allocate a part of the OFDM segments to a fixed receiving service, and allocate the remaining to a mobile receiving service. In the layer transmission, each layer is configured by one OFDM segment or a plurality of OFDM segments, and it is possible to set, for each layer, parameters such as a carrier modulation method, a coding rate of an inner code, or a length of time interleaving. Note that the number of layers may be set arbitrarily, for example, and the number of layers may be set up to three layers. FIG. 4B illustrates one example of a layer allocation of the OFDM segments in a case where the number of layers is three or two. In the example of (1) of FIG. 4B, the number of layers is three; an A layer is configured by one segment (segment 0); a B layer is configured by seven segments (segments 1 to 7); and a C layer is configured by five segments (segments 8 to 12). In the example of (2) of FIG. 4B, the number of layers is three; an A layer is configured by one segment (segment 0); a B layer is configured by five segments (segments 1 to

5); and a C layer is configured by seven segments (segments 6 to 12). In the example of (3) of FIG. 4B, the number of layers is two; an A layer is configured by one segment (segment 0); and a B layer is configured by 12 segments (segments 1 to 12). The number of OFDM segments and transmission path coding parameters of each layer are determined in accordance with organization information, and are transmitted by the TMCC signal that is control information for assisting the operation of the receiver.

[0187] Note that as one example of a usage example of segment layer allocation illustrated in (1), (2), and (3) of FIG. 4B, the following may be an example, for example.

[0188] For example, the layer allocation illustrated in (1) of FIG. 4B can be used in the dual-polarized terrestrial digital broadcasting according to the present embodiment, and the same segment layer allocation may be used for both a horizontal polarized wave and a vertical polarized wave. Specifically, the mobile receiving service of the existing terrestrial digital broadcasting may be transmitted by the one segment described above of the horizontal polarized wave as the A layer. (Note that in the mobile receiving service of the existing terrestrial digital broadcasting, the same service may be transmitted by the one segment described above of the vertical polarized wave. In this case, this is also treated as the A layer.) Further, the terrestrial digital broadcasting service in which video whose maximum resolution is 1920 horizontal pixels×1080 vertical pixels, which is the existing terrestrial digital broadcasting, is transmitted may be transmitted by the seven segments described above of the horizontal polarized wave as the B layer. (Note that the terrestrial digital broadcasting service in which the video whose maximum resolution is 1920 horizontal pixels×1080 vertical pixels is transmitted may transmit the same service by the seven segments described above of the vertical polarized wave. In this case, this is also treated as the B layer.) Moreover, it may be configured so as to transmit the advanced terrestrial digital broadcasting service capable of transmitting the video whose maximum resolution is the number of pixels that exceeds 1920 horizontal pixels×1080 vertical pixels by the five segments described above of each of both the horizontal polarized wave and the vertical polarized wave, that is, by total ten segments as the C layer. Details of the transmission will be described later. A transmission wave of the segment layer allocation can be received by the second tuner/demodulator **130T** of the broadcast receiving apparatus **100**, for example.

[0189] For example, the layer allocation illustrated in (2) of FIG. 4B can be used as an example different from that illustrated in (1) of FIG. 4B in the dual-polarized terrestrial digital broadcasting according to the present embodiment, and the same segment layer allocation may be used for both the horizontal polarized wave and the vertical polarized wave. Specifically, the mobile receiving service of the existing terrestrial digital broadcasting may be transmitted by the one segment described above of the horizontal polarized wave as the A layer. (Note that the mobile receiving service of the existing terrestrial digital broadcasting may transmit the same service by the one segment described above of the vertical polarized wave. In this case, this is also treated as the A layer.) Moreover, it may be configured so as to transmit the advanced terrestrial digital broadcasting service capable of transmitting the video whose maximum resolution is the number of pixels that exceeds 1920 horizontal pixels×1080 vertical pixels by the five segments described above of each of both the horizontal polarized wave and the vertical polarized wave, that is, by total ten segments as the B layer. Further, the terrestrial digital broadcasting service, which is the existing terrestrial digital broadcasting, in which the video whose maximum resolution is 1920 horizontal pixels×1080 vertical pixels is transmitted, may be transmitted by the seven segments described above of the horizontal polarized wave as the C layer. (Note that the terrestrial digital broadcasting service in which the video whose maximum resolution is 1920 horizontal pixels×1080 vertical pixels is transmitted may transmit the same service by the seven segments described above of the vertical polarized wave. In this case, this is also treated as the C layer.) Details of the transmission will be described later. A transmission wave of the segment layer allocation can be received by the second tuner/demodulator **130T** of the broadcast receiving apparatus **100** according

to the present embodiment, for example.

[0190] For example, the layer allocation illustrated in (3) of FIG. 4B can be used in the layered division multiplexing terrestrial digital broadcasting or the existing terrestrial digital broadcasting according to the present embodiment. Specifically, in a case where the layered division multiplexing terrestrial digital broadcasting is used, the mobile receiving service of the existing terrestrial digital broadcasting may be transmitted by one segment illustrated in FIG. 4B as the A layer. Moreover, it may be configured so as to transmit the advanced terrestrial digital broadcasting service capable of transmitting the video whose maximum resolution is the number of pixels that exceeds 1920 horizontal pixels×1080 vertical pixels by 12 segments illustrated in FIG. 4B as the B layer. A transmission wave of the segment layer allocation can be received by the third tuner/demodulator **130L** of the broadcast receiving apparatus **100** according to the present embodiment, for example. In a case where the existing terrestrial digital broadcasting is used, the mobile receiving service of the existing terrestrial digital broadcasting may be transmitted by one segment illustrated in FIG. 4B as the A layer, and the terrestrial digital broadcasting service, which is the existing terrestrial digital broadcasting, in which the video whose maximum resolution is 1920 horizontal pixels×1080 vertical pixels is transmitted, may be transmitted by 12 segments illustrated in FIG. 4B as the B layer. A transmission wave of the segment layer allocation can be received by the first tuner/demodulator **130C** of the broadcast receiving apparatus **100** according to the present embodiment, for example.

[0191] FIG. 4C illustrates one example of a system at the broadcast station side, which realizes a generating process for an OFDM transmission wave that is a digital broadcasting wave of the dual-polarized terrestrial digital broadcasting and the layered division multiplexing terrestrial digital broadcasting according to the present embodiment. An information source coder **411** codes each of video/audio/various kinds of data. A multiplexer/conditional access processor **415** multiplexes the video/audio/various kinds of data respectively coded in the information source coder **411**; further executes a process corresponding to conditional access appropriately; and outputs them as a packet stream. A plurality of information source coders **411** and a plurality of multiplexer/conditional access processors **415** can be provided parallelly, thereby generating a plurality of packet streams. A transmission path coder **416** remultiplexes the plurality of packet streams to create one packet stream, and executes a transmission path coding process to output it as the OFDM transmission wave. In the configuration illustrated in FIG. 4C, details of a method of information source coding and transmission path coding are different, but a configuration of realizing the generating process for the OFDM transmission wave is common to the ISDB-T method. Therefore, a part of the plurality of information source coders **411** and the plurality of multiplexer/conditional access processors **415** may be configured for the terrestrial digital broadcasting service with the ISDB-T method, another part thereof may be configured for the advanced terrestrial digital broadcasting service, and packet streams of a plurality of different terrestrial digital broadcasting services may be multiplexed in the transmission path coder **416**. In a case where the multiplexer/conditional access processor **415** is configured for the terrestrial digital broadcasting service with the ISDB-T method, the MPEG-2 TS that is a stream of a TSP (Transport Stream Packet) defined by MPEG-2 systems may be generated. Further, in a case where the multiplexer/conditional access processor **415** is configured for the advanced terrestrial digital broadcasting service, the MMT packet stream or a TLV stream including the MMT packet or the stream of the TSP defined by another system may be generated. Of course, all of the plurality of information source coders **411** and the plurality of multiplexer/conditional access processors **415** may be configured for the advanced terrestrial digital broadcasting service, and all of the packet streams to be multiplexed by the transmission path coder **416** may be a packet stream for the advanced terrestrial digital broadcasting service.

[0192] FIG. 4D illustrates one example of a configuration of the transmission path coder **416**.

[0193] (1) of FIG. 4D will first be described. (1) of FIG. 4D is a configuration of the transmission path coder **416** in a case where only the OFDM transmission wave of the digital broadcasting for

the existing terrestrial digital broadcasting service is generated. The OFDM transmission wave transmitted by this configuration has a segment configuration illustrated in (3) of FIG. 4B, for example. Redundancy of error correction is added to the packet stream inputted from the multiplexer/conditional access processor 415 and subjected to a remultiplexing process, and various kinds of interleaving processes such as byte interleaving, bit interleaving, time interleaving, or frequency interleaving are also executed for the packet stream. Then, a process by IFFT (Inverse Fast Fourier Transform) is executed together with a pilot signal, the TMCC signal, and the AC signal, and it becomes the OFDM transmission wave through orthogonal modulation after a guard interval is added. Note that the processes of an outer coding process, a power spreading process, the byte interleaving, an inner coding process, and a mapping process are configured so that the processes can be executed separately for each layer such as the A layer or the B layer. (Note that the digital broadcasting of the existing terrestrial digital broadcasting service has two layers in its operation, but it is possible to transmit up to three layers. For this reason, (1) of FIG. 4D illustrates an example of three layers.) The mapping process is a modulating process for a carrier. Further, in the packet stream inputted from the multiplexer/conditional access processor 415, information such as the TMCC information, a mode, or a guard interval ratio may be multiplexed. Note that as described above, the packet stream inputted into the transmission path coder 416 may be a stream of the TSP defined by the MPEG-2 systems. The OFDM transmission wave generated by the configuration illustrated in (1) of FIG. 4D can be received by the first tuner/demodulator 130C of the broadcast receiving apparatus 100 according to the present embodiment, for example.

[0194] Next, (2) of FIG. 4D will be described. (2) of FIG. 4D is a configuration of the transmission path coder 416 in a case where the OFDM transmission wave of the dual-polarized terrestrial digital broadcasting according to the present embodiment is generated. The OFDM transmission wave transmitted by this configuration has a segment configuration illustrated in (1) or (2) of FIG. 4B, for example. In (2) of FIG. 4D, various kinds of interleaving processes such as the byte interleaving, the bit interleaving, the time interleaving, or the frequency interleaving are also executed for the packet stream inputted from the multiplexer/conditional access processor 415 and subjected to the remultiplexing process in addition to addition of redundancy of error correction. Then, a process by the IFFT is executed together with the pilot signal, the TMCC signal, and the AC signal, and it becomes the OFDM transmission wave through the orthogonal modulation after a guard interval adding process is executed.

[0195] In the configuration example illustrated in (2) of FIG. 4D, processes of the outer coding process, the power spreading process, the byte interleaving, the inner coding process, the mapping process, and the time interleaving are configured so that the processes can be executed separately for each layer such as the A layer, the B layer, or the C layer. However, in the configuration example illustrated in (2) of FIG. 4D, not only the OFDM transmission wave of a horizontal polarized wave (H) but also the OFDM transmission wave of a vertical polarized wave (V) are generated, and a processing flow thus branches to two lines. When branching from a processing system of the horizontal polarized wave (H) to a processing system of the vertical polarized wave (V), it is possible to differentiate, for each layer in accordance with the segment configuration explained with reference to (1) or (2) of FIG. 4B, whether the same data as that of the processing system of the horizontal polarized wave (H) is caused to branch to the processing system of the vertical polarized wave (V), different data from that of the processing system of the horizontal polarized wave (H) is caused to branch to the processing system of the vertical polarized wave (V), or any data is caused not to branch to the processing system of the vertical polarized wave (V).

[0196] As the processes indicated in the configuration of (2) of FIG. 4D, such as the outer code, the inner code, or the mapping, it is possible to use a more advanced process that is not adopted in each process of the configuration of (1) of FIG. 4D in addition to the processes compatible with the configuration of (1) of FIG. 4D. Specifically, in the configuration of (2) of FIG. 4D, for a portion where the process is executed for each layer, the processes compatible with (1) of FIG. 4D are

executed in the layer in which the mobile receiving service of the existing terrestrial digital broadcasting or the existing terrestrial digital broadcasting service for transmitting the video whose maximum resolution is 1920 horizontal pixels×1080 vertical pixels is transmitted with respect to the processes such as the outer code, the inner code, or the mapping. On the other hand, in the configuration of (2) of FIG. 4D, for a portion where the process is executed for each layer, it may be configured so that more advanced process that is not adopted in each process of the configuration of (1) of FIG. 4D for the layer in which the advanced terrestrial digital broadcasting service capable of transmitting the video whose maximum resolution is the number of pixels that exceeds 1920 horizontal pixels×1080 vertical pixels is transmitted with respect to the processes such as the outer code, the inner code, or the mapping.

[0197] Note that in the dual-polarized terrestrial digital broadcasting according to the present embodiment according to the present embodiment, the TMCC information (will be described later) allows allocation of layer and the content terrestrial digital broadcasting service to be transmitted to be switched. For this reason, it is desirable that it is configured so that the process such as the outer code, the inner code, or the mapping to be subjected to each layer can be switched by the TMCC information.

[0198] Note that with respect to the layer for transmitting the advanced terrestrial digital broadcasting service in which the video whose maximum resolution is the number of pixels that exceeds 1920 horizontal pixels×1080 vertical pixels can be transmitted, as the byte interleaving, the bit interleaving, and the time interleaving, a process compatible with the existing terrestrial digital broadcasting service may be executed, or more advanced different process may be executed. Alternatively, with respect to the layer for transmitting the advanced terrestrial digital broadcasting service, part of the interleavings may be omitted.

[0199] Further, in the configuration illustrated in (2) of FIG. 4D, an input stream that becomes a source of the layer in which the mobile receiving service of the existing terrestrial digital broadcasting or the existing terrestrial digital broadcasting service for transmitting the video whose maximum resolution is 1920 horizontal pixels×1080 vertical pixels is transmitted may be a stream of the TSP defined by the MPEG-2 systems that is adopted by the existing terrestrial digital broadcasting among the packet streams inputted into the transmission path coder 416. An input stream that becomes a source of the layer for transmitting the advanced terrestrial digital broadcasting service with the configuration illustrated in (2) of FIG. 4D may be a stream defined by a system different from the stream of the TSP defined by the MPEG-2 systems such as the MMT packet stream or the TLV including MMT packet among the packet streams inputted into the transmission path coder 416. However, a stream of the TSP defined by the MPEG-2 systems in the advanced terrestrial digital broadcasting service may be adopted.

[0200] In the configuration of (2) of FIG. 4D explained above, in the layer in which the mobile receiving service of the existing terrestrial digital broadcasting or the existing terrestrial digital broadcasting service for transmitting the video whose maximum resolution is 1920 horizontal pixels×1080 vertical pixels is transmitted, a stream form compatible with the existing terrestrial digital broadcasting and a process are maintained until the OFDM transmission wave is generated from the input stream. As a result, even in a case where the receiving apparatus for the existing terrestrial digital broadcasting service receives one transmission wave of the OFDM transmission wave of the horizontal polarized wave generated in the configuration illustrated in (2) of FIG. 4D and the OFDM transmission wave of the vertical polarized wave, it becomes possible to receive and demodulate the broadcasting signal of the terrestrial digital broadcasting service correctly with respect to the layer in which the mobile receiving service of the existing terrestrial digital broadcasting or the existing terrestrial digital broadcasting service for transmitting the video whose maximum resolution is 1920 horizontal pixels×1080 vertical pixels is transmitted.

[0201] Further, in the configuration illustrated in (2) of FIG. 4D, it is possible to transmit the advanced terrestrial digital broadcasting service capable of transmitting the video whose maximum

resolution is the number of pixels that exceeds 1920 horizontal pixels×1080 vertical pixels in the layer in which segments of both the OFDM transmission wave of the horizontal polarized wave and the OFDM transmission wave of the vertical polarized wave are used, it becomes possible to receive and demodulate the broadcasting signal of the advanced terrestrial digital broadcasting service by the broadcast receiving apparatus **100** according to the embodiment of the present invention.

[0202] Namely, in the configuration illustrated in (2) of FIG. **4D**, it is also possible to generate a digital broadcasting wave the digital broadcasting can suitably receive and demodulate both in the broadcast receiving apparatus that is compatible with the advanced terrestrial digital broadcasting service and in the receiving apparatus for the existing terrestrial digital broadcasting service.

[0203] Next, (3) of FIG. **4D** will be described. (3) of FIG. **4D** is a configuration of the transmission path coder **416** in a case where the OFDM transmission wave of the layered division multiplexing terrestrial digital broadcasting according to the present embodiment is generated. In (3) of FIG. **4D**, various kinds of interleaving processes such as the byte interleaving, the bit interleaving, the time interleaving, or the frequency interleaving are also executed for the packet stream inputted from the multiplexer/conditional access processor **415** and subjected to the remultiplexing process in addition to addition of redundancy of error correction. Then, a process by the IFFT is executed together with the pilot signal, the TMCC signal, and the AC signal, and it becomes the OFDM transmission wave through the orthogonal modulation after the guard interval is added.

[0204] However, in the configuration of (3) of FIG. **4D**, each of a modulation wave transmitted by the upper layer and a modulation wave transmitted by the lower layer is generated, and the OFDM transmission wave that is a digital broadcasting wave after multiplexing is generated. A processing system indicated at an upper side of the configuration illustrated in (3) of FIG. **4D** is a processing system for generating the modulation wave to be transmitted by the upper layer, and a processing system indicated at a lower side is a processing system for generating the modulation wave to be transmitted by the lower layer. Data for transmitting the processing system for the generating the modulation wave to be transmitted by the upper layer illustrated in (3) of FIG. **4D** are the mobile receiving service of the existing terrestrial digital broadcasting and the existing terrestrial digital broadcasting service for transmitting the video whose maximum resolution is 1920 horizontal pixels×1080 vertical pixels, and various kinds of processes in the processing system for generating the modulation wave to be transmitted by the upper layer illustrated in (3) of FIG. **4D** are processes that are the same as various kinds of processed of (1) of FIG. **4D** or has compatibility therewith.

The modulation wave to be transmitted by the upper layer illustrated in (3) of FIG. **4D** has a segment configuration illustrated in (3) of FIG. **4B** as well as the transmission wave illustrated in (1) of FIG. **4D**, for example. Therefore, the modulation wave to be transmitted by the upper layer illustrated in (3) of FIG. **4D** is a digital broadcasting wave that has compatibility with the existing terrestrial digital broadcasting service for transmitting the mobile receiving service of the existing terrestrial digital broadcasting or the video whose maximum resolution is 1920 horizontal pixels×1080 vertical pixels. On the other hand, data for transmitting the processing system for generating the modulation wave to be transmitted by the lower layer illustrated in (3) of FIG. **4D** are the advanced terrestrial digital broadcasting service capable of transmitting the video whose maximum resolution is the number of pixels that exceeds 1920 horizontal pixels×1080 vertical pixels, for example, and it may be configured so that a more advanced process that is not adopted in each process of the configuration illustrated in (1) of FIG. **4D** is used with respect to the processes such as the outer code, the inner code, or the mapping.

[0205] The modulation wave to be transmitted by the lower layer illustrated in (3) of FIG. **4D** may be allocated to the advanced terrestrial digital broadcasting service capable of transmitting the video whose maximum resolution is the number of pixels that exceeds 1920 horizontal pixels×1080 vertical pixels by using all of the 13 segments as the A layer, for example. Alternatively, the mobile receiving service of the existing terrestrial digital broadcasting having a segment configuration

illustrated in (3) of FIG. 4B may be transmitted by the A layer of one segment, and the advanced terrestrial digital broadcasting service capable of transmitting the video whose maximum resolution is the number of pixels that exceeds $1920 \text{ horizontal pixels} \times 1080 \text{ vertical pixels}$ may be transmitted by the B layer of 12 segments. In case of the latter, as well as (2) of FIG. 4D, it may be configured so that the processes are switched for each layer such as the A layer or the B layer from the outer coding process to time interleaving process. In the layer for transmitting the mobile receiving service of the existing terrestrial digital broadcasting, it is similar to explanation for (2) of FIG. 4D in that it is necessary to maintain the process compatible with the existing terrestrial digital broadcasting.

[0206] In the configuration illustrated in (3) of FIG. 4D, the OFDM transmission wave, which is a terrestrial digital broadcasting wave obtained by multiplexing the modulation wave to be transmitted by the upper layer and the modulation wave to be transmitted by the lower layer, is generated. A technique of separating the modulation wave to be transmitted by the upper layer from the 4-D OFDM transmission wave is also equipped with the receiving apparatus for the existing terrestrial digital broadcasting service. For this reason, the broadcasting signal, which is contained in the modulation wave to be transmitted by the upper layer, of the mobile receiving service of the existing terrestrial digital broadcasting or the existing terrestrial digital broadcasting service for transmitting the video whose maximum resolution is $1920 \text{ horizontal pixels} \times 1080 \text{ vertical pixels}$ is correctly received and demodulated by the receiving apparatus of the existing terrestrial digital broadcasting service. On the other hand, the broadcasting signal, which is contained in the modulation wave to be transmitted by the lower layer, of the advanced terrestrial digital broadcasting service capable of transmitting the video whose maximum resolution is the number of pixels that exceeds $1920 \text{ horizontal pixels} \times 1080 \text{ vertical pixels}$ can be received and demodulated by the broadcast receiving apparatus **100** according to the embodiment of the present invention.

[0207] Namely, in the configuration illustrated in (3) of FIG. 4D, it is possible to generate the digital broadcasting wave by which the digital broadcasting can be received and demodulated suitably in both the broadcast receiving apparatus that is compatible with the advanced terrestrial digital broadcasting service and the receiving apparatus of the existing terrestrial digital broadcasting service. Further, in the configuration illustrated in (3) of FIG. 4D, unlike the configuration illustrated in (2) of FIG. 4D, it is possible to generate the OFDM transmission wave that can be received more easily without necessity to use the plurality of polarized waves.

[0208] In the 4-D OFDM transmission wave generating process illustrated in (1) of FIG. 4D, (2) of FIG. 4D, and (3) of FIG. 4D according to the present embodiment, three types of modes with the different number of carriers are prepared in consideration of the suitability of a distance between stations of the SFN and the resistance to Doppler shift in mobile reception. Note that another mode with the different number of carriers may further be prepared. In a mode with a large number of carriers, an effective symbol length becomes longer, and a guard interval length becomes longer if the guard interval ratio (guard interval length/effective symbol length) is the same. This makes it possible to have resistance to multipath with a long delay time difference. On the other hand, in case of a mode with the small number of carriers, a carrier interval becomes wide, and it is possible to reduce an influence of inter-carrier interference due to the Doppler shift that occurs in case of mobile reception or the like.

[0209] In the 4-D OFDM transmission wave generating process illustrated in (1) of FIG. 4D, (2) of FIG. 4D, and (3) of FIG. 4D according to the present embodiment, it is possible to set, for each layer configured by one OFDM segment or a plurality of OFDM segments, parameters such as a carrier modulation method, a code rate of an inner code, or a length of time interleaving. FIG. 4E illustrates one example of a transmission parameter for one segment unit of the OFDM segment identified in the mode of the system according to the present embodiment. Note that a carrier modulation method in FIG. 4E indicates a modulation method of a “data” carrier. An SP signal, a CP signal, a TMCC signal, and an AC signal adopt a modulation method different from the

modulation method of the “data” carrier. These signals are signals whose resistance to noise is more important than the quantity of information. For this reason, a modulation method of executing mapping to a constellation with a small number of states (BPSK or DBPSK, that is, two states) than the modulation method of “data” carrier (all are QPSK or more, that is, four states or more) is adopted, and this heightens the resistance to noise.

[0210] Further, each numerical value of the number of carriers is a value in a case where numerical values at a left side of an oblique line set QPSK, 16 QAM, or 64 QAM as the carrier modulation method, and is a value in a case where numerical values at a right side of the oblique line set DQPSK as the carrier modulation method. In FIG. 4E, underlined parameters are parameters each of which is not compatible with the mobile receiving service of the existing terrestrial digital broadcasting. Specifically, 256 QAM, 1024 QAM, and 4096 QAM in the modulation method of the “data” carrier are not adopted in the existing terrestrial digital broadcasting service. Therefore, in a process in the layer that requires compatibility with the existing terrestrial digital broadcasting service in an OFDM broadcast wave generating process according to (1) of FIG. 4D, (2) of FIG. 4D, and (3) of FIG. 4D of the present embodiment, 256 QAM, 1024 QAM, and 4096 QAM of the modulation method of the “data” carrier are not used. Further multi-level modulation method such as 256 QAM (the number of states is 256), 1024 QAM (the number of states is 1024), or 4096 QAM (the number of states is 4096) may be applied to the “data” carrier transmitted by the layer corresponding to the advanced terrestrial digital broadcasting service in addition to the modulation method that has compatibility with the existing terrestrial digital broadcasting service, such as QPSK (the number of states is 4), 16 QAM (the number of states is 16), or 64 QAM (the number of states is 64). Further, a modulation method different from these modulation methods may be adopted.

[0211] Note that as the modulation method for a pilot symbol (SP or CP) carrier, BPSK (the number of states is two) compatible with the existing terrestrial digital broadcasting service may be used. As the modulation method for an AC carrier and a TMCC carrier, DBPSK (the number of states is 2) compatible with the existing terrestrial digital broadcasting service may be used.

[0212] Further, as a method of the inner coding process, an LDPC code is not adopted in the existing terrestrial digital broadcasting service. Therefore, in the process for the layer necessary for compatibility with the existing terrestrial digital broadcasting service in the OFDM broadcast wave generating process illustrated in (1) of FIG. 4D, (2) of FIG. 4D, and (3) of FIG. 4D according to the present embodiment, the LDPC code is not used. The LDPC code may be applied, as the inner code, to the data to be transmitted by the layer that is compatible with the advanced terrestrial digital broadcasting service. Further, as a method of the outer coding process, a BCH code is not adopted in the existing terrestrial digital broadcasting service. Therefore, in the process for the layer necessary for compatibility with the existing terrestrial digital broadcasting service in the OFDM broadcast wave generating process illustrated in (1) of FIG. 4D, (2) of FIG. 4D, and (3) of FIG. 4D according to the present embodiment, the BCH code is not used. The BCH code may be applied, as the outer code, to the data to be transmitted by the layer that is compatible with the advanced terrestrial digital broadcasting service.

[0213] Further, FIG. 4F illustrates one example of a transmission signal parameter for one physical channel (6 MHz bandwidth) unit of the OFDM broadcast wave generating process illustrated in (1) of FIG. 4D, (2) of FIG. 4D, and (3) of FIG. 4D according to the present embodiment. In the OFDM broadcast wave generating process illustrated in (1) of FIG. 4D, (2) of FIG. 4D, and (3) of FIG. 4D according to the present embodiment, for compatibility with the existing terrestrial digital broadcasting service, parameters compatible with the existing terrestrial digital broadcasting service are basically adopted in the parameters illustrated in FIG. 4F as a general rule. However, in a case where all of the segments in the modulation wave to be transmitted by the lower layer illustrated in (3) of FIG. 4D are allocated to the advanced terrestrial digital broadcasting service, it is not necessary to maintain compatibility with the existing terrestrial digital broadcasting service

in the modulation wave. Therefore, in this case, parameters other than the parameters illustrated in FIG. 4F may be used with respect to the modulation wave to be transmitted by the lower layer illustrated in (3) of FIG. 4D.

[0214] Next, carriers of the OFDM transmission wave according to the present embodiment will be described. In addition to carriers by which data such as video or audio are transmitted, a carrier by which the pilot signal (SP, CP, AC1, or AC2) that becomes a reference of demodulation is transmitted, and a carrier by which the TMCC signal that is information on a modulation form and a convolution code rate of the carrier is transmitted are included in the carriers of the OFDM transmission wave according to the present embodiment. A number of carriers corresponding to 1/9 of the number of carriers for each segment is used for these types of transmission. Further, a concatenate code is adopted for the error correction; a shortened reed Solomon (204, 188) code is adopted for the outer code; a constraint length of 7 is adopted for the inner code; and a punctured convolution code with a code rate of $\frac{1}{2}$ as a mother code is adopted. Coding different from the above may be used for both the outer code and the inner code. Information rate differs depending upon parameters such as a carrier modulation form, a convolution code rate, or a guard interval ratio.

[0215] Further, 204 symbols are set as one frame, and the integer number of TSPs are included in one frame. Switching of transmission parameters is executed at the boundary of this frame.

[0216] There are an SP (Scattered Pilot), a CP (Continual Pilot), an AC (Auxiliary Channel) 1, and an AC2 as the pilot signals that becomes a reference of demodulation. FIG. 4G illustrates one example of an arrangement image of the pilot signals in the segments in case of synchronous modulation (such as QPSK, 16 QAM, 64 QAM, 256 QAM, 1024 QAM, or 4096 QAM). The SP is inserted into the segment of the synchronous modulation; is transmitted to 12 carriers in a carrier number (frequency axis) direction once; and is transmitted to four symbols in an OFDM symbol number (time axis) direction once. Since an amplitude and a phase of the SP are known, the SP can be used as a reference of synchronous demodulation. FIG. 4H illustrates one example of an arrangement image of the pilot signals in the segment in case of differential modulation (such as DQPSK). The CP is a continuous signal that is inserted at a left end of the segment for differential modulation, and is used for demodulation.

[0217] The AC1 and the AC2 carry information on the CP, and is also used for transmission of information for broadcasters in addition to a role of the pilot signal. They may be used for transmission of other information.

[0218] Note that each of the arrangement images illustrated in FIG. 4G and FIG. 4H is an example of in case of mode 3, and carrier numbers are 0 to 431. However, in case of each of mode 1 and mode 2, carrier numbers are 0 to 107 or 0 to 215. Further, the carrier for transmitting the AC1, the AC2, or the TMCC may be determined in advance for each segment. Note that the carrier for transmitting the AC1, the AC2, and the TMCC are arranged at random in a frequency direction in order to reduce an influence of periodic dip of transmission path characteristics by multipath.

[TMCC Signal]

[0219] The TMCC signal transmits information (TMCC information) regarding a layer configuration, a transmission parameter of the OFDM segment, and a demodulation operation of the receiver. The TMCC signal is transmitted by a carrier for TMCC transmission defined in each segment. FIG. 5A illustrates one example of bit allocation of the TMCC carrier. The TMCC carrier is configured by 204 bits (B0 to B203). The BC is a demodulation reference signal for a TMCC symbol, and has a predetermined amplitude and a predetermined phase reference. Each of the B1 to B16 is a synchronization signal, and is configured by words with 16 bits. As the synchronization signal, two types of w0 and w1 are defined, and the w0 and w1 are alternately sent out for each frame. The B17 to B19 are used for identification of a segment form, by which it is identified whether each segment is a differential modulator or a synchronous modulator. The TMCC information is described in the B20 to B121. The B122 to B203 are parity bits.

[0220] The TMCC information of the OFDM transmission wave according to the present embodiment may be configured so as to contain, as one example thereof, information for assisting demodulation and a decoding operation of the receiver, such as system identification, a transmission parameter switching indicator, a startup control signal (a startup flag for emergency warning broadcasting), current information, next information, frequency converting process identification, physical channel number identification, main signal identification, 4K signal transmission layer identification, or additional layer transmission identification, for example. The current information indicates a current layer configuration and a transmission parameter, and the next information indicates a layer configuration and a transmission parameter after switching. The switching of the transmission parameter is executed on a frame unit basis. FIG. 5B illustrates one example of bit allocation of the TMCC information. Further, FIG. 5C illustrates one example of a configuration of transmission parameter information contained in the current information/next information. Note that a connected transmission phase correction amount is control information used in case of terrestrial digital audio broadcasting ISDB-TSB (ISDB for Terrestrial Sound Broadcasting) having a common transmission method, and detailed explanation herein is omitted. [0221] FIG. 5D illustrates one example of bit allocation of the system identification. Two bits are allocated to a signal for the system identification. In case of an existing terrestrial digital television broadcasting system, "00" is set. In a case where the transmission method is a common terrestrial digital audio broadcasting system, "01" is set. Further, in case of an advanced terrestrial digital television broadcasting system such as the dual-polarized terrestrial digital broadcasting or the layered division multiplexing terrestrial digital broadcasting according to the present embodiment, "10" is set. In the advanced terrestrial digital television broadcasting system, it is possible to transmit a 2K broadcasting program (that is, a broadcasting program of video with 1920 horizontal pixels×1080 vertical pixels, a broadcasting program of video with lower resolution than it) and a 4K broadcasting program (that is, a broadcasting program of video with resolution that exceeds 1920 horizontal pixels×1080 vertical pixels) in the same service at the same time by means of broadcast wave transmission by the dual-polarized transmission method or the layered division multiplexing method.

[0222] The transmission parameter switching indicator is used to notify the receiver of switching timing by counting down when the transmission parameters are switched. This indicator is a value of "1111" at a normal time. In a case where the transmission parameters are switched, this indicator is subtracted by one for each frame from 15 frames before switching. The switching timing is next frame synchronization in which "0000" is sent out. A value of the indicator returns to "1111" after "0000". Countdown is executed in a case where any one or more of parameters such as the system identification of the TMCC information illustrated in FIG. 5B, the transmission parameter contained in the current information/next information, the frequency converting process identification, the main signal identification, the 4K signal transmission layer identification, or the additional layer transmission identification is switched. Countdown is not executed in a case where only the startup control signal of the TMCC information is switched.

[0223] The startup control signal (that is, the startup flag for the emergency warning broadcasting) is set to "1" when a startup control to the receiver is executed in the emergency warning broadcasting. The startup control signal is set to "0" when the startup control is not executed.

[0224] A partial reception flag for each of the current information/next information is set to "1" in a case where a segment in the center of the transmission band is set to partial reception. Otherwise, the partial reception flag thereof is set to "0". In a case where segment 0 is set for partial reception, the layer is defined as the A layer. In a case where the next information does not exist, the partial reception flag set to "1".

[0225] FIG. 5E illustrates one example of bit allocation for a carrier modulation mapping method (a modulation method for the data carrier) in each layer transmission parameter for each of the current information/next information. In a case where this parameter is "000", it indicates that the

modulation method is DQPSK. In a case where this parameter is “001”, it indicates that the modulation method is QPSK. In a case where this parameter is “010”, it indicates that the modulation method is 16 QAM. In a case where this parameter is “011”, it indicates that the modulation method is 64 QAM. In a case where this parameter is “100”, it indicates that the modulation method is 256 QAM. In a case where this parameter is “101”, it indicates that the modulation method is 1024 QAM. In a case where this parameter is “110”, it indicates that the modulation method is 4096 QAM. In a case where any unused layer or the next information does not exist, this parameter is set to “111”.

[0226] For setting of the code rate or the length of the time interleaving, each parameter may be set in accordance with the organization information of each layer for each of the current information/next information. The number of segments indicates the number of segments of each layer by a numerical value of four bits. In a case where any unused layer or the next information does not exist, the number of segments is set to “1111”. Note that since settings of a mode or a guard interval ratio are independently detected at the receiver side, transmission by the TMCC information may not be executed.

[0227] FIG. 5F illustrates one example of bit allocation of the frequency converting process identification. The frequency converting process identification is set to “0” in a case where a frequency converting process (in case of the dual-polarized transmission method) (will be described later) or a frequency conversion amplification process (in case of the layered division multiplexing transmission method) is executed in the converter **201T** or the converter **201L** illustrated in FIG. 2A. In a case where the frequency converting process or the frequency conversion amplification process is not executed, the frequency converting process identification is set to “1”. This parameter may be configured so as to be set to “1” when it is sent out from the broadcast station, for example. This parameter may be configured so as to be rewritten into “0” in the converter **201T** the converter **201L** when the frequency converting process or the frequency conversion amplification process is executed in the converter **201T** or the converter **201L**. By configuring it in this manner, in a case where it is received by the second tuner/demodulator **130T** or the third tuner/demodulator **130L** of the broadcast receiving apparatus **100** and a bit of the frequency converting process identification is “0”, it is possible to identify that the frequency converting process was executed after the OFDM transmission wave was sent out from the broadcast station.

[0228] In the dual-polarized terrestrial digital broadcasting according to the present embodiment, the setting or rewriting of the frequency converting process identification bit may be executed in each of the plurality of polarized waves. For example, in a case where both of the plurality of polarized waves is not subjected to frequency conversion in the converter **201T** of FIG. 2A, the frequency converting process identification bit contained in the OFDM transmission wave of each of the both may be left as “1”. Further, in a case where only one polarized wave of the plurality of polarized waves is subjected to frequency conversion in the converter **201T**, the frequency converting process identification bit included in the OFDM transmission wave of the polarized wave subjected to the frequency conversion may be rewritten into “0” in the converter **201T**. Further, in a case where both of the plurality of polarized waves is subjected to frequency conversion in the converter **201T**, the frequency converting process identification bit included in the OFDM transmission wave of each of both of polarized waves subjected to the frequency conversion may be rewritten into “0” in the converter **201T**. By configuring it in this manner, it is possible to identify presence or absence of the frequency conversion for each polarized wave of the plurality of polarized waves in the broadcast receiving apparatus **100**.

[0229] Note that since the frequency converting process identification bit is not defined in the existing terrestrial digital broadcasting, it is ignored in a terrestrial digital broadcasting receiving apparatus that has already been used by the user. However, the bit may be introduced into a new terrestrial digital broadcasting service obtained by improving the existing terrestrial digital

broadcasting, in which the video whose maximum resolution is 1920 horizontal pixels×1080 vertical pixels is transmitted. In this case, the first tuner/demodulator **130C** of the broadcast receiving apparatus **100** according to the embodiment of the present invention may also be configured as a first tuner/demodulator that is compatible with the new terrestrial digital broadcasting service.

[0230] Note that as a modification example, on the assumption that the frequency converting process or the frequency conversion amplification process is executed for the OFDM transmission wave in the converter **201T** or the converter **201L** illustrated in FIG. 2A, this parameter may be set to “0” in advance when it is sent out from the broadcast station. Note that in a case where the broadcast wave to be received is not the advanced terrestrial digital broadcasting service, this parameter may be configured so as to be set to “1”.

[0231] FIG. 5G illustrates one example of bit allocation of the physical channel number identification. The physical channel number identification is configured by a code with six bits, and identifies a physical channel number (13 ch to 52 ch) of the broadcast wave to be received. In a case where the broadcast wave to be received is not the advanced terrestrial digital broadcasting service, this parameter is set to “111111”. A bit of the physical channel number identification is not defined in the existing terrestrial digital broadcasting. Thus, the receiving apparatus of the existing terrestrial digital broadcasting could not obtain a physical channel number of a broadcast wave specified at the broadcast station side from the TMCC signal or the AC signal. In the broadcast receiving apparatus **100** according to the embodiment of the present invention, even though the carrier other than the TMCC signal or the AC signal is not demodulated using the bit of the physical channel number identification of the received OFDM transmission wave, it is possible to grasp the physical channel number that the broadcast station sets for the OFDM transmission wave. Note that the physical channels 13 ch to 52 ch have a bandwidth of 6 MHz per one ch, and are allocated to a frequency band from 470 to 710 MHz in advance. Therefore, the fact that the broadcast receiving apparatus **100** can grasp the physical channel number of the OFDM transmission wave on the basis of the bit of the physical channel number identification means that the frequency band in which the OFDM transmission wave is transmitted in the air as the terrestrial digital broadcasting wave can be grasped.

[0232] In the dual-polarized terrestrial digital broadcasting according to the present embodiment, in the OFDM transmission wave generating process at the broadcast station side, bits of the physical channel number identification may be arranged in each of the pair of polarized waves of the plurality of polarized waves in the bandwidth that originally constitutes one physical channel, and the same physical number may be given thereto. Here, depending upon installation environment of the broadcast receiving apparatus **100**, only frequency of one polarized wave of the plurality of polarized waves may be converted in the converter **201T** illustrated in FIG. 2A. As a result, in a case where the frequencies of the pair of polarized waves of the plurality of polarized waves are different from each other when they are received by the broadcast receiving apparatus **100**, the broadcast receiving apparatus side cannot demodulate the advanced terrestrial digital broadcasting by using both of the polarized waves of the dual-polarized terrestrial digital broadcasting if it is impossible to grasp, by any method, that the plurality of polarized waves with different frequencies were originally the pair of polarized waves. Even in such a case, by using the bits of the physical channel number identification described above, in a case where the transmission wave in which the bits of the physical channel number identification indicate that the same value exists in a plurality of different frequencies in the broadcast receiving apparatus **100**, it is possible to identify that it is a transmission wave transmitted as a pair of polarized waves that originally constitutes one physical channel at the broadcast station side. This makes it possible to realize demodulation of the advanced terrestrial digital broadcasting of the dual-polarized terrestrial digital broadcasting by using a plurality of transmission waves each indicating the same value.

[0233] FIG. 5H illustrates one example of bit allocation of the main signal identification. The

present embodiment is an example in which a bit of the main signal identification is arranged in bit B117.

[0234] In a case where the OFDM transmission wave to be transmitted is a transmission wave of the dual-polarized terrestrial digital broadcasting, this parameter is set to “1” in the TMCC information of a transmission wave to be transmitted by a main polarized wave. This parameter is set to “0” in the TMCC information of a transmission wave to be transmitted by a sub polarized wave. Note that the transmission wave to be transmitted by the main polarized wave indicates a polarization signal in the same polarization direction as the polarization direction used for transmission of the existing terrestrial digital broadcasting service of a vertical polarized signal and a horizontal polarized signal. Namely, in regions where transmission by the horizontal polarized wave is adopted in the existing terrestrial digital broadcasting service, in the dual-polarized terrestrial digital broadcasting service, the horizontal polarized wave is a main polarized wave, and the vertical polarized wave is a sub polarized wave. Further, in areas where transmission by the vertical polarized wave is adopted in the existing terrestrial digital broadcasting service, in the dual-polarized terrestrial digital broadcasting service, the vertical polarized wave is a main polarized wave, and the horizontal polarized wave is a sub polarized wave.

[0235] In the broadcast receiving apparatus **100** according to the embodiment of the present invention that receives the transmission wave of the dual-polarized terrestrial digital broadcasting, by using the bit of the main signal identification, it is possible to identify whether the received transmission wave is transmitted by the main polarized wave or is transmitted by the sub polarized wave at the time of transmission thereof. For example, by using an identifying process for the main polarized wave and the sub polarized wave, it becomes possible to execute a process in which at the time of an initial scan (will be described later), an initial scan is first executed for the transmission wave transmitted by the main polarized wave, and an initial scan is then executed for the transmission wave transmitted by the sub polarized wave after the initial scan for the transmission wave transmitted by the main polarized wave is completed.

[0236] Details of a configuration example of the digital broadcasting service for transmitting layers and segments of the dual-polarized terrestrial digital broadcasting according to the present embodiment will be described later. However, in a case where the existing terrestrial digital broadcasting service is transmitted by using a layer constituting a segment contained only in the main polarized wave and an advanced terrestrial digital service is transmitted by a layer containing a segment contained in both the main polarized wave and the sub polarized wave, the initial scan may first be executed for the transmission wave transmitted by the main polarized wave; the initial scan for the existing terrestrial digital broadcasting service may be completed; and then, the initial scan may be executed for the transmission wave transmitted by the sub polarized wave to execute the initial scan for the advanced terrestrial digital broadcasting service. By configuring it in this manner, it is possible to execute the initial scan for the advanced terrestrial digital broadcasting service after the initial scan for the existing terrestrial digital broadcasting service is completed. This makes it possible to reflect settings of the initial scan for the existing terrestrial digital broadcasting service to settings by the initial scan for the advanced terrestrial digital broadcasting service, whereby it is preferable.

[0237] Note that the definition of meanings of “1” and “0” of the bit of the main signal identification may be the reverse of the explanation described above.

[0238] Further, in place of the bit of the main signal identification, a polarization direction identification bit may be used as one parameter of the TMCC information. Specifically, the polarization direction identification bit may be set to “1” at the broadcast station side for the transmission wave to be transmitted by the horizontal polarized wave, and the polarization direction identification bit is set to “0” at the broadcast station side for the transmission wave to be transmitted by the vertical polarized wave. In the broadcast receiving apparatus **100** according to the embodiment of the present invention that receives the transmission wave of the dual-polarized

terrestrial digital broadcasting, by using the polarization direction identification bit, it is possible to identify whether the received transmission wave is transmitted in any polarization direction at the time of the transmission thereof. For example, by using an identifying process for the polarization direction, it becomes possible to execute a process in which at the time of an initial scan (will be described later), an initial scan is first executed for the transmission wave transmitted by the horizontal polarized wave, and an initial scan is then executed for the transmission wave transmitted by the vertical polarized wave after the initial scan for the transmission wave transmitted by the horizontal polarized wave is completed. With respect to explanation of effects of the process, the “main polarized wave” in the part regarding the initial scan in the explanation described above for the bit of the main signal identification may be read as the “horizontal polarized wave”, and the “sub polarized wave” may be read as the “vertical polarized wave”. For this reason, repeated explanation thereof is omitted.

[0239] Note that the definition of meanings of “1” and “0” of the polarization direction identification bit may be the reverse of the explanation described above.

[0240] Further, in place of the bit of the main signal identification described above, a first signal/second signal identification bit may be used as one parameter of the TMCC information. Specifically, one polarized wave of the horizontal polarized wave and the vertical polarized wave may be defined as a first polarized wave; a broadcasting signal of the transmission wave to be transmitted by the first polarized wave may be defined as a first signal; and the first signal/second signal identification bit may be set to “1” at the broadcast station side. Further, the other polarized wave may be defined as a second polarized wave; a broadcasting signal of the transmission wave to be transmitted by the second polarized wave may be defined as a second signal; and the first signal/second signal identification bit may be set to “0” at the broadcast station side. In the broadcast receiving apparatus **100** according to the embodiment of the present invention that receives the transmission wave of the dual-polarized terrestrial digital broadcasting, by using the first signal/second signal identification bit, it is possible to identify whether the received transmission wave is transmitted in any polarization direction at the time of the transmission thereof. Note that in the first signal/second signal identification bit, concepts of the “main polarized wave” and the “sub polarized wave” from the definition of the bit of the main signal identification described above are merely replaced by the “first polarized wave” and the “second polarized wave”. With respect to the process and the effects of the broadcast receiving apparatus **100**, the “main polarized wave” in the part regarding the process of the broadcast receiving apparatus **100** in the explanation described above for the bit of the main signal identification may be read as the “first polarized wave”, and the “sub polarized wave” may be read as the “second polarized wave”. For this reason, repeated explanation thereof is omitted.

[0241] Note that the definition of meanings of “1” and “0” of the first signal/second signal identification bit may be the reverse of the explanation described above.

[0242] Next, in the transmission wave of the layered division multiplexing terrestrial digital broadcasting according to the present embodiment, in place of the bit of the main signal identification described above, an upper/lower layer identification bit may be used as one parameter of the TMCC information. Specifically, the upper/lower layer identification bit described above may be set to “1” in the TMCC information of the modulation wave to be transmitted by the upper layer, and the upper/lower layer identification bit described above may be set to “0” in the TMCC information of the transmission wave to be transmitted by the lower layer. Further, in a case where the received broadcast wave is not the advanced terrestrial digital broadcasting service, this parameter may be set to “1”.

[0243] In the layered division multiplexing terrestrial digital broadcasting according to the present embodiment, here, in the generating process for the OFDM transmission wave at the broadcast station, with respect to a lower layer of a plurality of modulation waves originally transmitted by an upper layer and the lower layer of one physical channel, frequency conversion and signal

amplification may be executed by the converter **201L** illustrated in FIG. 2A depending upon installation environment of the broadcast receiving apparatus **100**. In a case where the broadcast receiving apparatus **100** receives the transmission wave of the layered division multiplexing terrestrial digital broadcasting, it is possible to identify, on the basis of the upper/lower layer identification bit described above, whether it is the modulation wave originally transmitted by the upper layer, or the modulation wave transmitted by the lower layer. For example, by means of the identifying process, it is possible to execute the initial scan for the advanced terrestrial digital broadcasting service to be transmitted by the lower layer after the initial scan for the existing terrestrial digital broadcasting service to be transmitted by the upper layer is completed, and this makes it possible to reflect the settings of the initial scan for the existing terrestrial digital broadcasting service to the settings by the initial scan for the advanced terrestrial digital broadcasting service. Further, in the third tuner/demodulator **130L** of the broadcast receiving apparatus **100**, it is possible to use the identification result for switching between the demodulator **133S** and the demodulator **133L**.

[0244] Note that in description of the dual-polarized transmission method according to each embodiment described below, unless otherwise specified, an example in which the horizontal polarized wave is the main polarized wave and the vertical polarized wave is the sub polarized wave will be described as one example. However, with respect to the horizontal polarized wave and the vertical polarized wave, the relation between the main and the sub may be reversed. FIG. 5I illustrates one example of bit allocation of the 4K signal transmission layer identification.

[0245] In a case where the broadcast wave to be transmitted is the transmission wave of the dual-polarized terrestrial digital broadcasting service according to the present embodiment, a bit of the 4K signal transmission layer identification may indicate for each of the B layer and the C layer whether transmission of the 4K broadcasting program is executed by using both the horizontal polarized signal and the vertical polarized signal or not. One bit is allocated to each of setting of the B layer and setting of the C layer. For example, in a case where the bit of the 4K signal transmission layer identification for each of the B layer and the C layer is “0”, it may indicate that the transmission of the 4K broadcasting program is executed by using both the horizontal polarized signal and the vertical polarized signal in the layer. In a case where the bit of the 4K signal transmission layer identification for each of the B layer and the C layer is “1”, it may indicate that transmission of the 4K broadcasting program using both the horizontal polarized signal and the vertical polarized signal in the layer is not executed. By configuring it in this manner, it is possible to identify whether the transmission of the 4K broadcasting program is executed by using both the horizontal polarized signal and the vertical polarized signal for each of the B layer and the C layer or not by using the bit of the 4K signal transmission layer identification in the broadcast receiving apparatus **100**.

[0246] Further, in a case where the broadcast wave to be transmitted is a broadcast wave of the layered division multiplexing terrestrial digital broadcasting service according to the present embodiment, the bit of the 4K signal transmission layer identification may indicate whether transmission of the 4K broadcasting program by the lower layer is to be executed or not. In a case where B119 of this parameter is “0”, transmission of the 4K broadcasting program by the lower layer is executed. In a case where B119 of this parameter is “1”, transmission of the 4K broadcasting program by the lower layer is not executed. By configuring it in this manner, it is possible to identify whether the transmission of the 4K broadcasting program by the lower layer by using the bit of the 4K signal transmission layer identification or not in the broadcast receiving apparatus **100**.

[0247] Note that in a case where this parameter is “0”, it is possible to adopt a modulation method of NUC (Non-Uniform Constellation) in addition to a basic modulation method illustrated in FIG. 5C as the carrier modulation mapping method. In this case, it is possible to transmit the current information/next information of transmission parameter additional information regarding the B

layer/C layer by using the AC1 and the like.

[0248] Further, in a case where the broadcast wave to be transmitted are not the advanced terrestrial digital broadcasting service, each of these parameters may be set to “1”.

[0249] Note that the definition of “0” and “1” of the bit of the 4K signal transmission layer identification explained above may be the reverse of the explanation described above.

[0250] FIG. 5J illustrates one example of bit allocation of the additional layer transmission identification. A bit of the additional layer transmission identification may indicate that the broadcast wave to be transmitted is the dual-polarized terrestrial digital broadcasting service according to the present embodiment and whether each of the B layer and the C layer of the transmission wave to be transmitted by the sub polarized wave is used as a virtual D layer or a virtual E layer or not.

[0251] For example, in the example of FIG. 5J, a bit arranged in B120 is a D layer transmission identification bit. In a case where this parameter is “0”, the B layer to be transmitted by the sub polarized wave is used as the virtual D layer. This means that to be expressed precisely, a group of segments each having the same segment number as the segment belonging to the B layer to be transmitted by the main polarized wave among the segments to be transmitted by the sub polarized wave is treated as the D layer that is a different layer from the B layer to be transmitted by the main polarized wave. In a case where this parameter is “1”, the B layer to be transmitted by the sub polarized wave is not used as the virtual D layer, but is used as the B layer.

[0252] Further, for example, a bit arranged in B121 is an E layer transmission identification bit. In a case where this parameter is “0”, the C layer to be transmitted by the sub polarized wave is used as the virtual E layer. This means that to be expressed precisely, a group of segments each having the same segment number as the segment belonging to the C layer to be transmitted by the main polarized wave among the segments to be transmitted by the sub polarized wave is treated as the E layer that is a different layer from the C layer to be transmitted by the main polarized wave. In a case where this parameter is “1”, the C layer to be transmitted by the sub polarized wave is not used as the virtual E layer, but is used as the C layer.

[0253] By configuring it in this manner, it is possible to identify presence or absence of the D layer and the E layer to be transmitted by the sub polarized wave by using the bit of the additional layer transmission identification (at least one of the D layer transmission identification bit or the E layer transmission identification bit) in the broadcast receiving apparatus **100**. Namely, in terrestrial digital broadcasting according to the present embodiment, by using a parameter of the additional layer transmission identification illustrated in FIG. 5J, it is possible to operate a new layer (in the example of FIG. 5J, the D layer and the E layer) beyond the number of layers that is limited to three layers including the A layer, the B layer, and the C layer in the existing terrestrial digital broadcasting.

[0254] Note that in a case where this parameter is “0”, it is possible to differentiate parameters such as the carrier modulation mapping method illustrated in FIG. 5C, the code rate, or the length of the time interleaving between the virtual D layer/virtual E layer and the B layer/C layer. In this case, in a case where the current information/next information of the parameters such as the carrier modulation mapping method regarding the virtual D layer/virtual E layer, the convolution code rate, or the length of the time interleaving is transmitted by using AC information (for example, the AC1), it is possible to grasp the parameters such as the carrier modulation mapping method regarding the virtual D layer/virtual E layer, the convolution code rate, or the length of the time interleaving at the broadcast receiving apparatus **100** side.

[0255] Note that as a modification example, in a case where the bit of the additional layer transmission identification (at least one of the D layer transmission identification bit or the E layer transmission identification bit) is “0”, it may be configured so that the transmission parameter of at least one of the B layer or the C layer of the current information/next information of the TMCC information to be transmitted by the sub polarized wave is switched to the meaning of the

transmission parameter of at least one of the virtual D layer or the virtual E layer. In this case, in a case where at least one of the virtual D layer or the virtual E layer is used, the A layer, the B layer, and the C layer may be used in the main polarized wave, the transmission parameter of these layers may be transmitted by the current information/next information of the TMCC information to be transmitted by the main polarized wave. Further, the A layer, the D layer, and the E layer may be used in the sub polarized wave, and the transmission parameter of these layers may be transmitted by the current information/next information of the TMCC information to be transmitted by the sub polarized wave. Even in this case, it is possible to grasp the parameters such as the carrier modulation mapping method regarding the virtual D layer/virtual E layer, the convolution code rate, or the length of the time interleaving at the broadcast receiving apparatus **100** side.

[0256] Further, in a case where the broadcast wave to be transmitted is not the advanced terrestrial digital broadcasting service or in a case where the layered division multiplexing transmission method is used even in the advanced terrestrial digital broadcasting service, it may be configured so that this parameter is respectively set to “1”.

[0257] Note that the parameter of the additional layer transmission identification may be stored in both the TMCC information of the main polarized wave and the TMCC information of the sub polarized wave. However, any of the processes described above can be realized so long as the parameter is stored at least in the TMCC information of the sub polarized wave.

[0258] Further, the definition of “0” and “1” of the bit of the additional layer transmission identification explained above may be the reverse of the explanation described above.

[0259] Note that in a case where a parameter of the 4K signal transmission layer identification described above indicates that transmission of the 4K broadcasting program is executed by the B layer, the broadcast receiving apparatus **100** may ignore the D layer transmission identification bit even though the D layer transmission identification bit described above indicates that the B layer is used as the virtual D layer. Similarly, in a case where the parameter of the 4K signal transmission layer identification indicates that the transmission of the 4K broadcasting program is executed by the C layer, the broadcast receiving apparatus **100** may be configured so as to ignore the E layer transmission identification bit even though the E layer transmission identification bit indicates that the C layer is used as the virtual E layer. By clarifying a priority order of the bits used in a determining process, it is possible to prevent conflict of the determining process in the broadcast receiving apparatus **100**.

[0260] Further, in the broadcast wave to be transmitted, all of the bit of the frequency converting process identification described above, the bit of the physical channel number identification, the bit of the main signal identification, a bit of 4K signal transmission identification, and the bit of the additional layer transmission identification may be set to “1” in principle in a case where the parameter of the system identification described above is not “10”. Even in a case where the parameter of the system identification is not “10” but exceptionally due to a certain problem, any of the bit of the frequency converting process identification, the bit of the physical channel number identification, the bit of the main signal identification, the bit of the 4K signal transmission identification, and the bit of the additional layer transmission identification is not “1”, the broadcast receiving apparatus **100** may be configured so as to ignore the bit that is not “1”, and determine that all of these bits are “1”.

[0261] FIG. 5K illustrates one example of a “code rate” bits illustrated in FIG. 5C, that is, bit allocation of code rate identification in the error correction.

[0262] Here, in a terrestrial digital broadcasting method for an existing 2K broadcasting service, an identification bit for transmitting the code rate dedicated to the “convolution code” is transmitted. However, in the digital broadcasting according to the present embodiment, it is possible to broadcast an advanced terrestrial digital broadcasting service of 4K broadcasting and the terrestrial digital broadcasting service of the 2K broadcasting so as to be mixed with each other. As have already been described, the LDPC code can be used as the inner code in the advanced terrestrial

digital broadcasting service of the 4K broadcasting.

[0263] Therefore, unlike the terrestrial digital broadcasting method of the existing 2K broadcasting service, a bit of the code rate identification in the error correction according to the present embodiment illustrated in FIG. 5K is configured so as not to correspond to the code rate identification bit dedicated for the convolution code, but is configured so as to correspond to the LDPC code.

[0264] Here, even in a case where the inner code of the target terrestrial digital broadcasting service is the convolution code or the LDPC code, by using bits arranged in a common range as the identification bit for code rate transmission, the number of bits can be saved. Moreover, even though the identification bits are the same as each other, by respectively setting the code rate independently between a case where the inner code of the target terrestrial digital broadcasting service is the convolution code and a case where the inner code is the LDPC code, it is possible to adopt a group of choices of the code rate suitable for each coding method as a digital broadcasting system.

[0265] Specifically, in the example illustrated in FIG. 5K, in a case where the identification bit is “000”, it indicates that the code rate is $\frac{1}{2}$ if the inner code is the convolution code, and the code rate is $\frac{2}{3}$ if the inner code is the LDPC code. In a case where the identification bit is “001”, it indicates that the code rate is $\frac{2}{3}$ if the inner code is the convolution code, and the code rate is $\frac{3}{4}$ if the inner code is the LDPC code. In a case where the identification bit is “010”, it indicates that the code rate is $\frac{3}{4}$ if the inner code is the convolution code, and the code rate is $\frac{5}{6}$ if the inner code is the LDPC code. In a case where the identification bit is “011”, it indicates that the code rate is $\frac{5}{6}$ if the inner code is the convolution code, and the code rate is $\frac{2}{16}$ if the inner code is the LDPC code. In a case where the identification bit is “100”, it indicates that the code rate is $\frac{7}{8}$ if the inner code is the convolution code, and the code rate is $\frac{6}{16}$ if the inner code is the LDPC code. In a case where the identification bit is “101”, it indicates that the code rate is undefined if the inner code is the convolution code, and the code rate is $\frac{10}{16}$ if the inner code is the LDPC code. In a case where the identification bit is “110”, it indicates that the code rate is undefined if the inner code is the convolution code, and the code rate is $\frac{14}{16}$ if the inner code is the LDPC code. In a case where there is unused layer or no next information, “111” is set to this parameter.

[0266] Note that identification of whether the inner code of the target terrestrial digital broadcasting service is the convolution code or the LDPC code may be executed by using a result of identifying whether the terrestrial digital broadcasting service is the existing terrestrial digital broadcasting service or the advanced terrestrial digital broadcasting service. The identification may be executed by using the identification bit explained with reference to FIG. 5D or FIG. 5I. Here, in a case where the target terrestrial digital broadcasting service is the existing terrestrial digital broadcasting service, it may be identified that the inner code is the convolution code. Further, in a case where the target terrestrial digital broadcasting service is the advanced terrestrial digital broadcasting service, it may be identified that the inner code is the LDPC code.

[0267] Further, as another example of the identification of whether the inner code of the target terrestrial digital broadcasting service is the convolution code or the LDPC code, it may be identified on the basis of an identification bit of an error correcting method, which will be described with reference to FIG. 6I.

[0268] According to the bit of the code rate identification of the error correction illustrated in FIG. 5K described above, it is suitable because it is possible to prevent the number of bits of the identification bit from increasing while supporting a plurality of inner code methods.

[0269] Further, in the advanced terrestrial digital broadcasting service of the dual-polarized transmission method, the TMCC information of the transmission wave to be transmitted by the horizontal polarized wave and the TMCC information of the transmission wave to be transmitted by the vertical polarized wave may be the same as each other, or may be different from each other. Similarly, in the advanced terrestrial digital broadcasting service of the layered division

multiplexing transmission method, the TMCC information of the transmission wave to be transmitted by the upper layer and the TMCC information of the transmission wave to be transmitted by the lower layer may be the same as each other, or may be different from each other. Further, the parameter of the frequency converting process identification described above, the parameter of the main signal identification, the parameter of the additional layer transmission identification may be described only in the TMCC information of the transmission wave to be transmitted by the sub polarized wave or the transmission wave to be transmitted by the lower layer.

[0270] Note that in the explanation described above, the example in which the parameter of the frequency converting process identification, the parameter of the main signal identification, the parameter of the polarization direction identification, the parameter of the first signal/second signal identification, the parameter of the upper/lower layer identification, the parameter of the 4K signal transmission layer identification, and the parameter of the additional layer transmission identification are transmitted so as to be included in the TMCC signal (or the TMCC carrier) has been explained. However, these parameters may be transmitted so as to be included in the AC signal (or the AC carrier). Namely, these parameters may be transmitted by a signal of a carrier (such as the TMCC carrier or the AC carrier) modulated by a modulation method that executes mapping with a smaller number of states than that of the modulation method for the data carrier.

[AC Signal]

[0271] The AC signal is an additional information signal regarding the broadcast, and includes additional information or seismic motion warning information regarding a transmission control of a modulation wave. Note that the seismic motion warning information is transmitted by using an AC carrier with segment 0. On the other hand, the additional information regarding the transmission control of the modulation wave can be transmitted by using an arbitrary AC carrier. FIG. 6A illustrates one example of bit allocation of the AC signal. The AC signal is configured by 204 bits (BC to B203). BC is a demodulation reference signal for an AC symbol, and has a predetermined amplitude and a phase reference. B1 to B3 are signals for identifying a configuration of the AC signal. B4 to B203 are used for transmission of the additional information or transmission of the seismic motion warning information regarding the transmission control of the modulation wave.

[0272] FIG. 6B illustrates one example of bit allocation of configuration identification of the AC signal. In a case where the seismic motion warning information is transmitted by using B4 to B203 of the AC signal, this parameter is set to “001” or “110”. A parameter of the configuration identification (“001” or “110”) in case of transmission of the seismic motion warning information has the same code as leading three bits (B1 to B3) of a synchronization signal of the TMCC signal, and is sent out alternately for each frame at the same timing as the TMCC signal. Further, in a case where this parameter is a value other than the value described above, it indicates that the additional information regarding the transmission control of the modulation wave is transmitted by using B4 to B203 of the AC signal. The additional information regarding the transmission control of the modulation wave may be transmitted by using B4 to B203 of the AC signal. In this case, as the parameter of the configuration identification of the AC signal, “000” and “111”, “010” and “101”, or “011” and “100” are sent out alternately for each frame.

[0273] B4 to B203 of the AC signal are used for transmission of the additional information or transmission of the seismic motion warning information regarding the transmission control of the modulation wave.

[0274] The transmission of the additional information regarding the transmission control of the modulation wave may be executed by various bit configurations. For example, the frequency converting process identification, the physical channel number identification, the main signal identification, the 4K signal transmission layer identification, and the additional layer transmission identification, which have been described in the explanation of the TMCC signal, the bit of the additional information regarding the transmission control of the modulation wave of the AC signal

may be allocated and transmitted in place of the TMCC signal or in addition to the TMCC signal. By configuring it in this manner, in the broadcast receiving apparatus **100**, it is possible to execute various kinds of identifying processes using these parameters, which have already been described in the explanation of the TMCC signal. Further, the current/next information of the transmission parameter additional information regarding the transmission layer of the 4K broadcasting program when any parameter of the 4K signal transmission layer identification is "0" or the transmission parameter regarding the virtual D layer/virtual E layer when any parameter of the additional layer transmission identification is "0" may be allocated. By configuring it in this manner, in the broadcast receiving apparatus **100**, it is possible to obtain the transmission parameter of each layer by using these parameters, and this makes it possible to control the demodulating process for each layer.

[0275] Transmission of the seismic motion warning information may be executed by bit allocation illustrated in FIG. **6C**. The seismic motion warning information is configured by a synchronization signal, a start/end flag, update flag, signal identification, seismic motion warning detailed information, CRC, and parity bits. The synchronization signal is configured by a code with 13 bits, and has the same code as 13 bits (B4 to B16) obtained by removing the leading three bits from the synchronization signal of the TMCC signal. In a case where the configuration identification of the AC signal indicates that the seismic motion warning information is to be transmitted, a code with 16 bits obtained by combining the configuration identification with the synchronization signal becomes the same synchronization words with 16 bits as the synchronization signal of the TMCC. The start/end flag is configured by a code with two bits as a flag of start timing/end timing of the seismic motion warning information. The start/end flag is changed from "11" to "00" when sending of the seismic motion warning information is started, and is changed from "00" to "11" when the sending of the seismic motion warning information is ended. The update flag is configured by a code with two bits, and is increased by "1" using "00" as an initial value every time the content of a series of seismic motion warning detailed information to be transmitted when the start/end flag is "00" is changed. After "11", it returns to "00". In a case where the start/end flag is "11", the update flag is also "11".

[0276] FIG. **6D** illustrates one example of bit allocation of the signal identification. The signal identification is configured by a code of three bits, and is used for identifying a type of the seismic motion warning detailed information. In a case where this parameter is "000", it means the "seismic motion warning detailed information (there is a corresponding region)". In a case where this parameter is "001", it means the "seismic motion warning detailed information (there is no corresponding region)". In a case where this parameter is "010", it means "a test signal of the seismic motion warning detailed information (there is a corresponding region)". In a case where this parameter is "011", it means "the test signal of the seismic motion warning detailed information (there is no corresponding region)". In a case where this parameter is "111", it means "no seismic motion warning detailed information". Note that in a case where the start/end flag is "00", the signal identification becomes any of "000", "001", "010", and "011". In a case where the start/end flag is "11", the signal identification becomes "111".

[0277] The seismic motion warning detailed information is configured by a code with 88 bits. In a case where the signal identification is any of "000", "001", "010", and "011", information such as information regarding a current time when the seismic motion warning information is sent out, information indicating a region that becomes a target of seismic motion warning, or latitude/longitude/earthquake intensity of a seismic center of earthquake that becomes a target of the seismic motion warning is transmitted as the seismic motion warning detailed information. FIG. **6E** illustrates one example of bit allocation of the seismic motion warning detailed information in a case where the signal identification is any of "000", "001", "010", and "011". Further, in a case where the signal identification is "111", it is possible to transmit a code or the like for identifying a broadcaster by using a bit of the seismic motion warning detailed information. FIG. **6F** illustrates

one example of bit allocation of the seismic motion warning detailed information in a case where the signal identification is “111”.

[0278] The CRC is a code generated by using a predetermined generator polynomial with respect to B21 to B111 of the seismic motion warning information. The parity bit is a code generated by a shortened code (187, 105) of a difference set cyclic code (273, 191) with reference to B17 to B121 of the seismic motion warning information.

[0279] In the broadcast receiving apparatus **100**, it is possible to execute various kinds of controls for addressing an emergency situation by using the parameters regarding the seismic motion warning, which have been described in FIG. 6C, FIG. 6D, FIG. 6E, and FIG. 6F. For example, it is possible to execute a presentation control of information regarding the seismic motion warning, a control to switch display content with low priority into display regarding the seismic motion warning, and a control to terminate display of an application and switch into display regarding the seismic motion warning or broadcasting program video.

[0280] FIG. 6G illustrates one example of bit allocation of the additional information regarding the transmission control of the modulation wave. The additional information regarding the transmission control of the modulation wave is configured by the synchronization signal, the current information, the next information, and the parity bit. The synchronization signal is configured by a code with 13 bits, and has the same code as 13 bits (B4 to B16) obtained by removing the leading three bits from the synchronization signal of the TMCC signal. In a case where the configuration identification of the AC signal indicates that the additional information regarding the transmission control of the modulation wave is to be transmitted, a code with 16 bits obtained by combining the configuration identification and the synchronization signal becomes synchronization words with 16 bits equivalent to the synchronization signal of the TMCC. The current information indicates current information of the transmission parameter additional information when the 4K broadcasting program is transmitted by the B layer or the C layer and the transmission parameter regarding the virtual D layer or the virtual E layer. The next information indicates information after switching of the transmission parameter additional information when the 4K broadcasting program is transmitted by the B layer or the C layer and the transmission parameter regarding the virtual D layer or the virtual E layer.

[0281] In the example of FIG. 6G, B18 to B30 of the current information are current information of the B layer transmission parameter additional information, and indicate current information of the transmission parameter additional information when the 4K broadcasting program is to be transmitted by the B layer. Further, B31 to B43 of the current information are current information of the C layer transmission parameter additional information, and indicate current information of the transmission parameter additional information when the 4K broadcasting program is to be transmitted by the C layer. Further, B70 to B82 of the next information are information after switching of the transmission parameters of the B layer transmission parameter additional information, and indicate information after switching of the transmission parameters of the transmission parameter additional information when the 4K broadcasting program is to be transmitted by the B layer. Further, B83 to B95 of the next information are information after switching of the transmission parameters of the C layer transmission parameter additional information, and indicate information after switching of the transmission parameters of the transmission parameter additional information when the 4K broadcasting program is to be transmitted by the C layer. Here, the transmission parameter additional information is a transmission parameter regarding modulation, which is added to the transmission parameter of the TMCC information illustrated in FIG. 5C to extend the specifications. The concrete content of the transmission parameter additional information will be described later.

[0282] In the example illustrated in FIG. 6G, B44 to B56 of the current information are current information of the transmission parameter for the virtual D layer in a case where the virtual D layer is operated. B57 to B69 of the current information are current information of the transmission

parameter for the virtual E layer in a case where the virtual E layer is operated. Further, B96 to B108 of the next information are information after switching of the transmission parameters for the virtual D layer in a case where the virtual D layer is operated. B109 to B121 of the current information are information after switching of the transmission parameters for the virtual E layer in a case where the virtual E layer is operated. The parameters stored in the transmission parameter for the virtual D layer and the transmission parameter for the virtual E layer may be the similar to those illustrated in FIG. 5C.

[0283] The virtual D layer and the virtual E layer are layers that do not exist in the existing terrestrial digital broadcasting. It is not easy to increase the number of bits of the TMCC information illustrated in FIG. 5B because it is necessary to maintain compatibility with the existing terrestrial digital broadcasting. Therefore, in the embodiment according to the present invention, the transmission parameters for the virtual D layer and the virtual E layer are not stored in the TMCC information, but are stored in the AC information as illustrated in FIG. 6G.

[0284] This makes it possible to transmit, to the receiving apparatus, information on modulation for the new virtual D layer and the new virtual E layer while maintaining compatibility of the TMCC information with the existing terrestrial digital broadcasting. As a result, in the broadcast wave of the dual-polarized terrestrial digital broadcasting service according to the present embodiment, in a case where the B layer/C layer of the transmission wave to be transmitted by the sub polarized wave is used as the virtual D layer/virtual E layer, it is possible to set the transmission parameter of the virtual D layer/virtual E layer of the transmission wave to be transmitted by the sub polarized wave so as to be differentiated from the transmission parameter of the B layer/C layer of the transmission wave to be transmitted by the main polarized wave.

[0285] Note that in a case where the virtual D layer or the virtual E layer is not used, information on the transmission parameter for the unused layer may be ignored by the broadcast receiving apparatus **100** without any problem. For example, in a case where the parameter of the additional layer transmission identification of the TMCC information illustrated in FIG. 5J indicates “1” for the virtual D layer or the virtual E layer (in a case where it indicates that the virtual D layer/virtual E layer is not used), the broadcast receiving apparatus **100** may be configured so as to ignore the information on the transmission parameter even though any value is contained in the transmission parameter illustrated in FIG. 6G for the unused virtual D layer or the unused virtual E layer.

[0286] Next, details of the transmission parameter additional information, which has been explained with reference to FIG. 6G, will be described.

[0287] FIG. 6H illustrates one concrete example of the transmission parameter additional information. It is possible to contain a parameter of an error correcting method, a parameter of a constellation form, and the like in the transmission parameter additional information.

[0288] The error correcting method indicates setting of what kind of coding method is used as the error correcting method for the inner code and the outer code when the 4K broadcasting program (the advanced terrestrial digital broadcasting service) is to be transmitted by the B layer or the C layer. FIG. 6I illustrates one example of bit allocation of the error correcting method. In a case where this parameter is “000”, a convolution code is used as the inner code and a shortened RS code is used as the outer code when the 4K broadcasting program is to be transmitted by the B layer or the C layer. In a case where this parameter is “001”, the LDPC code is used as the inner code and the BCH code is used as the outer code when the 4K broadcasting program is to be transmitted by the B layer and the C layer. Moreover, the other combinations may be set so that they can be selected.

[0289] Further, when the 4K broadcasting program is transmitted by the B layer and the C layer, it is possible to adopt not only a uniform constellation but also a non-uniform constellation (Non Uniform Constellation: NUC) as the carrier modulation mapping method. FIG. 6J illustrates one example of bit allocation of the constellation form. In a case where this parameter is “000”, the carrier modulation mapping method selected in the transmission parameter of the TMCC

information is applied to the uniform constellation. In a case where this parameter is any of “001” to “111”, the carrier modulation mapping method selected in the transmission parameter of the TMCC information is applied to the non-uniform constellation. Note that in a case where the non-uniform constellation is to be applied, the optimum value of the non-uniform constellation differs depending upon the type of the error correcting method and a code rate thereof. Therefore, in a case where the parameter of the constellation form is any of “001” to “111”, the broadcast receiving apparatus **100** according to the present embodiment may determine the non-uniform constellation used in the demodulating process on the basis of the parameter of the carrier modulation mapping method, the parameter of the error correcting method, and the parameter of the code rate thereof. This determination may be made by referring to a predetermined table that the broadcast receiving apparatus **100** stores in advance.

[Transmission Method 1 of Advanced Terrestrial Digital Broadcasting Service]

[0290] In order to realize 4K (3840 horizontal pixels×2160 vertical pixels) broadcasting while maintaining viewing environment of the existing terrestrial digital broadcasting service, the dual-polarized transmission method will be described as one example of the transmission method of the advanced terrestrial digital broadcasting service according to the embodiment of the present invention. The dual-polarized transmission method according to the embodiment of the present invention is a method that has specifications common to some specifications of the existing terrestrial digital broadcasting method. For example, 13 segments in about 6 MHz band corresponding to one physical channel are divided, and seven segments, five segments, and one segment are respectively allocated to transmission of a 2K (1920 horizontal pixels×1080 vertical pixels) broadcasting program, transmission of a 4K broadcasting program, and mobile reception (so-called one-segment broadcasting). Moreover, the five segments for the 4K broadcasting use not only the horizontal polarized signal but also the vertical polarized signal to secure a transmission capacity for total ten segments by a MIMO (Multiple-Input Multiple-Output) technique. Note that the 2K broadcasting program maintains image quality by optimizing the latest MPEG-2 Video compression technique, and can be received even by an existing television receiver. For the 4K broadcasting program, image quality is secured by optimization of a more high-efficient HEVC compression technique than MPEG-2 Video and multi-valued modulation. Note that the number of segments allocated for each broadcast may be different from the above.

[0291] FIG. 7A illustrates one example of the dual-polarized transmission method in the advanced terrestrial digital broadcasting service according to the embodiment of the present invention. A frequency band of 470 to 710 MHz is used for transmission of the broadcast wave of the terrestrial digital broadcasting service. The number of physical channels in the frequency band is 40 channels of 13 to 52 ch, each of the physical channels has a bandwidth of 6 MHz. In the dual-polarized transmission method according to the embodiment of the present invention, both the horizontal polarized signal and the vertical polarized signal are used in one physical channel.

[0292] FIG. 7A illustrates two examples of (1) and (2) as an allocation example of 13 segments. In the example of (1), transmission of a 2K broadcasting program is executed by using segments 1 to 7 (the B layer) of the horizontal polarized signal. Transmission of a 4K broadcasting program is executed by using total ten segments of segments 8 to 12 (the C layer) of the horizontal polarized signal and segments 8 to 12 (the C layer) of the vertical polarized signal. The segments 1 to 7 (the B layer) of the vertical polarized signal may be used for transmission of the same broadcasting program as the 2K broadcasting program to be transmitted by the segments 1 to 7 (the B layer) of the horizontal polarized signal. Alternatively, the segments 1 to 7 (the B layer) of the vertical polarized signal may be used for transmission of a different broadcasting program from the 2K broadcasting program to be transmitted by the segments 1 to 7 (the B layer) of the horizontal polarized signal. Alternatively, the segments 1 to 7 (the B layer) of the vertical polarized signal may be used for the other data transmission, or may not be used. Identification information of how to use the segments 1 to 7 (the B layer) of the vertical polarized signal can be transmitted to the

receiving apparatus side by the parameter of the 4K signal transmission layer identification or the parameter of the additional layer transmission identification of the TMCC signal, which has already been explained. In the broadcast receiving apparatus **100**, it is possible to identify handling of the segments 1 to 7 (the B layer) of the vertical polarized signal by these parameters. Further, the 2K broadcasting program to be transmitted by using the B layer of the horizontal polarized signal and the 4K broadcasting program to be transmitted by using the C layers of both horizontal/vertical polarized signals may be simultaneous broadcasting (simulcast) in which a broadcasting program with the same content is transmitted with different resolutions, or may be one in which a broadcasting program with different content is transmitted. In the segment 0 of both horizontal/vertical polarized signals, transmission of the same one-segment broadcasting program is executed.

[0293] The example illustrated in (2) of FIG. 7A is a modification example different from (1). In the example of (2), transmission of a 4K broadcasting program is executed by using total ten segments of segments 1 to 5 (the B layer) of the horizontal polarized signal and segments 1 to 5 (the B layer) of the vertical polarized signal. Transmission of a 2K broadcasting program is executed by using segments 6 to 12 (the C layer) of the horizontal polarized signal. Even in the example of (2), the segments 6 to 12 (the C layer) of the vertical polarized signal may be used for transmission of the same broadcasting program as the 2K broadcasting program to be transmitted by the segments 6 to 12 (the C layer) of the horizontal polarized signal. The segments 6 to 12 (the C layer) of the vertical polarized signal may be used for transmission of a different broadcasting program from the 2K broadcasting program to be transmitted by the segments 6 to 12 (the C layer) of the horizontal polarized signal. Further, the segments 6 to 12 (the C layer) of the vertical polarized signal may be used for the other data transmission, or may not be used. These kinds of identification information are similar to those in the example of (1). For this reason, repeated explanation thereof is omitted.

[0294] Note that in any of the examples illustrated in (1) and (2) of FIG. 7A, a case where the horizontal polarized wave is the main polarized wave has been described as an example, but the horizontal polarized wave and the vertical polarized wave may be reversed depending upon the operation thereof.

[0295] FIG. 7B illustrates one example of a configuration of a broadcasting system for the advanced terrestrial digital broadcasting service using the dual-polarized transmission method according to the embodiment of the present invention. This illustrates both a system at a transmission side and a system at a reception side of the advanced terrestrial digital broadcasting service using the dual-polarized transmission method. A configuration of the broadcasting system of the advanced terrestrial digital broadcasting service using the dual-polarized transmission method is basically similar to the configuration of the broadcasting system illustrated in FIG. 1. However, a radio tower **300T** that is facility of the broadcast station is a dual polarized transmission antenna capable of sending out the horizontal polarized signal and the vertical polarized signal at the same time. Further, in the example of FIG. 7B, only the tuning/wave detector **131H** and the tuning/wave detector **131V** of the second tuner/demodulator **130T** in the broadcast receiving apparatus **100** are extracted and illustrated, and the other operating units are omitted.

[0296] The horizontal polarized signal sent out from the radio tower **300T** is received by a horizontal polarized wave receiving element of the antenna **200T** that is a dual polarized reception antenna, and is inputted into the tuning/wave detector **131H** from a connector **100F1** via a coaxial cable **202T1**. On the other hand, the vertical polarized signal sent out from the radio tower **300T** is received by a vertical polarized wave receiving element of the antenna **200T**, and is inputted into the tuning/wave detector **131V** from a connector **100F2** via a coaxial cable **202T2**. An F type connector is generally used as a connector that connects an antenna (for example, a coaxial cable) and a television receiver.

[0297] Here, the user may mistakenly connect the coaxial cable **202T1** to the connector **100F2**, and

connect the coaxial cable **202T2** to the connector **100F1**. In this case, in the tuning/wave detector **131H** and the tuning/wave detector **131V**, there is a possibility that a defect such as being unable to identify whether the inputted broadcasting signal is the horizontal polarized signal or the vertical polarized signal. In order to prevent the defect described above, it is considered that one of the connectors each of which connects the antenna (coaxial cable) to the television receiver, for example, the connector between the coaxial cable **202T2** and the connector **100F2** via which the vertical polarized signal is to be transmitted is a connector with a different shape other than that of the F type connector of the connector between the coaxial cable **202T1** and the connector **100F1** via which the horizontal polarized signal is to be transmitted. Alternatively, each of the tuning/wave detector **131H** and the tuning/wave detector **131V** may refer to the main signal identification of the TMCC information of each input signal to control the operation thereof so as to identify whether the inputted broadcasting signal is the horizontal polarized signal or the vertical polarized signal. [0298] FIG. 7C illustrates one example of a configuration example different from the configuration of the broadcasting system described above for the advanced terrestrial digital broadcasting service using the dual-polarized transmission method according to the embodiment of the present invention. The configuration as illustrated in FIG. 7B in which the broadcast receiving apparatus **100** includes two connectors for inputting broadcasting signals and two coaxial cables are used to connect the antenna **200T** to the broadcast receiving apparatus **100** may not always be suitable for the cost of facility and handling at the time of cable wiring. Therefore, in the configuration illustrated in FIG. 7C, the horizontal polarized signal received by the horizontal polarized wave receiving element of the antenna **200T** and the vertical polarized signal received by the vertical polarized wave receiving element of the antenna **200T** are inputted into the converter (converting unit) **201T**, and connection between the converter **201T** and the broadcast receiving apparatus **100** is executed by a single coaxial cable **202T3**. A broadcasting signal inputted from a connector **100F3** is demultiplexed and inputted into the tuning/wave detector **131H** and the tuning/wave detector **131V**. The connector **100F3** may have a function to supply operating power to the converter **201T**.

[0299] The converter **201T** may belong to facility in an environment (for example, an apartment house or the like) in which the broadcast receiving apparatus **100** is installed. Alternatively, the converter **201T** may be configured as a device integrated with the antenna **200T**, and installed in a house or the like. The converter **201T** executes a frequency converting process for any one of the horizontal polarized signal received by the horizontal polarized wave receiving element of the antenna **200T** and the vertical polarized signal received by the vertical polarized wave receiving element of the antenna **200T**. By this process, the horizontal polarized signal and the vertical polarized signal transmitted from the radio tower **300T** to the antenna **200T** can be separated into different frequency bands from each other by using the horizontal polarized wave and the vertical polarized wave of the same frequency band, and may be transmitted to the broadcast receiving apparatus **100** by the single coaxial cable **202T3** at the same time. Note that, if necessary, the frequency converting process may be executed for both the horizontal polarized signal and the vertical polarized signal, but in this case, the frequency bands of the both after frequency conversion have to be different from each other. Further, the broadcast receiving apparatus **100** may include one connector **100F3** for inputting the broadcasting signal.

[0300] FIG. 7D illustrates one example of the frequency converting process. In this example, the frequency converting process is executed for the vertical polarized signal. Specifically, of a horizontal polarized signal and a vertical polarized signal transmitted by a frequency band of 470 to 710 MHz (that is, a band corresponding to 13 ch to 52 ch of UHF), a frequency band of the vertical polarized signal is converted from the frequency band of 470 to 710 MHz to a frequency band of 770 to 1010 MHz. By this process, it is possible to transmit signals transmitted by using the horizontal polarized wave and the vertical polarized wave with the same frequency band to the broadcast receiving apparatus **100** through the coaxial cable **202T3** at the same time without

interfering with each other. Note that the frequency converting process may be executed for the horizontal polarized signal.

[0301] Further, it is preferable that the frequency converting process is executed for the signal transmitted by the sub polarized wave in accordance with a result obtained by referring to the main signal identification of the TMCC information. As explained with reference to FIG. 5H, the signal transmitted by the main polarized wave is more likely to be transmitted so as to include the existing terrestrial digital broadcasting service compared with the signal transmitted by the sub polarized wave. Therefore, in order to maintain compatibility with the existing terrestrial digital broadcasting service more suitably, it is preferable that a frequency of the signal transmitted by the main polarized wave is not converted, but a frequency of the signal transmitted by the sub polarized wave is converted.

[0302] Further, in a case where the frequency of the signal transmitted by the sub polarized wave is converted, it is desirable that a frequency band of the signal transmitted by the sub polarized wave is set to be higher than a frequency band of the signal transmitted by the main polarized wave in the signals after conversion. As a result, by starting a scan from a low frequency side and causing the scan to proceed to a high frequency side in the initial scan of the broadcast receiving apparatus **100**, it is possible to execute the initial scan for the signal transmitted by the main polarized wave prior to that for the signal transmitted by the sub polarized wave. This makes it possible to execute a process of reflecting settings by the initial scan of the existing terrestrial digital broadcasting service to settings by the initial scan of the advanced terrestrial digital broadcasting service more suitably.

[0303] Further, the frequency converting process may be executed for all physical channels used in the advanced terrestrial digital broadcasting service, but may be executed only for a physical channel using signal transmission by the dual-polarized transmission method.

[0304] Note that it is preferable that the frequency band after conversion by the frequency converting process is between 710 and 1032 MHz. Namely, in a case where the terrestrial digital broadcasting service and a BS/CS digital broadcasting service are to be received at the same time, it is conceivable that the broadcasting signal of the terrestrial digital broadcasting service received by the antenna **200T** and the broadcasting signal of the BS/CS digital broadcasting service received by the antenna **200B** are mixed and transmitted to the broadcast receiving apparatus **100** with a single coaxial cable. In this case, a BS/CS-IF signal uses a frequency band of about 1032 to 2150 MHz. For this reason, by setting the frequency band after conversion by the frequency converting process to be between 710 and 1032 MHz, it becomes possible to avoid interference between the broadcasting signal of the terrestrial digital broadcasting service and the broadcasting signal of the BS/CS digital broadcasting service while avoiding interference between the horizontal polarized signal and the vertical polarized signal. Further, in case of considering reception of a retransmission broadcasting signal by a cable television (Community Antenna TV or Cable TV: CATV) station, a frequency band of 770 MHz or less (a band corresponding to 62 ch or less of the UHF) is used in television broadcasting distribution by the cable television station. For this reason, it is more preferable that the frequency band after conversion by the frequency converting process is set to be between 770 and 1032 MHz, which exceeds the band corresponding to 62 ch of the UHF.

[0305] Further, it is preferable that a bandwidth of a range between the frequency band before conversion by the frequency converting process and the frequency band after conversion (that is, “a” portion in FIG. 7D) is set to be an integral multiple of the bandwidth (6 MHz) of one physical channel. By configuring it in this manner, in the broadcast receiving apparatus **100**, there is an advantage that a frequency setting control becomes easy in a case where a frequency of the broadcasting signal of the frequency band before conversion by the frequency converting process and a frequency of the broadcasting signal of the frequency band after conversion are collectively scanned.

[0306] Note that as described above, in the dual-polarized transmission method according to the

embodiment of the present invention, both the horizontal polarized signal and the vertical polarized signal are used for transmission of the 4K broadcasting program. Therefore, in order to reproduce the 4K broadcasting program correctly, it is necessary for the reception side to correctly grasp a combination of physical channels between the broadcasting signal transmitted by the horizontal polarized wave and the broadcasting signal transmitted by the vertical polarized wave. Even in a case where the frequency converting process is executed and the broadcasting signal transmitted by the horizontal polarized wave and the broadcasting signal transmitted by the vertical polarized wave for the same physical channel are inputted into the receiving apparatus as signals of different frequency bands, in the broadcast receiving apparatus **100** according to the present embodiment, by appropriately referring to the parameters of the TMCC information illustrated in FIG. 5F to FIG. 5J (for example, the main signal identification and the physical channel number identification), it is possible to correctly grasp the combination of the broadcasting signal transmitted by the horizontal polarized wave and the broadcasting signal transmitted by the vertical polarized wave for the same physical channel. This makes it possible to suitably receive, demodulate, and reproduce the 4K broadcasting program in the broadcast receiving apparatus **100** according to the present embodiment.

[0307] Note that in any of the examples illustrated in FIG. 7B, FIG. 7C, and FIG. 7D, a case where the horizontal polarized wave is the main polarized wave has been described as an example, but the horizontal polarized wave and the vertical polarized wave may be reversed depending upon the operation thereof.

[0308] Note that as described above, the broadcast wave of the terrestrial digital broadcasting to be transmitted by the dual-polarized transmission method explained above can be received and reproduced by the second tuner/demodulator **130T** of the broadcast receiving apparatus **100**, but is received even by the first tuner/demodulator **130C** of the broadcast receiving apparatus **100**. In a case where the broadcast wave of the terrestrial digital broadcasting is received by the first tuner/demodulator **130C**, the broadcasting signal transmitted by the layer for the advanced terrestrial digital broadcasting service of the broadcasting signals of the broadcast wave of the terrestrial digital broadcasting is ignored, but the broadcasting signal transmitted by the layer for the existing terrestrial digital broadcasting service is reproduced.

<Pass-Through Transmission Method of Advanced Terrestrial Digital Broadcasting Service>

[0309] The broadcast receiving apparatus **100** can receive a signal transmitted by a pass-through transmission method. The pass-through transmission method is a method in which a broadcasting signal received by the cable television station or the like is sent out to a distribution system of a CATV at the same frequency or via frequency conversion by a signal method as it is.

[0310] The pass-through transmission method includes: (1) a method of executing transmission signal band extraction and level adjustment for each terrestrial digital broadcasting signal outputted from a terrestrial reception antenna, and transmitting the terrestrial digital broadcasting signal to a CATV facility at the same frequency as a transmission signal frequency; and (2) a method of executing the transmission signal band extraction and the level adjustment for each terrestrial digital broadcasting signal outputted from the terrestrial reception antenna, and transmitting the terrestrial digital broadcasting signal to the CATV facility at a frequency for any of a VHF band, a MID band, a SHB band, and a UHF band, which is set by a CATV facility manager. Equipment constituting a reception amplifier for executing signal processing of the first method or equipment constituting a reception amplifier and a frequency converter for executing signal processing of the second method is an OFDM signal processor (OFDM Signal Processor: OFDM-SP).

[0311] FIG. 7E illustrates one example of a system configuration in a case where the first method of the pass-through transmission method is applied to the advanced terrestrial digital broadcasting service of the dual-polarized transmission method. FIG. 7E illustrates a headend facility **400C** and the broadcast receiving apparatus **100** of the cable television station. Further, FIG. 7F illustrates one example of the frequency converting process at that time. A notation (H/V) in FIG. 7F

indicates a state where both a broadcasting signal transmitted by a horizontal polarized wave and a broadcasting signal transmitted by a vertical polarized wave are broadcasting signals that exist in the same frequency band. A notation (H) indicates the broadcasting signal transmitted by the horizontal polarized wave, and a notation (V) indicates the broadcasting signal transmitted by the vertical polarized wave. Notations in FIG. 7H and FIG. 7I described below respectively have the similar meanings.

[0312] In a case where the pass-through transmission of the first method is applied to the advanced terrestrial digital broadcasting service of the dual-polarized transmission method according to the embodiment of the present invention, the transmission signal band extraction and the level adjustment are executed for the broadcasting signal transmitted by the horizontal polarized wave in the headend facility **400C** of the cable television station, and the broadcasting signal is sent out at the same frequency as the transmission signal frequency. On the other hand, the transmission signal band extraction and the level adjustment are executed for the broadcasting signal transmitted by the vertical polarized wave in the headend facility **400C** of the cable television station, and the broadcasting signal is sent out after the similar frequency converting process to that described with reference to FIG. 7D (that is, the process of converting the broadcasting signal transmitted by the vertical polarized wave into a frequency band higher than the frequency band of 470 to 770 MHz, which is the band corresponding to 13 ch to 62 ch of the UHF) is executed. This process causes frequency bands of the broadcasting signal transmitted by the horizontal polarized wave and the broadcasting signal transmitted by the vertical polarized wave not to be overlapped. For this reason, signal transmission by one coaxial cable (or one optical fiber cable) becomes possible. The transmitted signal can be received by the broadcast receiving apparatus **100** according to the present embodiment. A process of receiving and demodulating the broadcasting signal transmitted by the horizontal polarized wave and the broadcasting signal transmitted by the vertical polarized wave included in the signal by the broadcast receiving apparatus **100** according to the present embodiment is similar to that described with reference to FIG. 7D. For this reason, repeated explanation thereof is omitted.

[0313] FIG. 7G illustrates one example of a system configuration in a case where the second method of the pass-through transmission method is applied to the advanced terrestrial digital broadcasting service of the dual-polarized transmission method. FIG. 7G illustrates the headend facility **400C** and the broadcast receiving apparatus **100** of the cable television station. Further, FIG. 7H illustrates one example of the frequency converting process at that time.

[0314] In a case where the pass-through transmission of the second method is applied to the advanced terrestrial digital broadcasting service of the dual-polarized transmission method according to the embodiment of the present invention, the transmission signal band extraction and the level adjustment are executed for the broadcasting signal transmitted by the horizontal polarized wave in the headend facility **400C** of the cable television station, and the broadcasting signal is sent out after the frequency converting process into the frequency set by the CATV facility manager is executed. On the other hand, the transmission signal band extraction and the level adjustment are executed for the broadcasting signal transmitted by the vertical polarized wave in the headend facility **400C** of the cable television station, and the broadcasting signal is sent out after the similar frequency converting process to that described with reference to FIG. 7D (that is, the process of converting the broadcasting signal transmitted by the vertical polarized wave into the frequency band higher than the frequency band of 470 to 770 MHz, which is the band corresponding to 13 ch to 62 ch of the UHF) is executed. The frequency converting process illustrated in FIG. 7H is different from that illustrated in FIG. 7F, and is one in which the frequency of the broadcasting signal transmitted by the horizontal polarized wave is converted so as to be rearranged in a range of 90 to 770 MHz, which is expanded to a lower frequency band without limiting the frequency band of 470 to 770 MHz that is the band of 13 ch to 62 ch of the UHF. This process causes frequency bands of the broadcasting signal transmitted by the horizontal polarized

wave and the broadcasting signal transmitted by the vertical polarized wave not to be overlapped. For this reason, signal transmission by one coaxial cable (or one optical fiber cable) becomes possible. The transmitted signal can be received by the broadcast receiving apparatus **100** according to the present embodiment. A process of receiving and demodulating the broadcasting signal transmitted by the horizontal polarized wave and the broadcasting signal transmitted by the vertical polarized wave included in the signal by the broadcast receiving apparatus **100** according to the present embodiment is similar to that described with reference to FIG. 7D. For this reason, repeated explanation thereof is omitted.

[0315] Further, as another modification example of the frequency converting process for the headend facility **400C** of the cable television station illustrated in FIG. 7G, the broadcasting signal at the time of pass-through output after the frequency conversion may be changed into any of states respectively illustrated in FIG. 7H and FIG. 7I. In this case, the transmission signal band extraction and the level adjustment may be executed for both the broadcasting signal transmitted by the horizontal polarized wave and the broadcasting signal transmitted by the vertical polarized wave, and the broadcasting signals may be sent out after the frequency converting process into the frequency set by the CATV facility manager is executed. In the example of FIG. 7I, frequency conversion is executed so that both the broadcasting signal transmitted by the horizontal polarized wave and the broadcasting signal transmitted by the vertical polarized wave are rearranged in a range of 90 to 770 MHz (that is, a range from VHF 1 ch to UHF 62 ch). Since any frequency band of a range that exceeds UHF 62 ch is not used, frequency band utilization efficiency of the broadcasting signal can be heightened compared with the case illustrated in FIG. 7H.

[0316] Further, the band in which the broadcasting signals are rearranged is wider than the frequency band of 470 to 710 MHz that is the band of 13 ch to 52 ch of the UHF at the time of reception by the antenna, as illustrated in the example of FIG. 7I, it is possible to alternately rearrange the broadcasting signal transmitted by the horizontal polarized wave and the broadcasting signal transmitted by the vertical polarized wave. At this time, as illustrated in the example of FIG. 7I, by alternately rearranging a pair of the broadcasting signal transmitted by the horizontal polarized wave and the broadcasting signal transmitted by the vertical polarized wave, which were in the same physical channel at the time of reception by the antenna, in the order of the physical channels at the time of reception by the antenna, in a case where the broadcast receiving apparatus **100** according to the present embodiment executes the initial scan from the low frequency side, it is possible to proceed with initial settings for the pair of the broadcasting signal transmitted by the horizontal polarized wave and the broadcasting signal transmitted by the vertical polarized wave, which are originally in the same physical channel, in order by originally the same physical channel unit, and this makes it possible to execute the initial scan efficiently.

[0317] Note that in any of the examples illustrated in FIG. 7E, FIG. 7F, FIG. 7G, FIG. 7H, and FIG. 7I, a case where the horizontal polarized wave is the main polarized wave has been described as an example, but the horizontal polarized wave and the vertical polarized wave may be reversed depending upon the operation thereof.

[0318] Note that as described above, the broadcast wave of the terrestrial digital broadcasting of the dual-polarized transmission method, to which the pass-through transmission method explained above was made, can be received and reproduced by the second tuner/demodulator **130T** of the broadcast receiving apparatus **100**, but it can be received even by the first tuner/demodulator **130C** of the broadcast receiving apparatus **100**. In a case where the broadcast wave of the terrestrial digital broadcasting is received by the first tuner/demodulator **130C**, the broadcasting signal transmitted by the layer for the advanced terrestrial digital broadcasting service of the broadcasting signals of the broadcast wave of the terrestrial digital broadcasting is ignored, but the broadcasting signal transmitted by the layer for the existing terrestrial digital broadcasting service is reproduced. [Transmission Method 2 of Advanced Terrestrial Digital Broadcasting Service]

[0319] In order to realize the 4K broadcasting while maintaining the viewing environment of the

existing terrestrial digital broadcasting service, as one example different from that described above of the transmission method of the advanced terrestrial digital broadcasting service according to the embodiment of the present invention, the layered division multiplexing transmission method will be described. The layered division multiplexing transmission method according to the embodiment of the present invention is a method that has specifications common to some specifications of the existing terrestrial digital broadcasting method. For example, a broadcast wave of a 4K broadcasting service whose signal level is a low level is multiplexed into the same channel as the broadcast wave of the existing 2K broadcasting service, and then is transmitted. Note that the 2K broadcasting is received as before by suppressing a reception level of the 4K broadcasting to a required C/N or less. With respect to the 4K broadcasting, a 2K broadcast wave is cancelled using a receiving technique compatible with an LDM (layered division multiplexing) technique while expanding a transmission capacity by multi-valued modulation or the like, and the 4K broadcasting is received through the remaining 4K broadcast wave.

[0320] FIG. 8A illustrates one example of the layered division multiplexing transmission method in the advanced terrestrial digital broadcasting service according to the embodiment of the present invention. An upper layer is configured by a modulation wave of the existing 2K broadcasting, and a lower layer is configured by a modulation wave of the 4K broadcasting. Then, the upper layer and the lower layer are multiplexed, and are outputted as a synthetic wave with the same frequency band. For example, they may be configured so that 64 QAM or the like is used as the modulation method in the upper layer and 256 QAM or the like is used as the modulation method in the lower layer. Note that a 2K broadcasting program to be transmitted using the upper layer and a 4K broadcasting program to be transmitted using the lower layer may be the simultaneous broadcasting in which the broadcasting program with the same content is transmitted at different resolutions, or may be one in which broadcasting programs with different content are transmitted. Here, the upper layer is transmitted with high electric power, and the lower layer is transmitted with low electric power. Note that a difference between a modulation wave level of the upper layer and a modulation wave level of the lower layer (that is, a difference between the electric powers) is called an injection level (IL: Injection Level), which is a value that is set at the broadcast station side. The injection level generally indicates the difference between the modulation wave levels (the difference between the electric powers) as a relative ratio of logarithmic representation (dB).

[0321] FIG. 8B illustrates one example of a configuration of a broadcasting system for the advanced terrestrial digital broadcasting service using the layered division multiplexing transmission method according to the embodiment of the present invention. The configuration of the broadcasting system of the advanced terrestrial digital broadcasting service using the layered division multiplexing transmission method is basically similar to the configuration of the broadcasting system illustrated in FIG. 1. However, a radio tower **300L** that is facility of the broadcast station is a transmission antenna configured to send out a broadcasting signal obtained by multiplexing the 2K broadcasting of the upper layer and the 4K broadcasting of the lower layer. Further, in the example of FIG. 8B, only the tuning/wave detector **131L** of the third tuner/demodulator **130L** in the broadcast receiving apparatus **100** is extracted and illustrated, and the other operating units are omitted.

[0322] A broadcasting signal received by the antenna **200L** is inputted from a connector **100F4** to the tuning/wave detector **131L** via a converter (converter) **201L** and a coaxial cable **202L**. Here, in the configuration, when the broadcasting signal is transmitted from the antenna **200L** to the broadcast receiving apparatus **100**, as illustrated in FIG. 8C, the converter **201L** may subject a frequency conversion amplification process to the broadcasting signal. Namely, in a case where the antenna **200L** is installed at the roof of an apartment building or the like and transmission of the broadcasting signal is executed to the broadcast receiving apparatus **100** of each room by the coaxial cable **202L** with a long cable length, it is conceivable that the broadcasting signal will be attenuated, and there is a possibility that a defect that the 4K broadcast wave of the lower layer

cannot particularly be received correctly by the tuning/wave detector **131L** occurs.

[0323] Therefore, in order to prevent the defect described above, the converter **201L** executes the frequency conversion amplification process for the 4K broadcasting signal of the lower layer. In the frequency conversion amplification process, a frequency band of the 4K broadcasting signal of the lower layer is converted from a frequency band of 470 to 710 MHz (that is, a band corresponding to 13 ch to 52 ch of the UHF) to a frequency band of 770 to 1010 MHz that exceed a band corresponding to 62 ch of the UHF, for example. Moreover, a process of amplifying the 4K broadcasting signal of the lower layer to a signal level at which an influence of attenuation by the cable does not cause a problem is executed. By executing such a process, it also becomes possible to avoid an influence of attenuation of the broadcasting signal during coaxial cable transmission while avoiding interference between the 2K broadcasting signal and the 4K broadcasting signal. Note that in a case where the influence of attenuation does not cause a problem, such as in a case where the cable length of the coaxial cable **202L** is short, the converter **201L** and the frequency conversion amplification process may be unnecessary.

[0324] Further, it is preferable that the frequency band after conversion by the frequency conversion amplification process is between 710 and 1032 MHz, which exceeds the band corresponding to 52 ch of the UHF, or between 770 and 1032 MHz, which exceeds the band corresponding to 62 ch of the UHF, (in case of retransmission by the cable television station); it is preferable that a bandwidth of a range between a frequency band before conversion by the frequency conversion amplification process and a frequency band after conversion is set to be an integral multiple of the bandwidth (6 MHz) of one physical channel; and the frequency conversion amplification process may be executed only for a physical channel using signal transmission by the layered division multiplexing transmission method. Since all of them are similar to those explained for the frequency conversion according to the present embodiment, which have already been described, repeated explanation thereof is omitted.

[0325] Note that the broadcast receiving apparatus **100** according to the present embodiment can identify whether the received broadcasting signal is the broadcasting signal transmitted by the lower layer or the broadcasting signal transmitted by the upper layer by using the upper/lower layer identification bit of the TMCC information, which has been explained with reference to FIG. 5H. Further, the broadcast receiving apparatus **100** according to the present embodiment can identify whether the received broadcasting signal is a broadcasting signal subjected to the frequency conversion after reception by the antenna or not by using the frequency converting process identification bit of the TMCC information, which has been explained with reference to FIG. 5F. Further, the broadcast receiving apparatus **100** according to the present embodiment can identify whether a 4K program is transmitted by the lower layer in the received broadcasting signal or not by using the 4K signal transmission layer identification bit of the TMCC information, which has been explained with reference to FIG. 5I. It is not impossible to execute these identifying processes by demodulating a data carrier and referring to control information contained in a stream. However, it is necessary to demodulate the data carrier, and this causes the process to become complicated. It is more easily and faster to identify them by referring to the parameters of the TMCC information described above. For this reason, it is possible to further speed up the initial scan of the broadcast receiving apparatus **100**, for example.

[0326] Note that as explained above, the tuning/wave detector **131L** of the third tuner/demodulator **130L** in the broadcast receiving apparatus **100** according to the embodiment of the present invention has the receiving function compatible with the LDM (layered division multiplexing) technique. For this reason, the converter **201L** illustrated in FIG. 8C is not always required between the antenna **200L** and the broadcast receiving apparatus **100**.

[0327] Note that as described above, the broadcast wave of the terrestrial digital broadcasting transmitted by the layered division multiplexing transmission method explained above can be received and reproduced by the third tuner/demodulator **130L** of the broadcast receiving apparatus

100, but can also be received by the first tuner/demodulator **130C** of the broadcast receiving apparatus **100**. In a case where the broadcast wave of the terrestrial digital broadcasting is received by the first tuner/demodulator **130C**, the broadcasting signal transmitted by the layer for the transmitted advanced terrestrial digital broadcasting service of the broadcasting signals of the broadcast wave of the terrestrial digital broadcasting is ignored, but the broadcasting signal transmitted by the layer for the existing terrestrial digital broadcasting service is reproduced. [MPEG-2 TS Method]

[0328] The broadcasting system according to the present embodiment is compatible with an MPEG-2 TS, which is adopted in the existing terrestrial digital broadcasting service, as a media transport method of transmitting data such as video or audio. Specifically, the method of the stream transmitted by the OFDM transmission wave illustrated in (1) of FIG. 4D is the MPEG-2 TS, and the method of the stream transmitted by the layer, by which the existing terrestrial digital broadcasting service is transmitted, of the OFDM transmission waves illustrated in (2) of FIG. 4D and (3) of FIG. 4D is the MPEG-2 TS. Further, the method of the stream obtained by demodulating the transmission wave by the first tuner/demodulator **130C** of the broadcast receiving apparatus **100** illustrated in FIG. 2A is the MPEG-2 TS. Further, the method of the stream, which corresponds to the layer by which the existing terrestrial digital broadcasting service is transmitted, among the streams obtained by demodulating the transmission wave by the second tuner/demodulator **130T** is the MPEG-2 TS. Similarly, the method of the stream, which corresponds to the layer by which the existing terrestrial digital broadcasting service is transmitted, among the streams obtained by demodulating the transmission wave by the third tuner/demodulator **130L** is the MPEG-2 TS.

[0329] The MPEG-2 TS is characterized in that components constituting a program such as video or audio is multiplexed into one packet stream together with a control signal and a clock. Since it is treated as one packet stream including the clock, it is suitable to transmit one content via one transmission path in which transmission quality is secured, and the MPEG-2 TS is adopted in many existing digital broadcasting systems. Further, it is possible to realize bidirectional communication via a bidirectional network such as a fixed network/portable network, and it is possible to be compatible with the broadcasting/communication cooperation system in which a digital broadcasting service is caused to cooperate with a function using a broadband network and acquisition of additional content via the broadband network, arithmetic processing by a server apparatus, and a presenting process in cooperation with portable terminal equipment are combined with the digital broadcasting service.

[0330] FIG. 9A illustrates one example of a protocol stack of a transmission signal in the broadcasting system using the MPEG-2 TS. In the MPEG-2 TS, PSI, SI, and the other control signals are transmitted by a section form.

[Control Signal of Broadcasting System Using MPEG-2 TS Method]

[0331] As control information of a MPEG-2 TS method, there are a table mainly used for program arrangement information and a table used for other than the program arrangement information. Each of the tables is transmitted in a section form, and descriptors are arranged in the corresponding table.

<Tables Used in Program Arrangement Information>

[0332] FIG. 9B illustrates a list of tables used in the program arrangement information of the broadcasting system of the MPEG-2 TS method. In the present embodiment, the following is used as the tables used in the program arrangement information. [0333] (1) PAT (Program Association Table) [0334] (2) CAT (Conditional Access Table) [0335] (3) PMT (Program Map Table) [0336] (4) NIT (Network Information Table) [0337] (5) SDT (Service Description Table) [0338] (6) BAT (Bouquet Association Table) [0339] (7) EIT (Event Information Table) [0340] (8) RST (Running Status Table) [0341] (9) TDT (Time and Date Table) [0342] (10) TOT (Time Offset Table) [0343] (11) LIT (Local Event Information Table) [0344] (12) ERT (Event Relation Table) [0345] (13) ITT (Index Transmission Table) [0346] (14) PCAT (Partial Content Announcement Table) [0347] (15)

ST (Stuffing Table) [0348] (16) BIT (Broadcaster Information Table) [0349] (17) NBIT (Network Board Information Table) [0350] (18) LDT (Linked Description Table) [0351] (19) AMT (Address Map Table)

(20) INT (IP/MAC Notification Table)

[0352] (21) table set by provider

<Tables Used in Digital Broadcasting>

[0353] FIG. 9C illustrates a list of tables used for other than the program arrangement information of the broadcasting system with the MPEG-2 TS method. In the present embodiment, the following is used as the tables used for other than the program arrangement information.

[0354] (1) ECM (Entitlement Control Message) [0355] (2) EMM (Entitlement Management Message) [0356] (3) DCT (Download Control Table) [0357] (4) DLT (DownLoad Table) [0358] (5) DIT (Discontinuity Information Table) [0359] (6) SIT (Selection Information Table) [0360] (7) SDTT (Software Download Trigger Table) [0361] (8) CDT (Common Data Table) [0362] (9) DSM-CC section [0363] (10) AIT (Application Information Table) [0364] (11) DCM (Download Control Message) [0365] (12) DMM (Download Management Message) [0366] (13) table set by provider

<Descriptors Used in Program Arrangement Information>

[0367] FIG. 9D, FIG. 9E, and FIG. 9F illustrate a list of descriptors used in the program arrangement information of the broadcasting system of the MPEG-2 TS method. In the present embodiment, the following is used as the descriptors used in the program arrangement information.

[0368] (1) limited reception type descriptor (Conditional Access Descriptor) [0369] (2) copyright descriptor (Copyright Descriptor) [0370] (3) network name descriptor (Network Name Descriptor) [0371] (4) service list descriptor (Service List Descriptor) [0372] (5) stuffing descriptor (Stuffing Descriptor) [0373] (6) satellite delivery system descriptor (Satellite Delivery System Descriptor) [0374] (7) terrestrial delivery system descriptor (Terrestrial Delivery System Descriptor) [0375] (8) bouquet name descriptor (Bouquet Name Descriptor) [0376] (9) service descriptor (Service Descriptor) [0377] (10) country reception availability descriptor (Country Availability Descriptor) [0378] (11) linkage descriptor (Linkage Descriptor) [0379] (12) NVOD reference service descriptor (NVOD Reference Descriptor) [0380] (13) time shifted service descriptor (Time Shifted Service Descriptor) [0381] (14) short form event descriptor (Short Event Descriptor) [0382] (15) extended form event descriptor (Extended Event Descriptor) [0383] (16) time shifted event descriptor (Time Shifted Event Descriptor) [0384] (17) component descriptor (Component Descriptor) [0385] (18) mosaic descriptor (Mosaic Descriptor) [0386] (19) stream identifier descriptor (Stream Identifier Descriptor) [0387] (20) CA identifier descriptor (CA Identifier Descriptor) [0388] (21) content descriptor (Content Descriptor) [0389] (22) parental rate descriptor (Parental Rating Descriptor) [0390] (23) hierarchical transmission descriptor (Hierarchical Transmission Descriptor) [0391] (24) digital copy control descriptor (Digital Copy Control Descriptor) [0392] (25) emergency information descriptor (Emergency Information Descriptor) [0393] (26) data coded method descriptor (Data Component Descriptor) [0394] (27) system management descriptor (System Management Descriptor) [0395] (28) local time offset descriptor (Local Time Offset Descriptor) [0396] (29) audio component descriptor (Audio Component Descriptor) [0397] (30) target region descriptor (Target Region Descriptor) [0398] (31) hyperlink descriptor (Hyperlink Descriptor) [0399] (32) data content descriptor (Data Content Descriptor) [0400] (33) video decode control descriptor (Video Decode Control Descriptor) [0401] (34) basic local event descriptor (Basic Local Event Descriptor) [0402] (35) reference descriptor (Reference Descriptor) [0403] (36) node relation descriptor (Node Relation Descriptor) [0404] (37) short form node information descriptor (Short Node Information Descriptor) [0405] (38) STC reference descriptor (STC Reference Descriptor) [0406] (39) partial reception descriptor (Partial Reception Descriptor) [0407] (40) series descriptor (Series Descriptor) [0408] (41) event group descriptor (Event Group Descriptor) [0409] (42) SI transmission parameter descriptor (SI Parameter Descriptor) [0410] (43) broadcaster name descriptor (Broadcaster Name Descriptor) [0411] (44) component group descriptor

(Component Group Descriptor) [0412] (45) SI prime TS descriptor (SI Prime TS Descriptor) [0413] (46) bulletin board information descriptor (Board Information Descriptor) [0414] (47) LDT linkage descriptor (LDT Linkage Descriptor) [0415] (48) connected transmission descriptor (Connected Transmission Descriptor) [0416] (49) TS information descriptor (TS Information Descriptor) [0417] (50) extended broadcaster descriptor (Extended Broadcaster Descriptor) [0418] (51) logo transmission descriptor (Logo Transmission Descriptor) [0419] (52) content availability descriptor (Content Availability Descriptor) [0420] (53) carousel compatible composite descriptor (Carousel Compatible Composite Descriptor) [0421] (54) limited reproduction method descriptor (Conditional Playback Descriptor) [0422] (55) AVC video descriptor (AVC Video Descriptor) [0423] (56) AVC timing HRD descriptor (AVC Timing and HRD Descriptor) [0424] (57) service group descriptor (Service Group Descriptor) [0425] (58) MPEG-4 audio descriptor (MPEG-4 Audio Descriptor) [0426] (59) MPEG-4 audio extension descriptor (MPEG-4 Audio Extension Descriptor) [0427] (60) registration descriptor (Registration Descriptor) [0428] (61) data broadcast identification descriptor (Data Broadcast Id Descriptor) [0429] (62) access control descriptor (Access Control Descriptor) [0430] (63) area broadcasting information descriptor (Area Broadcasting Information Descriptor) [0431] (64) material information descriptor (Material Information Descriptor) [0432] (65) HEVC video descriptor (HEVC Video Descriptor) [0433] (66) hierarchy coded descriptor (Hierarchy Descriptor) [0434] (67) communication cooperation information descriptor (Hybrid Information Descriptor) [0435] (68) scramble method descriptor (Scrambler Descriptor) [0436] (69) descriptor set by provider

<Descriptors Used in Digital Broadcasting>

[0437] FIG. 9G illustrates a list of descriptors used for other than the program arrangement information of the broadcasting system of the MPEG-2 TS method. In the present embodiment, the following is used as the descriptors other than the program arrangement information. [0438] (1) partial transport stream descriptor (Partial Transport Stream Descriptor) [0439] (2) network identification descriptor (Network Identification Descriptor) [0440] (3) partial transport stream time descriptor (Partial Transport Stream Time Descriptor) [0441] (4) download content descriptor (Download Content Descriptor) [0442] (5) CA EMM TS descriptor (CA EMM TS Descriptor) [0443] (6) CA contract information descriptor (CA Contract Information Descriptor) [0444] (7) CA service descriptor (CA Service Descriptor) [0445] (8) carousel identifier descriptor (Carousel Identifier Descriptor) [0446] (9) association tag descriptor (Association Tag Descriptor) [0447] (10) extended association tag descriptor (Deferred Association tags Descriptor) [0448] (11) network download content descriptor (Network Download Content Descriptor) [0449] (12) download protection descriptor (Download Protection Descriptor) [0450] (13) CA startup descriptor (CA Startup Descriptor) [0451] (14) descriptor set by provider

<Descriptor Used in INT>

[0452] FIG. 9H illustrates a list of descriptors used in INT of the broadcasting system of the MPEG-2 TS method. In the present embodiment, the following is used as the descriptors used in the INT. Note that the descriptors used in the program arrangement information described above and the descriptors used for other than the program arrangement information are not used in the INT. [0453] (1) target smartcard descriptor (Target Smartcard Descriptor) [0454] (2) target IP address descriptor (Target IP Address Descriptor) [0455] (3) target IPv6 address descriptor (Target IPv6 Address Descriptor) [0456] (4) IP/MAC platform name descriptor (IP/MAC Platform Name Descriptor) [0457] (5) IP/MAC platform provider name descriptor (IP/MAC Platform Provider Name Descriptor) [0458] (6) IP/MAC stream arrangement descriptor (IP/MAC Stream Location Descriptor) [0459] (7) descriptor set by provider

<Descriptors Used in AIT>

[0460] FIG. 9I illustrates a list of descriptors used in the AIT of the broadcasting system of the MPEG-2 TS method. In the present embodiment, the following is used as descriptors used in the AIT. Note that the descriptors used in the program arrangement information described above and

the descriptors used for other than the program arrangement information are not used in the INT. [0461] (1) application descriptor (Application Descriptor) [0462] (2) transport protocol descriptor (Transport Protocol Descriptor) [0463] (3) simple application location descriptor (Simple Application Location Descriptor) [0464] (4) application boundary authority setting descriptor (Application Boundary and Permission Descriptor) [0465] (5) startup priority information descriptor (Autostart Priority Descriptor) [0466] (6) cache information descriptor (Cache Control Info Descriptor) [0467] (7) stochastic application delay descriptor (Randomized Latency Descriptor) [0468] (8) external application control descriptor (External Application Control Descriptor) [0469] (9) video recording and reproducing application descriptor (Playback Application Descriptor) [0470] (10) simple video recording and reproducing application location descriptor (Simple Playback Application Location Descriptor) [0471] (11) application expiration date descriptor (Application Expiration Descriptor) [0472] (12) descriptor set by provider [MMT Method]

[0473] The broadcasting system according to the present embodiment can also address an MMT method as the media transport method of transmitting data such as video or audio. Specifically, the method of the stream to be transmitted by the layer by which the advanced terrestrial digital broadcasting service is transmitted in the OFDM transmission wave illustrate in (2) of FIG. 4D and (3) of FIG. 4D is the MMT method in principle. Further, the method of the stream corresponding to the layer by which the advanced terrestrial digital broadcasting service is transmitted in the stream obtained by demodulating the transmission wave by the second tuner/demodulator **130T** of the broadcast receiving apparatus **100** illustrated in FIG. 2A is an MMT in principle. Similarly, the method of the stream corresponding to the layer by which the advanced terrestrial digital broadcasting service is transmitted in the stream obtained by demodulating the transmission wave by the third tuner/demodulator **130L** is the MMT in principle. Note that as a modification example, the stream of the MPEG-2 TS may be operated by the advanced terrestrial digital broadcasting service. Further, the method of the stream obtained by demodulating the transmission wave by the fourth tuner/demodulator **130B** is the MMT.

[0474] The MMT method is a newly formulated media transport method because functions of the MPEG-2 TS method are limited with respect to a change in environment regarding content distribution such as diversification of content in the recent years, diversification of equipment that utilizes content, diversification of a transmission path for distributing content, or diversification of content storage environment.

[0475] A code of a video signal and an audio signal of a broadcasting program is an MFU (Media Fragment Unit)/MPU (Media Processing Unit); is put on MMT P (MMT Protocol) payload to be converted into an MMTP packet; and is transmitted by an IP packet. Further, data content and a signal of a subtitle related to the broadcasting program are also in an MFU/MPU form; are put on MMTP payload to be converted into an MMTP packet; and are transmitted by an IP packet.

[0476] UDP/IP (User Datagram Protocol/Internet Protocol) is used for transmission of the MMTP packet in a broadcast transmission path, and the UDP/IP or TCP/IP (Transmission Control Protocol/Internet Protocol) is used in a communication line. Further, in the broadcast transmission path, a TLV multiplex method may be used for efficient transmission of the IP packet.

[0477] FIG. 10A illustrates a protocol stack of the MMT in the broadcast transmission path. Further, FIG. 10B illustrates a protocol stack of the MMT in the communication line. In the MMT method, a mechanism for transmitting two types of control information including MMT-SI and TLV-SI is prepared. The MMT-SI is control information indicating a configuration of a broadcasting program. The MMT-SI is formed as a control message of the MMT; is put on the MMTP payload to become an MMTP packet; and is transmitted by the IP packet. The TLV-SI is control information regarding multiplexing of the IP packet, and provide information for channel selection and correspondence information between an IP address and a service.

[Control Signal of Broadcasting System Using MMT Method]

[0478] As described above, in the MMT method, the TLV-SI and the MMT-SI are prepared as the control information. The TLV-SI is configured by a table and descriptors. The table is transmitted in a section form, and the descriptors are arranged in the table. The MMT-SI is configured by three layers including a message that stores the table and the descriptors, a table having an element and an attribute that indicate specific information, and a descriptor that indicates more detailed information.

<Tables Used in TLV-SI>

[0479] FIG. 10C illustrates a list of tables used in the TLV-SI of the broadcasting system with the MMT method. In the present embodiment, the following is used as the tables of the TLV-SI. [0480] (1) network information table for TLV (Network Information Table for TLV) [0481] (2) address map table (Address Map Table) [0482] (3) table set by provider

<Descriptors Used in TLV-SI>

[0483] FIG. 10D illustrates a list of descriptors used in the TLV-SI of the broadcasting system with the MMT method. In the present embodiment, the following is used as the descriptors of the TLV-SI. [0484] (1) service list descriptor (Service List Descriptor) [0485] (2) satellite delivery system descriptor (Satellite Delivery System Descriptor) [0486] (3) system management descriptor (System Management Descriptor) [0487] (4) network name descriptor (Network Name Descriptor) [0488] (5) remote control key descriptor (Remote Control Key Descriptor) [0489] (6) descriptor set by provider

<Message Used in MMT-SI>

[0490] FIG. 10E illustrates a list of messages used in the MMT-SI of the broadcasting system with the MMT method. In the present embodiment, the following is used as the messages of the MMT-SI. [0491] (1) PA (Package Access) message [0492] (2) M2 section message [0493] (3) CA message [0494] (4) M2 short section message [0495] (5) data transmission message [0496] (6) message set by provider

<Table Used in MMT-SI>

[0497] FIG. 10F illustrates a list of tables used in the MMT-SI of the broadcasting system with the MMT method. In the present embodiment, the following is used as the tables of the MMT-SI. [0498] (1) MPT (MMT Package Table) [0499] (2) PLT (Package List Table) [0500] (3) LCT (Layout Configuration Table) [0501] (4) ECM (Entitlement Control Message) [0502] (5) EMM (Entitlement Management Message) [0503] (6) CAT (MH) (Conditional Access Table (MH)) [0504] (7) DCM (Download Control Message) [0505] (8) DMM (Download Management Message) [0506] (9) MH-EIT (MH-Event Information Table) [0507] (10) MH-AIT (MH-Application Information Table) [0508] (11) MH-BIT (MH-Broadcaster Information Table) [0509] (12) MH-SDTT (MH-Software Download Trigger Table) [0510] (13) MH-SDT (MH-Service Description Table) [0511] (14) MH-TOT (MH-Time Offset Table) [0512] (15) MH-CDT (MH-Common Data Table) [0513] (16) DDM table (Data Directory Management Table) [0514] (17) DAM table (Data Asset Management Table) [0515] (18) DCC table (Data Content Configuration Table) [0516] (19) EMT (Event Message Table) [0517] (20) table set by provider

<Descriptor Used in MMT-SI>

[0518] FIG. 10G, FIG. 10H, and FIG. 10I illustrate a list of descriptors used in the MMT-SI of the broadcasting system with the MMT method. In the present embodiment, the following is used as the descriptors of the MMT-SI. [0519] (1) asset group descriptor (Asset Group Descriptor) [0520] (2) event package descriptor (Event Package Descriptor) [0521] (3) background color specification descriptor (Background Color Descriptor) [0522] (4) MPU presentation region specification descriptor (MPU Presentation Region Descriptor) [0523] (5) MPU timestamp descriptor (MPU Timestamp Descriptor) [0524] (6) dependency descriptor (Dependency Descriptor) [0525] (7) access control descriptor (Access Control Descriptor) [0526] (8) scramble method descriptor (Scrambler Descriptor) [0527] (9) message authentication method descriptor (Message Authentication Method Descriptor) [0528] (10) emergency information descriptor (Emergency

Information Descriptor) [0529] (11) MH-MPEG-4 audio descriptor (MH-MPEG-4 Audio Descriptor) [0530] (12) MH-MPEG-4 audio extension descriptor (MH-MPEG-4 Audio Extension Descriptor) [0531] (13) MH-HEVC descriptor (MH-HEVC Descriptor) [0532] (14) MH-linkage descriptor (MH-Linkage Descriptor) [0533] (15) MH-event group descriptor (MH-Event Group Descriptor) [0534] (16) MH-service list descriptor (MH-Service List Descriptor) [0535] (17) MH-short form event descriptor (MH-Short Event Descriptor) [0536] (18) MH-extended form event descriptor (MH-Extended Event Descriptor) [0537] (19) video component descriptor (Video Component Descriptor) [0538] (20) MH-stream identifier descriptor (MH-Stream Identifier Descriptor) [0539] (21) MH-content descriptor (MH-Content Descriptor) [0540] (22) MH-parental rate descriptor (MH-Parental Rating Descriptor) [0541] (23) MH-audio component descriptor (MH-Audio Component Descriptor) [0542] (24) MH-target region descriptor (MH-Target Region Descriptor) [0543] (25) MH-series descriptor (MH-Series Descriptor) [0544] (26) MH-SI transmission parameter descriptor (MH-SI Parameter Descriptor) [0545] (27) MH-broadcaster name descriptor (MH-Broadcaster Name Descriptor) [0546] (28) MH-service descriptor (MH-Service Descriptor) [0547] (29) IP data flow descriptor (IP Data Flow Descriptor) [0548] (30) MH-CA startup descriptor (MH-CA Startup Descriptor) [0549] (31) MH-Type descriptor (MH-Type Descriptor) [0550] (32) MH-Info descriptor (MH-Info Descriptor) [0551] (33) MH-Expire descriptor (MH-Expire Descriptor) [0552] (34) MH-Compression Type descriptor (MH-Compression Type Descriptor) [0553] (35) MH-data coding method descriptor (MH-Data Component Descriptor) [0554] (36) UTC-NPT reference descriptor (UTC-NPT Reference Descriptor) [0555] (37) event message descriptor (Event Message Descriptor) [0556] (38) MH-local time offset descriptor (MH-Local Time Offset Descriptor) [0557] (39) MH-component group descriptor (MH-Component Group Descriptor) [0558] (40) MH-logo transmission descriptor (MH-Logo Transmission Descriptor) [0559] (41) MPU extended timestamp descriptor (MPU Extended Timestamp Descriptor) [0560] (42) MPU download content descriptor (MPU Download Content Descriptor) [0561] (43) MH-network download content descriptor (MH-Network Download Content Descriptor) [0562] (44) MH-application descriptor (MH-Application Descriptor) [0563] (45) MH-transport protocol descriptor (MH-Transport Protocol Descriptor) [0564] (46) MH-simple application location descriptor (MH-Simple Application Location Descriptor) [0565] (47) MH-application boundary authority setting descriptor (MH-Application Boundary and Permission Descriptor) [0566] (48) MH-startup priority information descriptor (MH-Autostart Priority Descriptor) [0567] (49) MH-cache information descriptor (MH-Cache Control Info Descriptor) [0568] (50) MH-stochastic application delay descriptor (MH-Randomized Latency Descriptor) [0569] (51) linked PU descriptor (Linked PU Descriptor) [0570] (52) locked cache specification descriptor (Locked Cache Descriptor) [0571] (53) unlocked cache specification descriptor (Unlocked Cache Descriptor) [0572] (54) MH-download protection descriptor (MH-DL Protection Descriptor) [0573] (55) application service descriptor (Application Service Descriptor) [0574] (56) MPU node descriptor (MPU Node Descriptor) [0575] (57) PU structure descriptor (PU Structure Descriptor) [0576] (58) MH-hierarchy coding descriptor (MH-Hierarchy Descriptor) [0577] (59) content copy control descriptor (Content Copy Control Descriptor) [0578] (60) content usage control descriptor (Content Usage Control Descriptor) [0579] (61) emergency news descriptor (Emergency News Descriptor) [0580] (62) MH-CA contract information descriptor (MH-CA Contract Info Descriptor) [0581] (63) MH-CA service descriptor (MH-CA Service Descriptor) [0582] (64) MH-external application control descriptor (MH-External Application Control Descriptor) [0583] (65) MH-video recording and reproducing application descriptor (MH-Playback Application Descriptor) [0584] (66) MH-simple video recording and reproducing application location descriptor (MH-Simple Playback Application Location Descriptor) [0585] (67) MH-application expiration date descriptor (MH-Application Expiration Descriptor) [0586] (68) related broadcaster descriptor (Related Broadcaster Descriptor) [0587] (69) multimedia service information descriptor (Multimedia Service Descriptor) [0588] (70) descriptor set by provider

<Relation between Data Transmission and Each Control Information in MMT Method>

[0589] FIG. **10J** illustrates a relation between data transmission and representative tables in the broadcasting system of the MMT method.

[0590] In the broadcasting system of the MMT method, data transmission can be executed via a plurality of routes such as the TLV stream via the broadcast transmission path or an IP data flow via the communication line. The TLV-SI such as a TLV-NIT or an AMT and the IP data flow that is a data flow of the IP packet are contained in the TLV stream. A video asset including a series of video MPUs and an audio asset including a series of audio MPUs are contained in the IP data flow. Moreover, any of a subtitle asset including a series of subtitle MPUs, a caption asset including a series of caption MPUs, and a data asset including a series of data MPUs may be contained therein. Various kinds of assets are associated with each other for package unit by an MPT (an MMT package table) stored in a PA message and to be transmitted. Specifically, a package ID and an asset ID of each asset included in a package may be described in the MPT so as to be associated with each other.

[0591] The assets constituting the package can be only the assets in the TLV stream. However, as illustrated in FIG. **10J**, the assets to be transmitted by the IP data flow of the communication line can also be included. This can be realized by containing location information of each asset included in the package in the MPT so that the broadcast receiving apparatus **100** can grasp a reference destination of each asset. As the location information of each asset, it is possible to specify various kinds of data to be transmitted by various transmission routes as follows: [0592] (1) data multiplexed into the same IP data flow as the MPT [0593] (2) data multiplexed into an IPv4 data flow [0594] (3) data multiplexed into an IPv6 data flow [0595] (4) data multiplexed into MPEG2-TS of broadcast [0596] (5) data multiplexed into the IP data flow in an MPEG2-TS form [0597] (6) data at a specified URL.

[0598] The broadcasting system with the MMT method further has a concept of an event. The event is a concept indicating a so-called program, which is treated by an MH-EIT that is transmitted while being included in an M2 section message. Specifically, in a package indicated by an event package descriptor stored in the MH-EIT, a series of data included in a period for a duration time from a disclosure time stored in the MH-EIT is data contained in the concept of the event. The MH-EIT can be used for various kinds of processes (for example, a generating process for a program guide, a control of video recording reservation or timer recording, a copyright management process such as temporary storage, and the like) for event unit in the broadcast receiving apparatus **100**.

[Channel Setting Process of Broadcast Receiving Apparatus]

<Initial Scan>

[0599] In the existing terrestrial digital broadcasting, the network ID is different for sending master unit, and it is general that information of the other stations is not described in an NIT. Therefore, with respect to the terrestrial digital broadcasting according to the embodiment of the present invention (advanced terrestrial digital broadcast or terrestrial digital broadcast in which the advanced terrestrial digital broadcasting and the existing terrestrial digital broadcasting are transmitted by the other layers at the same time), it is necessary for the broadcast receiving apparatus **100** according to the embodiment of the present invention, which has compatibility with the existing terrestrial digital broadcasting to have a function to search (or scan) all receivable channels at a reception spot and create a service list (that is, a receivable frequency table) based on a service ID. Note that in regions where the same network ID can be received on different physical channels by an MFN (Multi Frequency Network: multi frequency network), the broadcast receiving apparatus **100** may basically operate so as to select a channel with a good reception C/N or a good BER (Bit Error Rate) and store it in the service list.

[0600] Note that in the advanced BS digital broadcasting or the advanced CS digital broadcasting received by the fourth tuner/demodulator **130B** of the broadcast receiving apparatus **100** according

to the embodiment of the present invention, the broadcast receiving apparatus **100** may obtain and store the service list to be stored in the TLV-NIT, and it is thus unnecessary to create the service list. Therefore, the initial scan and rescan (will be described later) are not required for the advanced BS digital broadcasting or the advanced CS digital broadcasting received by the fourth tuner/demodulator **130B**.

<Rescan>

[0601] The broadcast receiving apparatus **100** according to the embodiment of the present invention has a rescan function prepared for a case of open of a new station, installation of a new relay station, or change in a reception spot of a television receiver. In a case where information that has already been set is to be changed, the broadcast receiving apparatus **100** can notify the user of that effect.

<Operation Example at Time of Initial/Rescan>

[0602] FIG. **11A** illustrates one example of an operation sequence of a channel setting process (initial/rescan) by the broadcast receiving apparatus **100** according to the embodiment of the present invention. Note that in FIG. **11A**, an example of a case where the MPEG-2 TS is adopted as the media transport method is illustrated, but it also becomes the similar process basically even in a case where the MMT method is adopted.

[0603] In the channel setting process, the receiving function controller **1102** first executes setting of a resident region (selection of a region where the broadcast receiving apparatus **100** is installed) on the basis of an instruction of the user (**S101**). At this time, in place of the instruction of the user, the setting of the resident region may be executed automatically on the basis of installation location information of the broadcast receiving apparatus **100**, which is obtained by a predetermined process. As an example of an obtaining process of the installation location information, such information may be obtained from a network to which the LAN communication unit **121** is connected, or information regarding an installation location may be obtained from external equipment to which the digital interface **125** is connected. Next, an initial value of a frequency range to be scanned is set, and a tuner/demodulator (described in this manner in a case where the first tuner/demodulator **130C**, the second tuner/demodulator **130T**, and the third tuner/demodulator **130L** are not distinguished. the same applies hereinafter.) is instructed so as to execute tuning to the set frequency (**S102**).

[0604] The tuner/demodulator executes the tuning on the basis of the instruction (**S103**). In a case where the lock to the set frequency is successful (**S103**: Yes), the processing flow proceeds to a process at **S104**. In a case where the lock is not successful (**S103**: No), the processing flow proceeds to a process at **S111**. In the process at **S104**, confirmation of a C/N is executed (**S104**). In a case where a predetermined number of C/N or more is obtained (**S104**: Yes), the processing flow proceeds to a process at **S105**, and a reception confirming process is executed. In a case where the predetermined number of C/N or more is not obtained (**S104**: No), the processing flow proceeds to a process at **S111**.

[0605] In the reception confirming process, the receiving function controller **1102** first obtains a BER of the received broadcast wave (**S105**). Next, an NIT is obtained and collated, whereby it is confirmed whether the NIT is effective data or not (**S106**). In a case where the NIT obtained in the process at **S106** is effective data, the receiving function controller **1102** obtains, from the NIT, information such as a transport stream ID or an original network ID. Further, the receiving function controller **1102** obtains, from a terrestrial delivery system descriptor, delivery system information regarding a physical condition of a broadcast transmission path corresponding to each transport stream ID/original network ID. Further, the receiving function controller **1102** obtains a list of service IDs from a service list descriptor.

[0606] Next, by confirming the service list stored in the receiving apparatus, the receiving function controller **1102** confirms whether the transport stream ID obtained in the process at **S106** has already been obtained or not (**S107**). In a case where the transport stream ID obtained in the

process at **S106** has not already been obtained (**S107: No**), the receiving function controller **1102** adds the various kinds of information obtained in the process at **S106** to the service list so as to be associated with the transport stream ID (**S108**). In a case where the transport stream ID obtained in the process at **S106** has already been obtained (**S107: Yes**), the receiving function controller **1102** compares the BER obtained in the process at **S105** with the BER when the transport stream ID, which has been described in the service list, is obtained (**S109**). As a result, in a case where the BER obtained in the process at **S105** is better (**S109: Yes**), the receiving function controller **1102** updates the service list on the basis of the various kinds of information obtained in the process at **S106** (**S110**). In a case where the BER obtained in the process at **S105** is not better (**S109: No**), the receiving function controller **1102** discards the various kinds of information obtained in the process at **S106**.

[0607] Further, at the time of the service list creating (adding/updating) process described above, a remote controller key ID may be obtained from a TS information descriptor, and a representative service for each transport stream may be associated with the remote controller key. By this process, it becomes possible to execute one-touch channel selection (will be described later).

[0608] When the reception confirming process is terminated, the receiving function controller **1102** confirms whether a current frequency setting is a final value in the frequency range to be scanned or not (**S111**). In a case where the current frequency setting is not the final value in the frequency range to be scanned (**S111: No**), the receiving function controller **1102** increases the frequency value set to the tuner/demodulator (**S112**), and repeats the processes at **S103** to **S110**. In a case where the current frequency setting is the final value in the frequency range to be scanned (**S111: Yes**), the processing flow proceeds to a process at **S113**.

[0609] In the process at **S113**, the service list created (or added or updated) in the processes described above is presented to the user as a result of the channel setting process (**S113**). Further, in a case where there is overlap of the remote controller keys, the user may be informed of that effect to encourage the user to change the setting for the remote controller key (**S114**). The service list created or updated in the processes described above is stored in the ROM **103** of the broadcast receiving apparatus **100** or a non-volatile memory such as the storage (accumulator) **110**.

[0610] FIG. **11B** illustrates one example of a data structure of the NIT. In FIG. **11B**, “transport_stream_id” corresponds to the transport stream ID described above, and “original_network_id” corresponds to the original network ID. Further, FIG. **11C** illustrates one example of a data structure of the terrestrial delivery system descriptor. In FIG. **11C**, “guard_interval”, “transmission mode”, “frequency”, and the like correspond to the delivery system information described above. FIG. **11D** illustrates one example of a data structure of the service list descriptor. In FIG. **11D**, “service_id” corresponds to the service ID described above. FIG. **11E** illustrates one example of a data structure of the TS information descriptor. In FIG. **11E**, “remote_control_key_id” corresponds to the remote controller key ID described above.

[0611] Note that the broadcast receiving apparatus **100** may control the receiving function controller **1102** so as to appropriately change the frequency range to be scanned described above in accordance with a broadcasting service to be received. For example, in a case where the broadcast receiving apparatus **100** receives the broadcast wave of the existing terrestrial digital broadcasting service, the broadcast receiving apparatus **100** controls the receiving function controller **1102** so as to scan a frequency range of 470 to 770 MHz (corresponding to 13 ch to 62 ch of the physical channel). Namely, the broadcast receiving apparatus **100** sets an initial value of the frequency range to 470 to 476 MHz (central frequency: 473 MHz); sets a final value of the frequency range to 764 to 770 MHz (central frequency: 767 MHz); and controls the receiving function controller **1102** so as to increase the frequency value by +6 MHz in the process at **S112**.

[0612] Further, in a case where the broadcast receiving apparatus **100** receives the broadcast wave including the advanced terrestrial digital broadcasting service, the broadcast receiving apparatus **100** controls the receiving function controller **1102** so as to scan a frequency range of 470 to 1010

MHz (because there is a possibility that the frequency converting process illustrated in FIG. 7D or the frequency conversion amplification process illustrated in FIG. 8C is executed). Namely, the broadcast receiving apparatus **100** sets an initial value of the frequency range to 470 to 476 MHz (central frequency: 473 MHz); sets a final value of the frequency range to 1004 to 1010 MHz (central frequency: 1007 MHz); and controls serviced the receiving function controller **1102** so as to increase the frequency value by +6 MHz in the process at **S112**. Note that even in a case where the broadcast receiving apparatus **100** receives the advanced terrestrial digital broadcasting service but it is determined that the frequency converting process or the frequency conversion amplification process described above is not executed, the broadcast receiving apparatus **100** may control the receiving function controller **1102** so as to scan only the frequency range of 470 to 770 MHz. The broadcast receiving apparatus **100** can execute a selection control of the frequency range to be scanned on the basis of the system identification of the TMCC information, the frequency converting process identification, and the like.

[0613] Further, in a case where the broadcasting system according to the embodiment of the present invention has the configuration illustrated in FIG. 7C, for example, and the broadcast receiving apparatus **100** receives the advanced terrestrial digital broadcasting service of the dual-polarized transmission method, one of the tuning/wave detector **131H** and the tuning/wave detector **131V** may scan the frequency range of 470 to 770 MHz, and the other may scan the frequency range of 770 to 1010 MHz (in a case where the frequency converting process is applied to the transmission wave by the polarized wave detected by the other tuning/wave detector). By controlling it in this manner on the basis of the system identification of the TMCC information and the frequency converting process identification, it is possible to omit scanning in an unnecessary frequency range, and this makes it possible to reduce the time required for channel setting. Moreover, in this case, both the tuning/wave detector **131H** and the tuning/wave detector **131V** may cause the operation sequence illustrated in FIG. 11A to proceed in parallel to synchronize the loop of frequency increase at **S112** in the operation sequence illustrated in FIG. 11A. At this time, in the loop of the same timing in the loop of the frequency increase in the operation sequence illustrated in FIG. 11A, in a case where the broadcast receiving apparatus **100** is configured so as to receive a pair of the horizontal polarized signal and the vertical polarized signal transmitted by the same physical channel in parallel, it becomes possible to decode the control information inside the packet stream of the advanced terrestrial digital service to be transmitted by the pair of the horizontal polarized signal and the vertical polarized signal to obtain the decoded control information during the loop process. This is preferable because the scanning and the creation of the service list proceed efficiently.

[0614] Similarly, in a case where the broadcast receiving apparatus **100** further has a configuration of so-called double tuners in which a plurality of tuners/demodulators (tuning/wave detectors) is provided (for example, a configuration in which a plurality of third tuner/demodulators **130L** is provided) in addition to the configuration illustrated in FIG. 8B and the broadcast receiving apparatus **100** receives the advanced terrestrial digital broadcasting service of the layered division multiplexing transmission method, one of the double tuners may scan the frequency range of 470 to 770 MHz, and the other may scan the frequency range of 770 to 1010 MHz (in a case where the frequency conversion amplification process is applied thereto). By controlling it in this manner, it becomes possible to reduce the time required for the channel setting in the similar manner to the above.

[0615] Note that as explained with reference to FIG. 8A, FIG. 8B, and FIG. 8C, the terrestrial digital broadcasting service transmitted by any one of the upper layer and the lower layer in the configuration illustrated in FIG. 8B is the existing terrestrial digital broadcasting service. Therefore, for example, the first tuner/demodulator **130C** may scan the frequency range in which the existing terrestrial digital broadcasting service is transmitted of the frequency range of 470 to 770 MHz and the frequency range of 770 to 1010 MHz, and the third tuner/demodulator **130L** may

scan the other frequency range in parallel. In this case, it is also possible to reduce the time required for the channel setting in the similar manner to the parallel scanning by the double tuners of the third tuner/demodulator **130L** described above. It is possible to identify whether the existing terrestrial digital broadcasting service is transmitted in any of the frequency range of 470 to 770 MHz and the frequency range of 770 to 1010 MHz and whether the advanced terrestrial digital broadcasting service is transmitted in any of the frequency range of 470 to 770 MHz and the frequency range of 770 to 1010 MHz by receiving total two points of one point from each frequency range, for example, two points of 470 to 476 MHz (central frequency: 473 MHz) and 770 to 776 MHz (central frequency: 773 MHz) before the operation sequence for the initial scan/rescan is started by means of the third tuner/demodulator **130L**, obtaining the TMCC information transmitted at each frequency, and referring to the parameters stored in the TMCC information (for example, the parameter of the system identification).

[0616] Note that in case of a channel that has a broadcasting program to be transmitted by using both the horizontal polarized signal and the vertical polarized signal like the 4K broadcasting program of the C layer illustrated in the hierarchical division example of (1) of FIG. 7A, for example, in the advanced terrestrial digital broadcasting service of the dual-polarized transmission method, the same transport ID is detected by scanning both the frequency range of 470 to 770 MHz and the frequency range of 770 to 1010 MHz. However, this is described in the service list as one channel. Further, in a case where the same broadcasting program is transmitted by the B layer of the horizontal polarized signal and the B layer of the vertical polarized signal in the 2K broadcasting program of the B layer illustrated in (1) of FIG. 7A, it may be stored in the service list as one channel even though the same transport ID is detected. Namely, in a case where the same broadcasting program is transmitted in the same layer transmitted by a different polarized wave, they are merged into one channel and recognized, and are not recognized as different channels. By configuring it in this manner, it is possible to avoid the user from being confused due to existence of exactly the same broadcasting program in the different channels in the channel selecting process using the service list.

[0617] On the other hand, in a case where different broadcasting programs are transmitted between the B layer of the horizontal polarized signal and the B layer of the vertical polarized signal in the advanced terrestrial digital broadcasting service of the dual-polarized transmission method (in a case where the B layer of the vertical polarized signal is treated as the virtual D layer), they are stored in the service list as different channels. It is possible to identify whether the same broadcasting program is transmitted by the B layer of the horizontal polarized signal and the B layer of the vertical polarized signal or not by referring to the additional layer transmission identification parameter of the TMCC information and determining such a manner by means of the broadcast receiving apparatus **100**.

[Channel Selecting Process of Broadcast Receiving Apparatus]

[0618] The broadcast receiving apparatus **100** according to the embodiment of the present invention has, as a program selecting function, functions such as one-touch channel selection by a one-touch key of a remote controller, channel up/down selection by a channel up/down key of the remote controller, or direct channel selection by a direct input of a triple-digit number using ten keys of the remote controller. Any channel selecting function may be executed by using information stored in the service list generated by the initial scan/rescan described above. Further, after channel selection, information on the selected channel (a triple-digit number used for direct channel selection, a branch number, a TS name, a service name, a logo, video resolution information (distinguishing among UHD, HD, and SD), presence or absence of video resolution up/down conversion, the number of audio channels, presence or absence of audio downmix, and the like) is displayed by a banner display or the like. By configuring it in this manner, the user visually obtains information of the channel after channel selection, and confirm whether the channel can be selected to a desired channel or not. Hereinafter, one example of a process in each

channel selecting method is described.

<Process Example of One-Touch Channel Selection>

[0619] (1) A service of “service id” specified by “remote_control_key_id” is selected by pressing the one-touch key of the remote controller. [0620] (2) A last mode is set, and channel information display after channel selection is executed.

<Process Example of Up/Down Channel Selection by Channel Up/Down Button>

[0621] (1) Channel selection is executed in the order of the triple-digit number using the direct channel selection by pressing the channel up/down key of the remote controller. [0622] (1-1) In a case where an up key is pressed, a service adjacent to an upper side of the triple-digit number is selected. However, in a case where a value of a current triple-digit number is the maximum value in the service list, a service with a number of the minimum value is selected. [0623] (1-2) In a case where a down key is pressed, a service adjacent to a lower side of the triple-digit number is selected. However, in a case where a value of a current triple-digit number is the minimum value of the service list, a service with a number of the maximum value is selected. [0624] (2) A last mode is set, and channel information display after channel selection is executed.

<Process Example of Direct Channel Selection>

[0625] (1) When direct channel selection is selected, it becomes a state of waiting for an input of triple-digit number. [0626] (2-1) In a case where the input of the triple-digit number is not completed within a predetermined time (about five seconds), the mode is returned to a normal mode, and channel information display of the currently selected service is executed. [0627] (2-2) In a case where an input of a triple-digit number is completed, it is determined whether the channel exists in the service list of a receivable frequency table or not. In a case where there is no channel, a message such as “this channel does not exist” is displayed. [0628] (3) In a case where the channel exists, a channel selecting process is executed, a last mode is set, and channel information display after channel selection is executed.

[0629] Note that the channel selection operation is executed on the basis of SI, and the broadcast receiving apparatus **100** may also have a function to inform the user of that effect by displaying the same in a case where it is determined that the broadcast is suspended.

<Remote Controller of Broadcast Receiving Apparatus>

[0630] FIG. **12A** illustrates one example of an appearance view of the remote controller (remote controller) used for inputting an operation instruction to the broadcast receiving apparatus **100** according to the embodiment of the present invention.

[0631] A remote controller **180R** includes a power source key **180R1** for turning on/off (standby on/off) of a power source of the broadcast receiving apparatus **100**, a cursor key (up, down, left, and right) **180R2** for moving a cursor up, down, right, and left, a determination key **180R3** for determining an item of a cursor position as a selection item, and a return key **180R4**.

[0632] Further, the remote controller **180R** also includes a network switching key (advanced terrestrial digital, terrestrial digital, advanced BS, BS, and CS) **180R5** for switching a broadcasting network received by the broadcast receiving apparatus **100**. Further, the remote controller **180R** also includes one-touch keys (1 to 12) **180R6** used for one-touch channel selection, a channel up/down key **180R7** used for channel selection by channel up/down, and ten keys used for inputting a triple-digit number at the time of direct channel selection. Note that in the example illustrated in FIG. **12A**, the ten keys are also used as the one-touch key **180R6**, and a triple-digit number can be inputted by operating the one-touch key **180R6** after pressing a direct key **180R8** at the time of direct channel selection.

[0633] Further, the remote controller **180R** also includes an EPG key **180R9** for displaying a program guide and a menu key **180RA** for displaying a system menu. The program guide and the system menu can be operated in detail by any of the cursor key **180R2**, the determination key **180R3**, and the return key **180R4**.

[0634] Further, the remote controller **180R** also includes a d key **180RB** used for a data

broadcasting service or a multimedia service, a cooperation key **180RC** for displaying a broadcasting/communication cooperation service and a list of corresponding applications, and a color key (blue, red, green, and yellow) **180RD**. In each of the data broadcasting service, the multimedia service, and the broadcasting/communication cooperation service, detailed operations can be carried out by using any of the cursor key **180R2**, the determination key **180R3**, the return key **180R4**, and the color key **180RD**.

[0635] Further, the remote controller **180R** also includes a video key **180RE** for selecting related video, an audio key **180RF** for switching of audio ES or switching of two languages, and a subtitle key **180RG** for switching of on/off of a subtitle or switching of a subtitle language. Further, the remote controller **180R** also includes a volume key **180RH** for increasing or decreasing volume of audio output, and a mute key **180RI** for switching on/off of the audio output.

<Process Example of Network Switching by Advanced Terrestrial Digital Key>

[0636] The remote controller **180R** of the broadcast receiving apparatus **100** according to the embodiment of the present invention includes an “advanced terrestrial digital key”, a “terrestrial digital key”, an “advanced BS key”, a “BS key”, and a “CS key” as the network switching key **180R5**. Here, with respect to the “advanced terrestrial digital key” and the “terrestrial digital key”, the broadcast receiving apparatus **100** may be configured so that in the advanced terrestrial digital broadcasting service, for example, in a case where the simultaneous broadcasting of a 4K broadcasting program and a 2K broadcasting program by different layers is conducted, selection of the 4K broadcasting program is prioritized at the time of channel selection in a state where the “advanced terrestrial digital key” is pressed; and selection of the 2K broadcasting program is prioritized at the time of channel selection in a state where the “terrestrial digital key” is pressed. By controlling the broadcast receiving apparatus **100** in this manner, in a case where there are many errors in the transmission wave of the 4K broadcasting program in a situation where the 4K broadcasting program can be received, for example, it becomes possible to control the broadcast receiving apparatus **100** so that the 2K broadcasting program can be selected forcibly by pressing the “terrestrial digital key”.

<Screen Display Example at Time of Channel Selection>

[0637] As described above, the broadcast receiving apparatus **100** according to the embodiment of the present invention has a function to display information on selected channel by a banner display or the like when channel selection by the one-touch channel selection, the channel up/down selection, or the direct channel selection is executed.

[0638] FIG. **12B** illustrates one example of the banner display at the time of channel selection. A banner display **192A1** is an example of a banner display that is displayed when a 2K broadcasting program is selected. For example, a program name, a start time and an end time of a program, a network type, a number of a direct channel selection key of a remote controller, a service logo, and a triple-digit number may be displayed. Further, a banner display **192A2** is an example of a banner display that is displayed when a 4K broadcasting program is selected. For example, in addition to similar information as those of the banner display **192A1** described above, a mark symbolizing “advanced”, which indicates that a program being received is a 4K broadcasting program, is further displayed. Further, in a case where a resolution converting process or a downmixing process is executed, a display indicating that effect may be executed. In the example of the banner display **192A2**, as one example, it is displayed that the down-converting process from UHD resolution to HD resolution and the downmixing process from 22.2 ch to 5.1 ch have been executed.

[0639] In the broadcast receiving apparatus **100**, by executing these displays, in a case where the same content is broadcast as broadcasting programs with different quality such as a 2K broadcasting program or a 4K broadcasting program at the same time by the simultaneous broadcasting or the like, the user can suitably grasp which broadcasting program is displayed.

[0640] According to the system of the advanced digital broadcasting service, which includes a part or all of the functions of each function according to the embodiment of the present invention

described above, it becomes possible to provide a transmission technique and a reception technique for an advanced digital broadcasting service with higher functionality in consideration of compatibility with an existing digital broadcasting service. Namely, it is possible to provide a technique for more suitably transmitting or receiving the advanced digital broadcasting service.

Second Embodiment

[System Configuration of Clock Synchronization and Presentation Synchronization of Broadcast Receiving Apparatus]

[0641] In the present embodiment, one example of a presentation switching process for simultaneous broadcasting, clock synchronization, and presentation synchronization in a system of the advanced digital broadcasting service, which has been described in the first embodiment, will be described. Note that in order to simplify explanation thereof, in the present embodiment below, only different points from those according to the first embodiment will be described. Since points, which are not mentioned in the present embodiment, are similar to those according to the first embodiment, repeated explanation will be omitted.

[0642] In an MPEG2-TS method, a synchronous control is executed by a timestamp based on time information called PCR (Program Clock Reference). FIG. 9A illustrates a protocol stack of the MPEG2-TS method. A time system based on this PCR consists of a counter for upper 33 bits, which operates at 90 kHz, and a counter for lower 9 bits, which operates at 27 MHz. 300 counts of the lower 9 bits corresponds to one count of the upper 33 bits. This counter for the upper 33 bits makes a round for about 26.5 hours, and the counter returns to zero. On the other hand, in an MMT method, a synchronous control is executed by a timestamp based on time information called an NTP (Network Time Protocol). FIG. 10A and FIG. 10B illustrates a protocol stack of the MMT method. The clock synchronized with this NTP consists of upper 32 bits (in seconds) and lower 32 bits (a portion after the decimal point). Therefore, the clock based on the NTP has about 136 years as a period that can be expressed, which is completely different from the PCR system as a system for expressing time.

[0643] FIG. 13 is one example of a system clock unit **40000** used by a broadcast receiving apparatus **100** according to the present embodiment for clock synchronization and presentation synchronization. The system clock unit **40000** includes an NTP clock counter **40001** based on the NTP and a PCR clock counter **40002** based on the PCR.

[0644] Time information for the PCR is transmitted by an extension portion (adaptation field) of a TS packet header (FIG. 14). The time information is configured by upper 33 bits, lower 9 bits, and reserve 6 bits.

[0645] FIG. 15A illustrates one example of a configuration of the time information to be transmitted by the NTP. Further, FIG. 15B illustrates one example of a data structure of an MPU timestamp descriptor. The NTP is transmitted by this data format. However, as illustrated in FIG. 10B, it is also possible to match with the NTP included in a signal received by a communication line path (for example, a communication line that can be received by the LAN communication unit **121** illustrated in FIG. 2A).

[0646] In a broadcasting system according to the present embodiment, which handles MMT content, UTC is transmitted in the form of an NTP timestamp with a 64-bit length from a broadcast transmitting system to a receiver (the broadcast receiving apparatus **100** according to the present embodiment or the like). In an NTP timestamp format, “more than a second” of the UTC is represented by 32 bits, and “less than a second” thereof is represented by 32 bits. However, in fact, it is difficult to reproduce one second with 32-bit accuracy. For this reason, as an NTP system clock for synchronizing the MMT content or an NTP system clock for operating a clock of an NTP format, the NTP clock counter **40001** as illustrated in FIG. 13, which uses a frequency of “2 to the power 24” Hz (about 16.8 MHz) may be used, for example. Note that considering that a PCR system clock in a broadcasting system that handles MPEG2-TS content is 27 MHz and a hardware configuration of a receiver can be constructed easily, it is desirable to adopt a frequency of the

power of 2 such as about “2 to the power 24” to “2 to the power 28” as the NTP system clock. [0647] Note that in the broadcast transmitting system side that handles the MMT content or the receiver side, in a case where the NTP system clock is set to the frequency of the power of 2 such as about “2 to the power 24” to “2 to the power 28” as described above, lower 8 to 4 bits in the NTP timestamp format to be transmitted from the broadcast transmitting system side to the receiver side, which are not referred to by a PLL (Phase Locked Loop) system for reproducing the NTP system clock or the clock of the NTP format, may be fixed to “0” or “1”. Namely, if the NTP system clock is “2 to the power n” Hz (in the example illustrated in FIG. 13, $n=24$), lower “32-n” bits of the NTP timestamp format may be fixed to “0” or “1”. Alternatively, the receiver may execute processing so as to ignore the lower “32-n” bits of the NTP timestamp format.

[0648] In the broadcast transmitting system side, when the time information of the NTP format is obtained from the outside, a PLL system is configured with a (32+n)-bit counter by a VCO (Voltage Controlled Oscillator) of the “2 to the power n” Hz, and realizes a transmission system clock that synchronizes with the time information given from the outside. Further, the entire signal processing system is operated in synchronization with the NTP system clock of the “2 to the power n” Hz. Moreover, an output of the transmission system clock is periodically transmitted to the receiver side via a broadcast transmission path as the time information of an NTP length format.

[0649] The receiver side receives the time information of the NTP length format via the broadcast transmission path, and reproduces a reception system clock by the PLL system based on the VCO of the “2 to the power n” Hz in the similar manner to that of the broadcast transmitting system side. As a result, the reception system clock is a clock that synchronizes with the broadcast transmitting system side. Further, by operating the signal processing system of the receiver in synchronization with the NTP system clock with the “2 to the power n” Hz, clock synchronization between the broadcast transmitting system side and the receiver side is realized, whereby stable signal reproduction is possible. Further, in the broadcast transmitting system side, a decoding time and a presentation time for each presentation of a video signal and an audio signal are set on the basis of the time information of the NTP format. Here, the MPU timestamp descriptor illustrated in FIG. 15B described above is stored in an MPT stored in a PA message to be transmitted by a broadcasting signal. An “mpu_sequence number (an MPU sequence number)” parameter in the MPU timestamp descriptor illustrated in FIG. 15B indicates a sequence number of an MPU describing a timestamp, and an “mpu_presentation_time (MPU presentation time)” parameter indicates a presentation time for the MPU in a 64-bit NTP timestamp format. Therefore, the receiver can refer to the MPU timestamp descriptor stored in the MPT, and control timing of the presentation (display, output, and the like) of each MPU such as a video signal, an audio signal, a subtitle, and a caption.

[0650] Note that in a case where a control of decoding timing and presentation timing for each presentation of the video signal and the audio signal described above is focused on, synchronization between the video signal and the audio signal can be ensured by with a clock of about “2 to the power 16” Hz (about 65.5 KHz). In this case, it is not necessary to refer to lower 16 bits of the NTP timestamp format described in the MPU timestamp descriptor. Namely, in a case where a clock of “2 to the power m” Hz generated by dividing the NTP system clock by the control of the decoding timing and the presentation timing, it is not necessary to refer to lower “32-m” bits of the NTP timestamp format described in the MPU timestamp descriptor or the like. Therefore, the lower “32-m” bits of the NTP timestamp format described in the MPU timestamp descriptor or the like may be fixed to “0” or “1”.

<Simultaneous Broadcasting Switching>

[0651] In this method, as described in the first embodiment, it is also possible to set 4K broadcasting and 2K broadcasting in the same physical channel to simultaneous broadcasting. In a case where it becomes difficult to receive, demodulate, or display the 4K broadcasting due to any reason (for example, a defect of control information of the 4K broadcasting), it may automatically

or manually be switched into 2K simultaneous broadcasting as a backup. Moreover, in a case where a situation is improved and it becomes possible to receive the 4K broadcasting, it may automatically or manually be switched into 4K simultaneous broadcasting. If they are within the same physical channel, switching is executed smoothly. Further, the simultaneous broadcasting may be IP simultaneous broadcasting (Internet broadcasting) through a network (a communication transmission line). When it becomes possible to receive broadcast by radio waves in case of the Internet broadcasting, it is desirable to immediately switch into reception by the radio waves in order to alleviate traffic congestion of the network.

[0652] In a case where there is a plurality of options for the simultaneous broadcasting as the backup, it may automatically be switched in a priority order defined in advance. For example, in a case where it becomes difficult to receive the 4K broadcasting due to any reason while viewing the 4K broadcasting transmitted on a broadcast transmission path, switching into simultaneous broadcasting of the 2K broadcasting transmitted by the same physical channel of the broadcast transmission path may be treated as first priority, and switching into simultaneous broadcasting via a communication network may be treated as second priority. In general, broadcasting reception via the broadcast transmission path can be switched at a higher speed than reception of IP broadcasting via the communication transmission line. Therefore, it can be said that it is suitable to raise of switching into the simultaneous broadcasting of the broadcast transmission path, and reduce the priority of switching into the simultaneous broadcasting of the communication transmission. Further, as another process example of switching into the simultaneous broadcasting in a case where there is the plurality of options in the simultaneous broadcasting as the backup, the options may be presented to a user to cause the user to manually select one. Further, the backup may be VOD (Video On Demand) content. In case of switching into the VOD, it is desirable to reproduce the VOD content from the middle thereof so that the viewing content continues. FIG. 16 illustrates an example in which switching options into simultaneous broadcasting (alternative broadcast) are displayed by banner display **40020** in a case where a reception state of the 4K broadcasting deteriorates. FIG. 16 illustrates an example in which switching into terrestrial digital 2K broadcasting, which is the simultaneous broadcasting in the same physical channel, is executed when a blue button of a remote controller is pressed; and switching into the Internet broadcasting with the same content is executed when a red button is pressed. This option presentation of content switching and the selection thereof may be executed by a portable information terminal **700** on which a cooperation application with the broadcast receiving apparatus **100** is operating (FIG. 17). Note that in any of the above cases, the presentation of the options of the content switching may be limited to options that are available at that time.

[0653] Further, in a case where a reception status of 4K broadcasting deteriorates while recording the 4K broadcasting, it may be switched into 2K broadcasting of simultaneous broadcasting to continue the recording. Note that in a case where the 2K broadcasting is to be recorded in the middle thereof as a backup, 2K video may be up-converted into 4K video and recorded in order to make a video format continuous.

[0654] If the simultaneous broadcasting is executed in the same physical channel in this manner, program switching due to a change in the reception status is executed more smoothly. Further, by being combined with the Internet broadcasting, the options of the backup expand.

<Broadcast at the Time of Emergency Broadcast>

[0655] In a case where simultaneous broadcasting of 4K broadcasting and 2K broadcasting is being executed in a system for an advanced digital broadcasting service and emergency occurs, a transmitting side may switch each of the 4K broadcasting and the 2K broadcasting, by which the simultaneous broadcasting is executed, into emergency broadcast. Further, the transmitting side may switch only the 2K broadcasting into the emergency broadcast. Note that in a case where the transmitting side switches each of the 4K broadcasting and the 2K broadcasting, by which the simultaneous broadcasting is executed, into the emergency broadcast or in a case where the

transmitting side switches only the 2K broadcasting into the emergency broadcast, the broadcast receiving apparatus may forcibly switch a broadcasting program of the 4K broadcasting into viewing (reception and display) of a screen of the 2K broadcasting (an emergency broadcasting program) even while viewing (receiving and displaying) the broadcasting program of the 4K broadcasting. The switching process can be executed on the basis of emergency warning information received via the 4K broadcasting. The 2K broadcasting, which is technically easier than technically advanced 4K broadcasting, is more likely to be stable broadcast, and it is possible to transmit emergency information to viewers more reliably by preferentially receiving and displaying the 2K broadcasting at the time of the emergency. Further, after the broadcast receiving apparatus switches into the emergency broadcast, it may be possible to switch into 4K broadcasting that is not the emergency broadcast in accordance with determination of a viewer. Further, note that during emergency broadcast in which the emergency broadcast is executed by any one of 4K broadcasting and 2K broadcasting, which are related to simultaneous broadcasting, switching into the other broadcasting that is not the emergency broadcast may be prohibited.

[0656] Moreover, in a case where the viewer switches into and views the Internet broadcasting via the communication transmission line, at the same time as start of the emergency broadcast, it may forcibly be switched into viewing of broadcast that has been switched into the emergency broadcast among the broadcasts by the radio waves that are the broadcast transmission paths. In particular, in a case where the emergency broadcast is transmitted by a plurality of different transport stream formats, forcibly switching into viewing of emergency broadcast of the MPEG2-TS method, which can be received more stably, may be executed.

[0657] By forcibly switching into the emergency broadcast as described above, it is possible to transmit the emergency information to the viewers surely.

<Standby Synchronization>

[0658] As described above, in a case where switching of the content is executed depending upon a reception state of broadcast, it is switched into another simultaneous broadcasting due to deterioration of the reception state, and it is returned to original simultaneous broadcasting due to improvement of the reception state. At this time, if 4K broadcasting is an MMT method and 2K broadcasting is an MPEG2-TS method, switching between the simultaneous broadcasting is switching between the MMT method and the MPEG2-TS method. The switching between the simultaneous broadcasting is different from normal channel selection, and programs have the same content. Therefore, it is desirable that interruption at the time of switching is as short as possible. For this reason, in a case where display of a program or recording (although it is not displayed) is started, in order to quickly execute display or recording, the clock counter (**40001** or **40002**) of a corresponding method for the simultaneous broadcasting, which is not viewed (or displayed) currently, of the 4K broadcasting of the MMT method and the 2K broadcasting of the MPEG2-TS method may be synchronized (standby synchronization). By doing so, it is possible to shorten irruption time of video, or eliminate the irruption as compared with a case where the standby synchronization is not executed. In a case where standby synchronization is not executed, synchronization of an internal clock counter is started or restarted at the start time of the display or the recording using the corresponding internal clock counter (**40001** or **40002**). In this case, it takes time to establish the synchronization of the internal clock counter (**40001** or **40002**).

[0659] Further, the standby synchronization may be executed at all time in a state where time information of an NTP used in the MMT method or PCR used in the MPEG2-TS method can be received, or may be executed in a case where switching is predicted. For example, at a time point when it becomes a situation where it is predicted that a reception state of 4K broadcasting deteriorates and it will be difficult to continue the reception while the 4K broadcasting is being received, the standby synchronization of the PCR clock counter **40002** may be started by using the PCR of the corresponding 2K simultaneous broadcasting. Further, at a time point when it becomes a situation where it is predicted that the deteriorated reception status of the 4K broadcasting is

recovered and the reception will be able to be restarted, the standby synchronization of the NTP clock counter **40001** may be started by using the NTP of the 4K broadcasting for which the viewing is to be restarted.

[0660] In a case where the reception state of the 4K broadcasting is recovered after it becomes difficult to view the 4K broadcasting due to deterioration of the reception state and it is switched into the 2K simultaneous broadcasting, in a standby synchronization mode, the synchronization of the NTP clock counter **40001** can be recovered quickly, and return to the 4K broadcasting can be executed quickly. The return to the 4K broadcasting may be executed automatically when the reception status is improved, or the viewer may be informed that the return to the 4K broadcasting is possible after the synchronization of the NTP clock counter is recovered while viewing the 2K broadcasting as the backup, and may wait for a return instruction.

[0661] Even in a case where it is switched into the 2K broadcasting of the simultaneous broadcasting due to deterioration of the reception status of the 4K broadcasting and recording is continued while recording the 4K broadcasting, the synchronization of the internal clock counter (**40001** or **40002**) is established before switching by setting the standby synchronization mode when it is switched into the 2K broadcasting or when it is returned to the recording of the 4K broadcasting due to recover of the reception status of the 4K broadcasting. Therefore, it is possible to shorten the irruption time of video, or eliminate the irruption.

[0662] By executing the standby synchronization of the system clock in this manner, it is possible to improve viewing environment at the time of the switching across transmission methods.

<Adjustment of Leap Second>

[0663] Unlike the PCR, a time system by the NTP is an absolute time, and leap second adjustment occurs. As the leap second adjustment, there are a case where a leap second is inserted into a time by an NTP at a broadcast station side, and a case where the leap second is deleted from the time by the NTP at the broadcast station side. In a case where the leap second is to be inserted, 8:59:59 continues for two seconds. In a case where the leap second is to be deleted, one second after 8:59:58 becomes 9:00:00.

[0664] In the broadcast receiving apparatus **100**, any of two kinds of methods is used as a method of reflecting the leap second adjustment of an NTP system clock while using the NTP system clock (the NTP clock counter **40001** illustrated in FIG. 13) in a receiver by display or recording (referred to as “while viewing” in this embodiment).

(1) First Adjustment Operation

[0665] The leap second adjustment at the broadcast station side is reflected to the NTP system clock in the receiver by synchronizing with the time of the received NTP as it is. Namely, at a time point when the leap second adjustment is executed, it operates so that 8:59:59 continues for two seconds in a case where the leap second is to be added; and 8:59:59 does not exist in a case where the leap second is to be deleted.

(2) Second Adjustment Operation

[0666] A second adjustment operation is an adjustment operation different from the first adjustment operation. In the second adjustment operation, even in a case where the leap second adjustment is executed at the broadcast station side, the leap second adjustment is not reflected as the NTP system clock in the receiver, and a step of the time is continuously executed independently for one second. As a result, at a moment of the leap second adjustment, the receiver does not generate time overlapping, time discontinuity, or the like. For that reason, after the leap second adjustment at the broadcast station side, deviation of one second occurs between the time of the NTP transmitted by a broadcasting wave and the NTP system clock in the receiver. This state is continued until a predetermined operation such as a channel selection operation or switching into simultaneous broadcasting. During this period, it becomes a state where one second is left deviated between the time of the NTP and the NTP system clock in the receiver and the leap second adjustment at the broadcast station side is not reflected thereto. In this state, the NTP system clock in the receiver

may be a so-called free-run state where synchronization with the time of the NTP is not executed. [0667] Further, it may become the state where one second in seconds may be left deviated between the time of the NTP and the NTP system clock in the receiver, and may be continued as a synchronized state for units less than one second. Then, when the predetermined operation described above is executed, the leap second adjustment of the time of the NTP may be reflected to the NTP system clock in the receiver (deviation of one second is eliminated). By doing so, it is possible to prevent distortion of video based on these from being presented to the user without causing overlapping or discontinuity of the time of the NTP system clock in the receiver to occur during the user viewing. An operation of switching into full-screen EPG display may be included in the predetermined operation described above. The same effect can be obtained even though another operation of switching a video decoded screen of MMT using the NTP system clock in the receiver into another screen is included in the predetermined operation described above. Note that in a case where the NTP system clock in the receiver is not used after the channel selection or switching of simultaneous broadcasting and the standby synchronization described above is not executed, in the first place, it is not necessary to synchronize the time of the NTP with the NTP system clock in the receiver.

[0668] Next, a case where the NTP system clock in the receiver is not used (referred to as “during standby” in this embodiment) and leap second adjustment of the NTP system clock in the receiver is executed will be described. This case is a case where the standby synchronization described above is executed in preparation for start of display or start of recording of the MMT content. At this time, it is desirable to always synchronize with the time of the NTP in preparation for start of using the NTP system clock in the receiver. Therefore, the first adjustment operation described above is executed.

[0669] The broadcast transmitting system transmits information on the leap second adjustment (information indicating a type of a process for the leap second adjustment) as a descriptor “time_leap_indicator” together with information to be presented, and the broadcast receiving apparatus **100** adjusts the NTP system clock in the receiver on the basis of the information on the leap second adjustment. At the time of execution of the leap second adjustment for a system clock of the transmitting system, a value set to the “time_leap_indicator” indicates one in a case where a presentation time of the information to be presented is from 9:00:00 am on the day before insertion of a leap second to first 8:59:59 am on the day of the insertion of the leap second in Japan time; the value set to the “time_leap_indicator” indicates two in a case where the presentation time is from 9:00:00 am on the day before deletion of a leap second to 8:59:58 am on the day of the deletion of the leap second; and the value set to the “time_leap_indicator” indicates zero in the other cases.

[0670] Leap second correction states can be classified into some states depending upon the value of the descriptor for this leap second adjustment and the value of the NTP system clock in the receiver as follows. [0671] a: There is leap second correction within 24 hours (the value of the descriptor is one or two); [0672] b: There is no leap second correction within 24 hours (the value of the descriptor is zero); [0673] c: The leap second adjustment for the NTP system clock in the receiver is completed (in a synchronization state); [0674] d: It is 8:59:59, which is the two seconds after addition of a time; [0675] e: Asynchronization state (advance); and [0676] f: Asynchronization state (delay).

[0677] Here, the states c to f are states immediately after the time of the leap second correction, and normally, the leap second correction does not occur frequently. Therefore, they are also the state b. FIG. **18A** and FIG. **18B** illustrate operations of the leap second adjustment for the NTP system clock in the receiver together with these states. FIG. **18A** is a case where a time is to be added, and FIG. **18B** is a case where a time is to be deleted. They show a case where leap second adjustment is executed during viewing and a case where leap second adjustment is executed during standby (or during waiting). Note that “channel selection, etc.” represents a predetermined event such as switching into simultaneous broadcasting, a process of changing a viewed service by turning on the

power, or switching into full-screen EPG display, in addition to the channel selection. The subsequent states are shown assuming a case where a clock synchronization operation is executed. Moreover, a time and a time width of the “channel selection, etc.” are merely examples.

[0678] As illustrated in FIG. 18A and FIG. 18B, during viewing, not only a first adjustment operation but also a second adjustment operation may be adopted. In a case where the second adjustment operation is executed, deviation occurs between an NTP time of the transmitting system and a time of the NTP system clock in the receiver. A “mpu_presentation_time” descriptor, which is a presentation time based on a time of the transmitting system, is set to an MPU for video or the like. Therefore, the broadcast receiving apparatus 100 corrects a value of this descriptor so as to match with the time of the NTP system clock in the receiver. On the other hand, since it is not necessary to consider an influence on the viewing of the user during standby, the first adjustment operation may be adopted. Therefore, when the leap second adjustment is executed at the broadcast station side in the broadcast receiving apparatus 100 that adopts the second adjustment operation as an adjustment operation during viewing, the second adjustment operation is executed in a case where the broadcast receiving apparatus 100 is in a state of viewing a program to be reproduced using the NTP system clock in the receiver; and the first adjustment operation is executed in a case where the broadcast receiving apparatus 100 is in a state of the standby when any program to be reproduced using the NTP system clock in the receiver is not being viewed. By doing so, it is possible to execute the leap second adjustment quickly while suitably reducing an influence on the viewing of the user.

<Process In a Case Where There Are Plural Types of Descriptors for Leap Second Adjustment>

[0679] In the above, the number of the descriptor “time_leap_indicator” for the leap second adjustment is one type. However, multiple types of descriptors may be provided in accordance with types of information to be presented. For example, two types including one type for subtitle display (“reference_start_time_leap_indicator”) and one type for presentation of video and audio (“mpu_presentation_time_leap_indicator”) may be provided.

[0680] In a case where a different NTP system clock in the receiver is used for each content display, leap second adjustment for the NTP system clock in the receiver is executed using a descriptor for the leap second adjustment corresponding to each NTP system clock. For example, it is configured so as to include one type of NTP system clock in the receiver for presentation of video and audio, and to separately include one type of NTP system clock in the receiver for subtitle display. In this case, in general, the NTP system clock for presentation of video and audio in the receiver may use the “mpu_presentation_time_leap_indicator” as the descriptor for the leap second adjustment. Further, in general, the NTP system clock for subtitle display in the receiver may use the “reference_start_time_leap_indicator” as the descriptor for the leap second adjustment.

[0681] On the other hand, the NTP system clocks in the receiver may be shared by one. A value of a descriptor for leap second adjustment should be the same value regardless of a type of information to be presented. Therefore, the leap second adjustment for the NTP system clock in the receiver may be executed using an arbitrary descriptor for the leap second adjustment.

[0682] However, so long as the types are separated, there is a possibility that values of these descriptors for leap second adjustment are different values depending upon the types by any false operation of the transmitting system. In that case, in the broadcast receiving apparatus, one method is to comply with a value of a descriptor for leap second adjustment compatible with information for which higher accuracy of presentation synchronization is required. In this case, the “mpu_presentation_time_leap_indicator” for presentation of video and audio may be prioritized, and the descriptor may be used for subtitle display, which is presentation processing other than video and audio.

[0683] Further, in a case where complexity of the processing at the transmitting system side is considered, the descriptor for the leap second adjustment corresponding to simpler information is less likely to become an incorrect value. Therefore, it is one method to comply with that descriptor.

In this case, the “reference_start_time_leap_indicator” is prioritized.

[0684] Moreover, the information on the descriptor for leap second adjustment may be obtained from a time when the leap second correction is to be executed to a time when correction of the leap second is executed. In a case where the information is accurate and it becomes a value at which the leap second correction is to be executed, the value has a property of not changing until the leap second correction is executed. Therefore, a method of monitoring the values of the descriptors for the leap second adjustment regardless of the type and processing a value that appears most frequently among the monitored values as an accurate value is also effective.

[0685] By the method for time management described above, it becomes possible to control quick display switching and accurate display.

Third Embodiment

[Time Managing Method for Managing Content]

[0686] In the present embodiment, one example of a method for time management, which is used for a program guide generating process, control of recording reservation or viewing reservation, a copyright managing process such as temporary accumulation, or the like, will be described. Note that in order to simplify explanation thereof, in the present embodiment below, only different points from those according to the first embodiment will be described. Since points, which are not mentioned in the present embodiment, are similar to those according to the first embodiment, repeated explanation will be omitted.

[0687] A broadcast receiving apparatus according to the present embodiment has two types of time managing functions. A first time managing function is a time managing function based on an NTP or PCR, and is used for controlling content decoding or presentation synchronization as explained in the embodiments described above. A second time managing function is a time managed by “Time_Offset_Table”, and is used for the program guide generating process, the control of record reservation or view reservation, the copyright managing process such as temporary accumulation, or the like. Time information on the “Time_Offset_Table” is transmitted by the TOT illustrated in FIG. 9B in an MPEG2-TS method, or is transmitted by the MH-TOT (MH-Time_Offset_Table) illustrated in FIG. 10F in an MMT method. Note that since the both transmit the same information, the TOT in the MPEG2-TS method and the MH-TOT in the MMT method are collectively referred to as “TOT” below in the present embodiment so long as no particular distinction is required.

[0688] As described in the above embodiments, since the NTP or the PCR is used for presentation synchronization of the content, it is possible to manage a time less than a second. However, the information transmitted by the TOT is as follows. It is assumed that a broadcast receiving apparatus **100** can obtain a current date and Japan standard time by the TOT. FIG. 19A illustrates one example of a data structure of the TOT. The broadcast receiving apparatus **100** can obtain a current date and a current time from a “JST_time” parameter of the TOT. As illustrated in FIG. 19B, the “JST_time” parameter contains information on lower 16 bits of coded data of the current date by Modified Julian Date (MJD) and information on 24 bits obtained by representing Japan Standard Time (JST) by six 4-bit binary-coded decimals (BCD). The current date can be calculated by applying a predetermined operation to 16-bit coded data of the MJD. The six 4-bit binary-coded decimals represent “hours” with double digits in decadal system by two 4-bit binary-coded decimal, represent “minutes” with double digits in decadal system by next two 4-bit binary-coded decimal, and represent “seconds” with double digits in decadal system by the last two 4-bit binary-coded decimal.

[0689] In the broadcast receiving apparatus **100** according to the present embodiment, the time managing function based on “JST_time” containing 24-bit information represented by the six 4-bit binary-coded decimals of the TOT is used for a process of presenting a current time to a user or each process of handling an event information table. The event information table is transmitted by the EIT illustrated in FIG. 9B in the MPEG2-TS method, or is transmitted by the MH-EIT (MH-Event_Information_Table) illustrated in FIG. 10F in the MMT method. Note that since the both

transmit information having the same structure, the EIT in the MPEG2-TS method and the MH-EIT in the MMT method are collectively referred to as “EIT” below in the present embodiment so long as no particular distinction is required. FIG. 20 illustrates one example of a data structure of this EIT. A “start time” illustrated in FIG. 20 indicates a start time of an event, and is stored in the same format as that illustrated in FIG. 19B. Further, a “duration” illustrated in FIG. 20 indicates a duration time of the event, and is stored in the format of “hours”, “minutes”, and “seconds” each having double digits in decadal system as the 24-bit information represented by the six 4-bit binary-coded decimal.

[0690] Generally, in the process of presenting the current time to the user in the broadcast receiving apparatus 100, accuracy to less than a second is rarely required. Further, as described above, each time information described in the EIT has good consistency with the format of the time information in the TOT. Specifically, the processes using the EIT include the program guide generating process, the control of record reservation or view reservation, and the copyright managing process such as temporary accumulation. It is rare that any process requires the accuracy of less than a second, and accuracy of one second unit is sufficient. Thus, in consideration of the required accuracy of the time management and the data format, usage of the time managing function based on the TOT for the process of presenting the current time or each process using the EIT reduces the waste of processing and reduces the cost.

[0691] Here, in the broadcast receiving apparatus 100 according to the present embodiment, both a program of the MPEG2-TS method and a program of the MMT method may be received by one physical channel. At this time, for the TOT, the TOT of the MPEG2-TS method and the MH-TOT of the MMT method can be received, but the TOT for the program being viewed is used.

<Modification Example of Time Management by TOT>

[0692] However, considering a 2K program of an MPEG2-TS method and a 4K program of an MMT method are broadcasted as simultaneous broadcasting and a reception status of the 4K program deteriorates for any reason to be switched into the 2K program as a backup, time management for a broadcast receiving apparatus 100 may be executed using a TOT transmitted by the more stable MPEG2-TS method regardless of which program is being viewed.

[0693] Moreover, in a case where it is switched into viewing of the VOD content via the Internet as the content for a backup, the cause is the deterioration of the reception status of the 4K program in the first place. Therefore, as a TOT used for the time management of the broadcast receiving apparatus 100, it is preferable to use the TOT transmitted by the MPEG2-TS method. A method of executing time management by an MH-TOT transmitted by the MMT method while viewing the 4K program, and switching into time management by the TOT transmitted by the MPEG2-TS method after switching into viewing of the VOD content for the backup is also acceptable.

[0694] As explained above, by using the TOT transmitted by the MPEG2-TS method, which can be received more stably by the physical channel being received, for the time management of the broadcast receiving apparatus 100, more stable time management can be realized.

[0695] According to the system for the advanced digital broadcasting service having a part or all of the functions according to the embodiments of the present invention that have been described above, it becomes possible to provide a transmission technique and a reception technique for the advanced digital broadcasting service with higher functionality in consideration of compatibility with the existing digital broadcasting service. Namely, it is possible to provide a technique of transmitting or receiving the advanced digital broadcasting service more suitably.

[0696] As described above, the examples of the embodiments according to the present invention have been described using the first embodiment to the third embodiment. However, the configuration for realizing the techniques according to the present invention is not limited to the first and second embodiments, and various modification examples can be considered. For example, a part of the configuration of one embodiment can be replaced by the configuration of the other embodiment, and the condition of one embodiment can also be added to the configuration of the

other embodiment. All of these belong to the scope of the present invention. Further, numerical values or messages appearing in the specification and the drawings are merely one example, and the effects of the present invention may not be impaired even though different ones are used. [0697] The functions of the present invention described above may be realized by hardware, for example, by designing a part or all of them by an integrated circuit. Further, a microprocessor unit or the like interprets and executes operating programs for realizing the respective functions, whereby the functions may be realized by software. Hardware and software may be used together. [0698] Note that the broadcast receiving apparatus **100** may become a state where the software for controlling the broadcast receiving apparatus **100** is stored in at least one of the ROM **103** or the storage **110** of the broadcast receiving apparatus **100** in advance at the time of product shipment. The broadcast receiving apparatus **100** may be configured so as to obtain the software from the other application server on Internet **800** via the LAN communication unit **121** after the product shipment. Further, the broadcast receiving apparatus **100** may obtain the software stored in a memory card or an optical disc via the extension interface **124**. Similarly, the portable information terminal **700** become a state where the software for controlling the portable information terminal **700** is stored in at least one of a ROM or a storage of the portable information terminal **700** in advance at the time of product shipment. The portable information terminal **700** may be configured so as to obtain the software from the other application server on the Internet **800** via a LAN communication unit or a mobile phone network communication unit of the portable information terminal **700** after the product shipment. Further, the portable information terminal **700** may obtain the software stored in a memory card or an optical disc via an extension interface of the portable information terminal **700**.

[0699] Further, the control lines and the information lines illustrated in the drawings indicate what are considered necessary for explanation thereof, and do not necessarily indicate all of the control lines and the information lines on a product. In fact, it may be considered that almost all configurations are connected mutually.

REFERENCE SIGNS LIST

[0700] **100**: broadcast receiving apparatus, **101**: main controller, **102**: system bus, **103**: ROM, **104**: RAM, **110**: storage (accumulator), **121**: LAN communication unit, **124**: extension interface, **125**: digital interface, **130C**, **130T**, **130L**, **130B**: tuner/demodulator, **140S**, **140U**: decoder, **180**: operation input unit, **191**: video selector, **192**: monitor, **193**: video output unit, **194**: audio selector, **195**: speaker, **196**: audio output unit, **180R**: remote controller, **200**, **200T**, **200L**, **200B**: antenna, **300**, **300T**, **300L**: radio tower, **400C**: headend of cable television station, **400**: broadcast station server, **500**: service provider server, **600**: mobile phone communication server, **600B**: base station, **700**: portable information terminal, **800**: Internet, **800R**: router device.

Claims

1. A broadcast receiving apparatus comprising: a receiver configured to receive a 4K broadcasting service and a 2K broadcasting service and demodulate the 4K broadcasting service and the 2K broadcasting service; a display; a remote control receiver configured to receive a signal from a remote controller; and a controller, wherein the receiver is configured to receive the 4K broadcasting service transmitted by a first transmission method and demodulate the 4K broadcasting service, and the controller is configured to control the display to display a broadcasting program of the 4K broadcasting service, wherein the receiver is configured to receive the 2K broadcasting service transmitted by a second transmission method and demodulate the 2K broadcasting service, and the controller is configured to control the display to display a broadcasting program of the 2K broadcasting service, wherein the controller is configured to control clock synchronization processing for the 4K broadcasting service and clock synchronization processing for the 2K broadcasting service, and wherein the controller is

- configured to control: the clock synchronization processing for the 2K broadcasting service to continue even when the broadcasting program of the 4K broadcasting service is displayed on the display; and the clock synchronization processing for the 4K broadcasting service to continue even when a broadcasting program of the 2K broadcasting service is displayed on the display.
2. The broadcast receiving apparatus according to claim 1, wherein the first transmission method is an MMT method.
 3. The broadcast receiving apparatus according to claim 1, wherein the second transmission method is an MPEG2-TS.
 4. A broadcast receiving apparatus comprising: a receiver configured to receive a 4K broadcasting service and a 2K broadcasting service and demodulate the 4K broadcasting service and the 2K broadcasting service; a display; a remote control receiver configured to receive a signal from a remote controller; a controller, a non-volatile memory configured to store operating software; and a random access memory configured to be used for the operating software, wherein the receiver is configured to receive the 4K broadcasting service transmitted by a first transmission method and demodulate the 4K broadcasting service, and the controller is configured to control the display to display a broadcasting program of the 4K broadcasting service, wherein the receiver is configured to receive the 2K broadcasting service transmitted by a second transmission method and demodulate the 2K broadcasting service, and the controller is configured to control the display to display a broadcasting program of the 2K broadcasting service, wherein the controller is configured to control clock synchronization processing for the 4K broadcasting service and clock synchronization processing for the 2K broadcasting service, and wherein the controller is configured to control: the clock synchronization processing for the 2K broadcasting service to continue even when the broadcasting program of the 4K broadcasting service is displayed on the display; and the clock synchronization processing for the 4K broadcasting service to continue even when a broadcasting program of the 2K broadcasting service is displayed on the display.
 5. The broadcast receiving apparatus according to claim 4, wherein the first transmission method is an MMT method.
 6. The broadcast receiving apparatus according to claim 4, wherein the second transmission method is an MPEG2-TS.
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